Amazon L5 Applied Scientist

Onsite interview question bank

01. Loop Format and Hiring Bar: Amazon L5 Applied Scientist (Applied Scientist I I)

- Format notes (what to expect, 2025-2026)
- Questions
- References

02. ML Breadth Round: Classical ML, Statistics, Probability, Optimization, Deep Learning, NLP/CV

- Format notes
- A. Linear models
- B. Trees, ensembles and other classical models
- C. Bias-variance, overfitting, regularization and model selection
- D. Evaluation metrics, calibration and class imbalance
- E. Unsupervised learning and dimensionality reduction
- F. Probability and statistics
- G. Optimization
- H. Deep learning fundamentals
- I. Sequence models and classic NLP
- J. Transformers and LLMs (now a breadth staple)
- K. Computer vision fundamentals
- L. Applied breadth: production, ranking, forecasting and data issues
- Prioritization for an L5 loop (based on report frequency)
- References

03. ML Depth and Science Application: Research and Project Deep Dives, and Applying Science to Amazon Problems

- Format notes
- A. Opening the deep dive: framing, scope and ownership
- B. Problem formulation and data
- C. Modeling choices and alternatives
- D. Loss, optimization and training dynamics
- E. Evaluation and scientific rigor
- F. Deployment, scale and post-launch
- G. Impact, business value and reflection
- H. Research talk, job talk or paper round (when the loop includes one)
- I. Depth in your specialty: NLP and LLMs (common for 2025-2026 L5 loops)
- J. Depth in your specialty: search, ranking and recommendations
- K. Depth in your specialty: vision and multimodal
- L. Depth in your specialty: forecasting, optimization and causal inference
- M. Science application: reported Amazon problems
- N. Science application: likely Amazon problems by org (L5, 2025-2026)
- O. Applying your own research to Amazon
- P. LP probes woven into the depth round
- References

04. Coding Round for Applied Scientists: DSA Questions Reported at Amazon

- Format notes for this round
- A. Arrays, strings and hashing
- B. Intervals and sorting
- C. Two pointers and sliding window
- D. Linked lists and stacks
- E. Trees and BSTs
- F. Graphs and grids
- G. Heaps, top-k and streaming
- H. Binary search
- I. Dynamic programming

J. Randomization and sampling
K. Design-style data structures
L. Practical data-processing algorithms
References

05. ML Coding: Implementing Algorithms and Metrics from Scratch
Format notes for this round
A. Evaluation metrics from scratch
B. Clustering, neighbors and similarity search
C. Linear models, optimization and estimation
D. Trees and ensembles
E. Neural networks and backpropagation
F. Attention, transformers and LLM components
G. Probability, sampling and statistics in code
H. Recommendation, search and NLP building blocks
I. Data manipulation and ML pipeline coding
J. Round-level probes (how the interviewer pushes at L5)
Prep priorities (derived from the tags above)
References

06. ML and Science System Design: Recommendations, Search, Ads, Fraud, Forecasting, Alexa/Rufus LLMs, Pricing and Supply Chain
Format notes for this round
A. Generic framing (the questions every design round opens or pivots with)
B. Recommendations and personalization
C. Product search and ranking
D. Advertising (Sponsored Products, DSP, Ads Trust)
E. Fraud, abuse, trust and payments
F. Computer vision and catalog quality
G. Forecasting, supply chain and inventory (SCOT)
H. Pricing, causal inference and experimentation
I. Alexa, Rufus and LLM-powered products
J. Scaling, MLOps and lifecycle (probes across any design)
K. L5-bar probes and LP crossovers inside the design round
L. Team-specific "warm-up" prompts to rehearse (inferred from postings and Amazon Science)
References

07. LLM and GenAI Questions: Transformers, Fine-Tuning, RLHF/DPO, RAG, Evaluation, Hallucination, Efficiency, and Amazon Product Context (Bedrock, Nova, Rufus, Alexa+)
Format notes for this round
A. Transformer fundamentals
B. Inference, decoding and long context
C. Pretraining, fine-tuning and PEFT
D. Alignment: RLHF, DPO and beyond
E. Retrieval-augmented generation (RAG)
F. Agents and tool use
G. Evaluation
H. Hallucination and grounding
I. Efficiency, serving and cost
J. Multimodal and speech (team-dependent)
K. Applying GenAI to Amazon problems (science-application flavor)
L. Depth probes on your own LLM project (depth-round style)
M. GenAI behavioral and LP crossovers
Coverage notes for prep
References

08. Statistics, Causal Inference and A/B Testing: Hypothesis Tests, Power, Variance Reduction, Weblab, Bandits and Forecasting

- Format notes for this round
- A. Probability and statistical foundations
- B. Hypothesis testing and p-values
- C. A/B test design, power and sample size
- D. Metrics: primary, guardrail and long-term
- E. Variance reduction and sensitivity
- F. Multiple testing, peeking and sequential testing
- G. Pitfalls at Amazon scale: interference, novelty and marketplaces
- H. Weblab, launch decisions and post-launch monitoring
- I. Causal inference without clean randomization
- J. Bandits and adaptive experimentation
- K. Forecasting and time series
- L. Applied stats case questions (mixed, interview-style)
- References

09. Leadership Principles Bank: Behavioral Questions for the L5 Applied Scientist Loop, with L5 Probes and Bar Raiser Favorites

- Format notes for the behavioral portion
- Questions
- Bar raiser favorites and pressure probes
- L5 calibration probes (any interviewer, layered on any story)
- Generic deepening probes interviewers attach to any LP story
- References

10. Bar Raiser Round: Reported Questions, Mixed Technical and Behavioral Probing, and Follow-up Patterns

- Format notes
- Questions
- References

11. Amazon Science Team Context: What Each Hiring Org Asks, and the Recent Amazon Science Work to Know

- Format notes for this round
- A. Cross-team "why this team" and team-fit questions
- B. Search (Stores Foundational AI / Search Science, formerly M5)
- C. Advertising (Sponsored Products, DSP, Ads Trust, Ads measurement)
- D. Alexa AI, AGI and Amazon Nova (foundation models)
- E. AWS AI (Bedrock, SageMaker, Q / coding agents, AI research)
- F. Rufus (generative shopping assistant)
- G. Personalization and recommendations (homepage, Amazon Music, explanations)
- H. SCOT (Supply Chain Optimization Technologies), forecasting and inventory
- I. Pricing and causal inference
- J. Prime Video (and other Amazon content and devices teams)
- K. Payments, trust, fraud and abuse
- L. "Have you read our work?" Recent Amazon Science publications as interview questions
- M. L5-bar probes that recur across all teams
- References

12. ML System Design Reference Diagrams: Six Canonical Amazon Prompts and a 45-Minute Whiteboard Template

- How the [Reported] / [Likely] tag system works
- Format notes for this round
- Part 1. The whiteboard template (draw this first, for any prompt)
- Part 2. Recommendations: candidate generation -> ranking -> re-ranking
- Part 3. Product search: retrieval -> ranking

Part 4. Sponsored ads: CTR prediction + auction
Part 5. Fraud detection with streaming features
Part 6. SCOT demand forecasting
Part 7. Rufus / Alexa-style RAG assistant
Part 8. Cross-cutting probes on any diagram (feedback, monitoring, retraining)
Part 9. L5-bar probes inside the design round
References

13. ML System Design: Product Tagging, Catalog Attribute Extraction and PDF/Document Understanding

Format notes for this round
Reference diagram 1: Product tagging and attribute extraction (whiteboard scaffold)
Reference diagram 2: PDF and document understanding pipeline (whiteboard scaffold)
A. Framing and scoping (both systems)
B. Product taxonomy and product-type classification
C. Multi-label tagging and attribute schema
D. Attribute value extraction (classical and LLM)
E. LLM verification and LLM-as-labeler
F. Cold-start tagging
G. Human-in-the-loop labeling and review queues
H. Metrics and evaluation for tagging
I. Document ingestion, OCR and layout
J. Key-value, table and field extraction
K. Document classification, splitting and routing
L. Confidence thresholds, straight-through processing and human review
M. Multimodal VLM and LLM extraction for documents
N. Data, labeling and privacy for documents
O. Failure modes, monitoring and drift (both systems)
P. Trade-offs, cost and the L5 bar
References

14. End-to-End GenAI System Design: Whiteboard Diagrams and Probe Questions for RAG, Agents, Evaluation and Guardrails, and Fine-Tune-to-Serve

How the [Reported] / [Likely] tag system works
Format notes for this round
Part 0. The universal GenAI whiteboard skeleton
Part 1. RAG shopping assistant (Rufus / Alexa for Shopping style)
Part 2. LLM agent with tool use
Part 3. LLM evaluation and guardrail pipeline
Part 4. Fine-tuning to serving infrastructure (Bedrock / Nova)
Part 5. Online A/B and human-evaluation loop (closing the system)
Part 6. L5-level cross-cutting probes (any of the four designs)
Part 7. Additional probes (October 2026 review pass)
References

01. Loop Format and Hiring Bar: Amazon L5 Applied Scientist (Applied Scientist II)

> Scope: how the L5 Applied Scientist (AS II) process is structured, who sits in each seat, how the bar raiser and debrief work, how L5 is calibrated against L4 and L6, and the questions each seat tends to ask. Later chapters cover the technical content (ML breadth, depth, coding, ML design, LPs) in full. This chapter covers the format-defining questions and probes.
>
> Verification status (2026-10-14 review): a re-check of the cited URLs could not be completed. The egress proxy rejected every one of the 43 reference hosts, and no web search was available. The [Reported] tags and citations are therefore unchanged from the original research pass and have not been re-verified. No new questions were added, because none could be sourced.
>
> Research note: everything here comes from Amazon's own pages, candidate self-reports (Glassdoor, Blind, LeetCode, 1point3acres, Medium) and prep-vendor guides (IGotAnOffer, Interview Kickstart, Dataford and others). The live page fetcher could not reach these sites from this environment, so the content was read through search-engine excerpts of each page. Vendor guides are marketing material and are weighted below Amazon's own pages and first-person reports.

Format notes (what to expect, 2025-2026)

Pipeline

Stage	What's reported	Sources
Recruiter call	Logistics, level target, team, and sometimes a light background check on your ML experience.	[22] [24]
Technical phone screen(s)	Amazon says there are one or two screens of about 60 minutes each, run by "a senior leader on our team" (in practice usually a senior or principal scientist, sometimes the hiring manager). Candidates report a mix of ML fundamentals, a project walk-through, 1-2 LP questions, and sometimes one LeetCode easy/medium or a pandas/data-manipulation task. Some screens have no coding at all.	[1] [9] [20] [27] [33]
Onsite "loop" (now usually virtual on Chime)	Amazon's official page says "four 55-minute interviews ... with members of our science community." Candidate reports for AS II more often describe 5 rounds of 55-60 minutes, sometimes spread over 1-3 days. Some teams add a job talk of about 45-50 minutes plus Q&A.	[1] [6] [10] [17] [28]
Debrief	Interviewers submit written feedback and a vote independently. The bar raiser chairs the debrief, and the hiring manager and bar raiser make the final decision.	[3] [4] [21] [30]

Stage	What's reported	Sources
Team match (sometimes)	Some candidates pass the loop and are then told they need a team match before an offer letter. Others are "recycled" to another team's opening without re-interviewing. Some are asked to do another loop.	[11] [18] [34]

Typical L5 AS loop composition (as candidates report it)

- **ML/Science Breadth** (about 1 round): fast survey of classical ML, deep learning, statistics, and, increasingly, transformers, LLMs and GenAI. Reported examples include bias-variance, bagging vs boosting, transformer attention, BERT vs GPT, DPO vs RLHF, and how SAM works. [6][7][26]
- **ML/Science Depth** (about 1 round): a deep dive on one project or publication from your resume, where the interviewer keeps drilling down on your choices. [10][22][28]
- **Science Application / ML System Design** (about 1 round, often the hiring manager): a team-specific, open-ended problem. Blind and vendor sources say L5 loops usually include it, while L4 loops may not. [6][12][27]
- **Coding** (1 round, sometimes 2): usually LeetCode easy to medium, sometimes pandas/numpy, ML-from-scratch, or OOD. A Blind commenter says the bar is about SDE I level. [10][12][20][27]
- **Bar Raiser** (1 round): mostly LP/behavioral, run by a trained interviewer from outside the hiring org. It may also include technical probing. Candidates report BRs from non-tech orgs and Principal TPMs. [6][11][13]
- **Leadership Principles in every round:** Amazon's prep page says each interviewer typically asks 2-3 behavioral questions. Candidates report about 15 minutes of each round, and one L5 report put behavioral questions at over 60% of the loop. [1][19][27]

Who owns what

- Each interviewer is pre-assigned a competency (breadth, depth, application, coding) plus about 2-3 Leadership Principles, so the loop covers the full LP set without repeating. Amazon's page confirms that interviewers are assigned competencies "based off of our Leadership Principles". The per-LP split comes from prep guides, not from Amazon. [1][36][23]
- **Hiring manager:** usually owns the Science Application / design round and team fit. With the BR, the HM makes the final hire/no-hire call. [3][37]
- **Bar raiser:** a volunteer interviewer who is nominated, trained, and shadows veteran BRs for several months. The BR comes from outside the hiring team, chairs the debrief, and holds a de facto veto, which former Amazonians say is rarely used explicitly. The BR's job is to judge whether the candidate is better than at least half of current employees at that level. That 50% bar is folklore quoted by one BR, not a published standard. [3][4][5][29][37]

What the L5 bar is (calibration)

- **Level map:** AS I = L4, AS II = L5, Senior AS = L6. A PhD is often credited as about 2-3 years of experience, and new PhDs are frequently interviewed at L5. One example

L5 posting asks for a Master's, or a PhD plus 5+ years of practical ML. Levels.fyi data puts typical L5 experience at 5-7 years. [15][31][32]

- **L5 vs L4:** L5 is expected to own projects end to end, including design, to make the technical calls, and to influence peers on the team, with some spillover to sister teams. Vendor debriefs say L5 LP answers face 2-3 layers of follow-up, against 1 layer at L4. [15][27]
- **L5 vs L6:** L6 needs proof of large scope, influence outside the team, mechanisms, and multiple strong examples. Blind commenters say the coding bar is similar at L5 and L6, and that the LP narratives are what differentiate them. [15][16][25]
- **Leveling outcomes:** downleveling from L6 to L5 and from L5 to L4 is common. A BR can push the level down even when the HM is happy (one L5 candidate's BR thought they were "at the cusp of L4 and L5"). One Blind comment says that a weak coding round with strong science can lead to a Research Scientist offer instead. [11][12][16]

Questions

Tag legend:

- **[Reported]**: a candidate or source attributes this question, or a close paraphrase, to Amazon AS (or, where noted, Amazon DS or Senior AS) interviews.
- **[Likely]**: an inference grounded in the format, the LP rubric and the team's domain. It has not been seen verbatim.

A. Recruiter call and technical phone screen

A1. Recruiter / logistics screen

1. Walk me through your background and why Amazon, why Applied Science, and why this team. [Likely]
    - Which Amazon products or science areas have you followed, and why?
    - Are you open to other teams if this one isn't the right fit (team match / recycle)?
2. How many years of hands-on applied ML experience do you have post-degree, and how much of it shipped to production? [Likely]
    - Which of your projects would you consider "L5 scope," and why?
3. Are you interviewing at other companies, and what is your timeline? [Likely]

A2. Technical phone screen (senior scientist or HM, about 60 min)

4. Walk me through the research or project on your resume you're most proud of. [Reported] [33][39]
    - Why that modeling approach over the obvious baseline?
    - What was the business metric, and how did the offline metric map onto it?
5. Explain the bias-variance trade-off and how it shows up in a model you've built. [Reported] [6][20]
    - How did you diagnose which side you were on?
6. How does hyperparameter tuning work, and how do you evaluate models? [Reported] [20]

    - How do you avoid leakage between tuning and evaluation?
7. Explain decision trees, batch normalization and word embeddings. [Reported] [39]
    - Why does batch norm behave differently at train and inference time?
8. Write out the matrix form of OLS and derive the closed-form solution. [Reported] [20]
    - What happens when X^T X is singular, and how does ridge fix it?
9. Load this dataset in pandas, clean it, engineer features, and fit a model. Which breadth questions does each step raise? [Reported] [9][20]
    - How would you handle missing values and class imbalance here?
10. Solve one LeetCode-style easy/medium problem (one candidate reported a graph problem, others arrays or trees). [Reported] [10][20]
    - What is the time and space complexity, and can you do better?
11. Tell me about a time you made a technical decision without complete data. This is also reported as a bar raiser question. [Reported] [26]
    - What would have changed your mind?
12. (Ads, 2024-25 report) Explain how transformers and attention work. One candidate with strong classical ML answers was rejected after not preparing for this. [Reported] [43]
    - How does a decoder-only LLM generate text at inference time?

B. Job talk / research presentation (team-dependent)

13. Present a recent piece of your work, covering only your own contributions, in about 45-50 minutes with about 10 minutes of Q&A. [Reported] [17][34]
    - Which parts were yours, and which were your collaborators'?
    - What are the impact numbers, and how were they measured?
14. Why did this problem matter to the business or customer, and how did you know? [Likely]
    - If you had to ship a version in four weeks, what would you cut?
15. What baselines did you compare against, and why are they the right ones? [Likely]
    - Which ablation would most likely break your conclusion?
16. Which paper references does this build on, and what is novel versus engineering? [Reported] [17]
17. How would this work transfer to one of our team's problems? [Likely]
    - What data would you need from us that you don't see yet?

C. ML/Science Breadth round

18. Compare bagging and boosting. When would each fail? [Reported] [6][7]
    - Why does boosting overfit noisy labels more readily?
19. Explain L1 and L2 regularization and their effect on the learned model. [Reported] [24]
20. Explain the transformer attention mechanism. [Reported] [7]
    - Why is standard self-attention expensive for long contexts, and how does Longformer-style attention help? [Reported] [7]
21. How does BERT differ from GPT? [Reported] [7]
22. Compare DPO and RLHF. [Reported] [7]
    - What failure modes of PPO-based RLHF does DPO avoid, and which does it introduce?

23. How does KV caching work during autoregressive inference? [Reported] [7]
24. Describe how SAM (Segment Anything Model) works and its key architectural components. [Reported] [6][26]
    - How would you improve CLIP for a product-image retrieval use case? [Reported] [26]
25. Compare gradient descent with Adam. When might a second-order optimizer help? [Reported] [22]
26. State and use Hoeffding's inequality. [Reported] [22]
27. Explain hypothesis testing and design an A/B test for a model change. [Reported] [9] [24]
    - If treatment improves the target metric but a population-level metric drops, what do you do? [Reported] [24]
28. Cover supervised vs unsupervised learning, dimensionality reduction, neural network basics and evaluation metrics, in rapid fire. [Reported] [28]
29. Pick a technique from a field adjacent to your PhD and explain it at grad-course depth. [Likely]
    - Where does your knowledge stop, and how would you find out more?

D. ML/Science Depth round

30. Walk through a project or publication from your resume, end to end. [Reported] [10] [22]
    - Why this loss and this architecture? What did you try that didn't work?
    - What did you personally build, versus what the team built?
31. You proposed an algorithmic fix just now. Go one level deeper: derive it or justify it mathematically. [Reported] [7]
32. Which alternatives did you consider, and what trade-offs drove the final choice? [Reported] [22]
33. What was the business impact, and how did you attribute it to the model rather than to confounders? [Reported] [28]
34. If you restarted this project today with LLMs and foundation models available, what would change? [Likely]
35. What was the hardest unknown problem in that project, and how did you get unstuck? [Likely]
    - How did you know your solution was correct?
36. What did the data look like? How were labels produced, and how noisy were they? [Likely]
    - How did label noise bound your achievable accuracy?
37. Did the model hold up after launch? How did you monitor drift? [Likely]
    - Did the gains hold six months later? [Likely]

E. Science Application / ML system design (often the hiring manager)

38. Design an agentic recommendation system for Amazon, including component architecture, model choices, and quality and safety evaluation. [Reported] [26]
39. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [26]
40. Design a GitHub Copilot-style coding agent (Senior AS, Boston report). [Reported] [7]

41. Design an enterprise RAG service. [Reported] [35]
42. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [24]
43. Here's a vague business goal from our team. Frame it as an ML problem, choose metrics, and outline v1. [Likely]
    - What's your online/offline metric pair, and how do they disagree?
    - What's the cheapest baseline you would ship first?
44. Evaluate an LLM agent with metrics beyond accuracy: tool use, hallucination, calibration. [Reported] [7]
45. Walk me through how you'd scale this from a prototype to Amazon traffic. [Likely]
    - Where do latency and cost bite first?
46. What would make you kill this project? [Likely]
47. (Product-thinking variant, Seattle 2026 report) Pick an agentic product we could build. Who's the customer, and how do you measure success? [Reported] [6]

F. Coding round

48. String compression (in place, run-length). [Reported] [6][26]
49. Top-K elements in a stream or array, in Python. [Reported] [27]
    - Heap versus quickselect: which would you choose, and when?
50. Design a rental system (object-oriented design). [Reported] [27]
51. A LeetCode medium on trees, followed by ML questions in the same round. [Reported] [7]
52. Implement a simple ML algorithm or training loop from scratch (e.g., logistic regression with gradient descent, k-means). [Reported] [12]
53. A dynamic programming problem, such as an edit-distance-style or knapsack-style problem. [Reported] [7]
54. Compute or minimize a mathematical function in code (e.g., numerically). [Likely]
55. Write code on LiveCode without running it, then test it by hand. [Likely]
    - What edge cases would break this, and what tests would you write?

G. Bar raiser round (LP-heavy, outside interviewer)

56. Tell me about a time you had a conflict with a coworker or manager and how you approached it (Have Backbone; Disagree and Commit). [Reported] [23][41][42]
    - What did you do once the decision went against you?
57. Tell me about a time you had a major conflict with a Principal Engineer. How did you resolve it? [Reported] [25]
58. Tell me about a project failure and how you managed the aftermath. [Reported] [25]
    - What would you do differently? [Reported] [22]
59. Tell me about a time your model underperformed and you had to diagnose the issue. [Reported] [24]
    - What did you learn, and how did you pivot? [Reported] [26]
60. (Pushback probe) I think your approach was wrong. Convince me otherwise. Candidates report BRs deliberately telling them they were wrong. [Reported] [41][42]
    - What data would make you concede?
61. How was that measured? [Reported] [41][42]
62. What was your exact individual contribution, stated without "we"? [Reported] [41][42]

63. Tell me about a time you took initiative without waiting for permission. One rejection was reportedly explained as "seemed to wait for permission." [Likely]
64. Tell me about a time you raised the quality bar on a deliverable even though it delayed you (Insist on the Highest Standards). [Likely]
65. Tell me about a time you simplified a complex scientific approach so it could ship (Invent and Simplify). [Likely]
66. Tell me about a time you had to decide quickly with incomplete information and the decision turned out wrong (Bias for Action / Are Right, A Lot). [Likely]
67. Tell me about a time you dove into data or code to find a root cause that others missed (Dive Deep). [Likely]
68. Tell me about the most ambitious science bet you have proposed (Think Big). [Likely]
69. Tell me about a time you did more with less: compute, labels or headcount (Frugality). [Likely]
70. Tell me about a time you mentored someone or raised the bar of your team's science (Hire and Develop the Best). [Likely]
71. Tell me about a time you considered the downstream harm or fairness impact of a model (Success and Scale Bring Broad Responsibility). [Likely]
72. Tell me about a time you made your team's work environment better (Strive to be Earth's Best Employer). [Likely]
73. (TPM- or non-science BR) How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [13]
    - When did a science timeline slip, and how did you tell stakeholders?
74. (TPM BR) A few stats/math checks, such as interpreting a p-value or confidence interval. [Reported] [13]

H. LP questions embedded in technical rounds (each interviewer owns about 2-3 LPs)

75. Tell me about a time you started from the customer and worked backwards to the science problem (Customer Obsession). [Likely]
    - Who was the customer, and how did you get their signal?
76. Tell me about a time you took ownership of something outside your formal scope (Ownership). [Likely]
77. Tell me about a time you delivered a model under a hard deadline (Deliver Results). [Likely]
    - What did you descope, and what did that cost?
78. Tell me about a time you learned a new area fast to unblock a project (Learn and Be Curious). [Likely]
79. Tell me about a time you earned the trust of an engineering or product partner who doubted your science (Earn Trust). [Likely]
80. Tell me about a time you were wrong about a technical call. How did you find out? (Are Right, A Lot). [Likely]
81. Tell me about a time you pushed back on a stakeholder's metric or launch decision using data. [Likely]
    - Did you commit after being overruled?

I. Level-calibration probes (L5 vs L4 vs L6 signal)

82. What was the largest scope you have owned end to end, from problem framing to production? [Likely]
    - Who else depended on it, beyond your own team?
83. Tell me about a time your work influenced a sister team's roadmap or technical direction. [Likely]
84. Tell me about a time you set the technical direction for other scientists or engineers, not just executed it. [Likely]
    - How did you get buy-in without authority?
85. What mechanism did you build (review, monitoring, tooling) that kept working without you in the room? [Likely]
86. Give me a second example: a different project where you showed the same thing. L6 rejections often cite a lack of multiple strong examples. [Likely]
87. How do you balance publishing or novelty against shipping on your current team? [Likely]
88. (Depth round, applied to leveling) Where did you have to make a judgment call that your manager didn't make for you? [Likely]
89. (HM) What would you expect to own in your first six months here? [Likely]
    - What would you need from me to get there?

J. Questions to put to the recruiter or HM about format (candidate-side calibration)

90. Is the loop four or five rounds? Is there a job talk, and how long is it? [Likely]
91. Which rounds are breadth, depth, application/design and coding, and which one is the bar raiser? [Likely]
92. Is the coding round DSA (LeetCode), ML-from-scratch, or data manipulation (pandas/SQL)? [Likely]
93. Is the loop calibrated to L5, or is it an L5/L6 "up-level-able" loop? Can an L6 be considered if signal is strong? [Likely]
94. If the loop is positive but this team's headcount closes, does the result carry to other teams (recycle/team match)? [Likely]
95. How soon after the loop does the debrief happen, and when should I expect feedback? [Likely]
96. If I'm not successful, what is the cool-down period, and does it apply org-wide? Reports range from none ("recycle") to 6-24 months. [Likely]
97. Who on the panel is likely to be from outside the team? [Likely]
98. Does the team expect GenAI/LLM depth even if the role is not LLM-titled? [Likely]

K. Debrief-driven follow-up (second loops, extra rounds)

99. (Extra round after a split debrief) Let's revisit coding or design. Can you solve this follow-up problem? [Reported] [14]
100. (New HM after team switch) Tell me about your project again, framed for our domain. [Reported] [12]
101. (Down-level conversation) Would you accept an L4 / AS I offer? What would you need to see to reach L5 quickly? [Likely]

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02. ML Breadth Round: Classical ML, Statistics, Probability, Optimization, Deep Learning, NLP/CV

Format notes

- **Official loop shape.** Amazon's Applied Scientist interview-prep page describes a 60-minute technical phone screen with a senior scientist or leader, then a loop of four 55-minute interviews with members of the science community. Each loop interview covers technical areas plus Leadership Principles (LPs), and each interviewer is assigned different areas [1].
- **What candidates actually see.** Reported loops usually include one dedicated "Science / ML Breadth" interview next to Science Depth (your own project or paper), coding, a team-specific science application or ML design round, and a Bar Raiser. One Applied Scientist II (L5) loop was reported as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [7]. Other loops had five 60-minute interviews over three days [18], or five rounds including a separate "Scalable ML" round [19]. Interviews are about 55-60 minutes, and LP questions are usually built into each one [1][11][26].
- **Who interviews.** Usually an Applied Scientist, Senior Applied Scientist or Principal Applied Scientist from the hiring org or a partner team [1][15]. The breadth round often starts from one of your projects or from something the team works on, then branches out. A Blind commenter on an L5 virtual onsite said breadth "can go anywhere the interviewer wants to take it" [2].
- **Style.** Expect rapid, quiz-like questions with "why" follow-ups. Blind commenters describe it as a quiz-style series of questions on general ML that tests how deep you can go, and say interviewers care about the intuition behind basic concepts, not just the formula [4]. Candidates report being asked to write things out mathematically: losses, regularization, and recurrent shapes in matrix form [13][16].
- **Scope at L5 (2025-2026).** The round reliably covers classical ML (from linear regression to ensembles), bias-variance and regularization, metrics, and probability and statistics [3][4][7][18]. **Transformers and attention are now treated as core breadth even when the team does not use LLMs.** An L5 candidate reported this [2], and an Ads candidate was rejected at the phone screen after being unprepared for attention, transformers and LLMs [5]. Questions from 2025-2026 go into decoder-only inference, KV caching, long-context attention, retrieval encoders and LoRA [6][10][14].
- **L5-specific bar.** Blind commenters say L5 questions resemble L4 questions, but L5 candidates are vetted more for whether they "raise the bar" [27]. A prep debrief says L5 answers should hold up through two or three layers of follow-up, against one layer at L4 [26]. In practice, the L5 signal is depth on demand: deriving results, naming

trade-offs and failure modes, and tying a technique to a product decision. The official page and prep guides also stress being able to explain *why* algorithms work [1][23].

- **Method note.** This chapter comes from about 45 web searches. Direct page fetches to Glassdoor, Blind, LeetCode, Medium and similar sites were blocked by this environment's egress policy, so every question marked [Reported] rests on search-result extracts of the cited page. Several sources are prep-site aggregators or older posts (2020-2022). Where a question appears only in a prep guide, a promotional prep service or an unattributed review, and not in a clearly attributable candidate account, it is tagged [Likely] with a "see [n]" pointer.
- **Review note (October 2026).** An adversarial review pass could not re-fetch any cited URL: every fetch failed at DNS resolution, and the session's web-search budget was exhausted. Because of that, the review (a) downgraded [Reported] tags whose only support is an aggregator, a promotional prep service or a review that could not be matched to the question; (b) merged duplicate items; and (c) added 20 new L5-depth questions. All new questions are tagged [Likely] because no new candidate report could be verified. The URLs below have not been re-checked in this pass and should be re-verified when network access is available.

Tag legend: [Reported] means a candidate or source reports it for Amazon Applied Scientist or a closely related Amazon science role (cited). [Likely] means a well-grounded inference from the role, recent loop trends or Amazon products.

A. Linear models

A1. Linear regression

1. What is the closed-form (normal-equation) solution for linear regression, and when would you use gradient descent instead? [Likely] (see [28])
    - What breaks in the closed form when features are highly collinear or when n and d are very large?
    - What is the computational complexity of each approach?
2. What are the assumptions of linear regression, and what happens to your estimates and inference when each is violated? [Reported] [4][5]
    - How would you detect heteroscedasticity or autocorrelated residuals?
3. What is the difference between MLE and MAP for linear regression? [Reported] [15]
    - Under what noise assumption does least squares equal MLE?
    - When does MLE equal MAP?
4. What is the probabilistic interpretation of Bayesian linear regression, and how would you model uncertainty in its predictions? [Reported] [9]
    - How does the posterior predictive variance behave far from the training data?

A2. Logistic regression

5. How does logistic regression work? Write down the model and its loss. [Reported] [9] [21]
    - Why is the decision boundary linear in the original feature space?

    - What does maximizing the likelihood mean for logistic regression? [Reported] [15]
6. What does logistic regression assume about P(y|x)? [Reported] [15]
7. What happens if you put a prior over the weights of logistic regression, and why would you assume one? [Reported] [15]
    - Which prior gives you L1 and which gives L2?
8. What are the differences between L1 and L2 regularization in logistic regression? [Reported] [9][21]
    - Why does L1 produce sparse solutions, geometrically and from a gradient or subgradient view?
9. What is the difference between logistic regression and linear regression, and why not use linear regression for classification? [Reported] [15]
10. Write the loss function for logistic regression and show that it is convex. [Likely] (see [13], which reports "derive loss functions for each model")
    - Derive the gradient. Why does it have such a simple form?
    - What happens to the weights on linearly separable data with no regularization?
11. How would you classify Amazon product reviews when you do not have a well-labeled dataset? [Likely] (Amazon reviews are a core data source; a similar older Glassdoor report is described in search extracts, but its page could not be verified.)

B. Trees, ensembles and other classical models

12. How is a decision tree built? How are splits chosen? [Reported] [15][19]
    - Compare Gini impurity, entropy or information gain, and variance reduction.
    - How do you stop a tree from overfitting (pruning, depth, min-samples)?
13. How is a random forest different from a single decision tree, and why is the forest better? [Reported] [19][11]
    - The interviewer pressed repeatedly: *why* does averaging decorrelated trees reduce variance? [Reported] [19]
14. What is the difference between bagging and boosting? [Reported] [7][15][26]
    - How does the bias-variance decomposition apply to bagging versus boosting? [Reported] [12]
15. How does gradient boosting (GBDT) work? [Reported] [15]
    - What exactly is being fit at each stage (pseudo-residuals or negative gradients)?
    - Why do the learning rate (shrinkage) and the number of trees interact?
16. What is the computational difference between XGBoost and random forest? [Reported] [21][11]
    - What makes XGBoost fast (histogram splits, sparsity-aware splits, second-order terms)?
    - How does it handle missing values?
17. Compare random forest and gradient-boosted trees in terms of bias and variance. When would you pick each? [Likely] (see [29], a prep article whose author calls this common in Google, Amazon and TikTok interviews; not an Amazon candidate report)
18. Walk through KNN, Naive Bayes and SVM: the assumptions behind each, its complexity, and when it fails. [Reported] [18]

    - What is the "naive" assumption, and why does Naive Bayes still classify well when it is violated?
    - What do the C parameter and the kernel choice do in an SVM? [Likely] (see [18])
    - What are the failure modes of k-nearest neighbors (scale sensitivity, curse of dimensionality, query cost), and how would you make it fast at scale? [Likely] (see [28])
19. How would you handle high-cardinality categorical features (for example, seller ID or ASIN) in a tree ensemble? What are the leakage risks of target encoding, and how do you avoid them? [Likely]
20. What is your favorite ML algorithm, and what are its drawbacks? [Reported] [12]

C. Bias-variance, overfitting, regularization and model selection

21. Explain the bias-variance trade-off and how it affects model performance. [Reported] [3][7][11]
    - Write the decomposition of expected squared error.
22. How do you detect and reduce overfitting? [Reported] [3][9][11]
    - What do you do about underfitting?
23. You have two models, each with fit and eval functions. How would you tell which one is overfitting, especially on genuinely new test data? [Reported] [19]
    - Why is "train error low, test error high" not a sufficient answer?
24. Explain regularization mathematically. [Reported] [13]
    - Show that L2 corresponds to a Gaussian prior and L1 to a Laplace prior (MAP view). [Likely] (see [15])
25. Is dropout a form of regularization? Why? [Reported] [19]
    - How are weights handled at inference time with dropout? [Likely] (see [21], a prep aggregator)
26. How do you handle data imbalance, collinearity, feature selection and regularization in one model? [Likely] (see [28])
27. How do you approach feature selection? [Reported] [3][20]
    - Compare filter, wrapper and embedded methods.
28. How do you tune hyperparameters, and how do you avoid leaking validation data into model selection? [Reported] [3]
29. Why is standardization of features preferred, and which models need it? [Reported] [15]
30. When would you use k-fold, stratified, grouped or time-based cross-validation? What goes wrong with a random split on time-series data or on data with many rows per customer? [Likely]

D. Evaluation metrics, calibration and class imbalance

31. How do you evaluate a model? Which metrics would you choose, and why? [Reported] [3][18]
32. How do you handle severe class imbalance? Would you evaluate with precision, recall and F1, or with ROC-AUC? [Likely] (see [26], a promotional prep-service debrief)
    - Why can ROC-AUC look optimistic under heavy imbalance, and when is PR-AUC more informative?
    - Compare resampling, class weighting and threshold tuning. Which do you try first?
33. Explain ROC-AUC. What does it measure, and how would you compute it by hand? [Likely] (see [24], which describes a "write a program to calculate AUC" item)
    - Why does AUC equal the probability that a random positive outranks a random negative?
34. How would you train and evaluate a real-time fraud or abuse model with severe imbalance? [Likely] (see [25]; an Ads Trust loop is reported in [18])
35. How would you pick a decision threshold when false positives and false negatives have different business costs? [Likely]
36. What does it mean for a classifier to be calibrated? How would you measure calibration (reliability diagram, expected calibration error) and fix it (Platt scaling, isotonic regression, temperature scaling)? [Likely]
    - Why does downsampling negatives miscalibrate predicted probabilities, and how do you correct it?
37. Which ranking metrics would you use for search or recommendations (NDCG, MRR, recall@k), and how do they relate to online metrics? [Likely] (Amazon search, ads and recommendation teams; see [32])
38. Your training labels are clicks logged by the current production ranker. What biases does that introduce (position bias, selection bias), and how would you evaluate a new policy offline (inverse propensity scoring, counterfactual evaluation)? [Likely] (see [32])

E. Unsupervised learning and dimensionality reduction

39. Explain k-means clustering. [Reported] [19]
    - How do you choose k? What objective does it minimize, and does it converge to the global optimum?
    - When would you use a GMM or DBSCAN instead?
40. Derive the EM algorithm for a Gaussian mixture model. Why does each iteration never decrease the likelihood, and how does k-means arise as a limiting case? [Likely]
41. What methods can reduce dimensionality, and how do they work? [Reported] [18] (see also [28])
    - Distinguish feature selection from feature extraction.
42. Explain PCA. How does covariance relate to PCA? [Reported] [12]
    - Does PCA keep the "most important features"? (Expect a correction: it builds new axes.)
43. What is the difference between eigenvalue decomposition (EVD) and SVD? [Reported] [15]
    - What is the computational complexity of SVD? [Reported] [12]

    - Would you compute PCA by eigendecomposing the covariance matrix or by SVD of the data matrix? Why?
44. What is the difference between Pearson correlation and covariance? [Reported] [12]
45. What is the difference between a VAE and a plain autoencoder? [Reported] [15]
    - Where does the KL term come from in the ELBO?
46. What is mutual information, and how would you use it? [Reported] [15]
47. What are the differences among supervised, unsupervised and reinforcement learning? [Reported] [9][12]
    - How does reinforcement learning differ from supervised learning in what feedback the learner gets? [Reported] [12]
    - What is a policy gradient, at a high level? [Likely]

F. Probability and statistics

48. What is Bayes' rule? Explain posterior, likelihood and prior. [Reported] [15]
49. What is maximum likelihood estimation, and how does it differ from Bayesian inference? [Likely] (see [23])
    - Derive the MLE for the probability of heads from k heads in n tosses. [Likely] (see [21])
    - What is the difference between a frequentist confidence interval and a Bayesian credible interval?
50. Work a coin-flip probability problem on the whiteboard. [Reported] [4] (the exact wording was not captured; practice variants such as the expected number of tosses until two consecutive heads)
    - How would you get a fair bit from a biased coin? [Likely]
51. A rare disease has prevalence 1 in 1,000, and a test is 98% sensitive with a 1% false-positive rate. Given a positive result, what is P(disease)? [Likely] (attributed to Amazon on prep sites; unverified)
52. State the Central Limit Theorem. Why does it matter for A/B tests? [Reported] [5]
53. What is a p-value? Interpret it in the context of an A/B test. [Reported] [10][20]
    - What is a common misinterpretation of a 95% confidence interval?
54. Walk through hypothesis testing: which statistical tests would you use, and when? [Reported] [5][20]
    - When would you use a t-test, a z-test, chi-squared, or a non-parametric test?
55. How would you design an A/B test for a model launch? [Reported] [20]
    - How do you choose the randomization unit, primary and guardrail metrics, and sample size or power?
    - How would you handle many simultaneous variants (multiple testing)? [Likely] (see [24])
    - How do you measure a recommender when network or interference effects break user-level randomization? [Likely] (see [24])
56. How can you reduce the variance of an A/B-test estimate so the test needs less traffic (for example, CUPED or stratification using pre-experiment covariates)? [Likely]
57. Explain the bootstrap. How would you build a confidence interval for a metric with no simple variance formula, such as AUC or a median order value? [Likely]

58. Give an example of Simpson's paradox. When you only have observational data, how would you estimate the causal effect of a feature or a promotion (propensity scores, difference-in-differences), and what assumptions does each method need? [Likely]
59. What is KL divergence, and how does it relate to cross-entropy loss in practice? [Reported] [9][20][21]
    - Is KL symmetric? When does forward versus reverse KL matter?
    - Relate entropy, cross-entropy and KL to information gain in decision trees. [Likely]

G. Optimization

60. Describe SGD and Adam, and say when you would use each. [Reported] [3] (see also [24][25])
    - What do momentum and the second-moment estimate each fix?
    - Why might Adam generalize worse than SGD in some settings? What does AdamW change?
61. Which loss functions and optimizers did you use in your project, and why those, given the alternatives? [Reported] [16][3]
62. What changes if you swap cross-entropy for MSE in a classification loss? [Reported] [8]
    - Why do sigmoid or softmax with MSE give weak gradients when the prediction is confidently wrong?
63. What is the role of the learning rate, and how do warmup and decay schedules help? [Likely]
64. What happens to convergence with very large batch sizes, and how do you compensate? [Likely] (see [23])
65. How would you diagnose a training loss that diverges or goes NaN? [Likely] (gradient explosion is reported in [11])
    - What does gradient clipping do?
66. Define convexity. Which common ML objectives are convex and which are not? Why do deep nets still train well? [Likely]
67. Compare first-order methods with second-order or quasi-Newton methods (Newton's method, L-BFGS). Why are second-order methods common for logistic regression but rare for deep networks? [Likely]

H. Deep learning fundamentals

68. Explain backpropagation for an MLP. [Reported] [4][5]
    - Derive the gradient for one hidden layer with a sigmoid or ReLU.
69. Why use a sigmoid activation, and what are its problems in deep networks? [Reported] [12]
    - Compare ReLU, GELU and tanh. What is the dying-ReLU problem?
70. Explain the vanishing and exploding gradient problems and how to address them. [Reported] [16][26]

71. Explain residual connections. Why do they make very deep networks trainable, and how do they interact with normalization (pre-norm versus post-norm)? [Likely]
72. Explain batch normalization. Why does it help, and how does it behave differently at train and inference time? [Reported] [17]
    - Why do Transformers use LayerNorm instead? What is RMSNorm? [Likely] (a 2024 Glassdoor review mentions "LLM architecture and various normalization, regularization etc methods"; see [6])
73. What is label smoothing, and what is the intuition behind it? [Likely] (see [23])
    - How does it affect calibration?
74. How do you initialize weights, and why does it matter (Xavier or He)? [Likely]
75. How does knowledge distillation work? What do the soft targets and the temperature contribute, and when does a distilled student beat a student trained on hard labels? [Likely]
76. A model is too slow or too expensive to serve at Amazon scale. Compare post-training quantization, quantization-aware training, pruning and distillation. What accuracy risks does each carry? [Likely] (see [22])

I. Sequence models and classic NLP

77. Explain LSTMs and how they differ from a vanilla neural network or RNN. [Reported] [16]
    - Why is an LSTM better than a plain RNN? [Reported] [16]
    - How does backpropagation through time work? [Reported] [16]
78. How does the shape of the input change as data moves through a recurrent network? Write it in matrix form. [Reported] [16]
79. What is the difference between LSTMs and GRUs? [Reported] [15]
    - How do GRU cells address the vanishing gradient problem? How does a BiLSTM work? [Likely] (see [21])
80. Explain the difference between LSTMs and Transformers mathematically. [Reported] [13]
    - What are the compute and parallelism trade-offs? How do they handle long-term dependencies?
81. Which text representations are there? What is the difference between Word2Vec, GloVe and FastText? [Reported] [15]
82. What is Word2Vec, how is it trained, what are the alternatives, and what are its drawbacks? [Reported] [17]
    - Compare skip-gram and CBOW, and explain negative sampling.
83. How is BERT trained? What are its constraints and advantages, and which other members of the BERT family do you know? [Reported] [17]
    - Why might BERT not work in a given use case? [Reported] [17]
84. What are embeddings, and how do you get sentence or token embeddings from BERT? [Reported] [12]
    - CLS token or mean pooling? Why do raw BERT embeddings make poor sentence similarity features?
85. Compare subword tokenizers (BPE, WordPiece, SentencePiece or unigram). How does the choice of tokenizer affect multilingual text, product titles full of model numbers,

and sequence length? [Likely] (multilingual Alexa work; see [31])
86. What is statistical language modeling, and what does perplexity measure? [Likely] (see [21]; language modeling is reported in Glassdoor extracts summarized in [7])

J. Transformers and LLMs (now a breadth staple)

87. Explain the Transformer architecture: encoder, decoder, and what each component does. [Reported] [3][8][10]
    - Why scale the dot product by sqrt(d_k)?
    - Why multi-head attention?
88. Why do Transformers need positional encoding? Compare sinusoidal, learned and rotary (RoPE). [Likely] (positional encoding appears in candidate-report extracts for 2025 transformer-heavy screens; see [14])
89. What happens if you set Wq = Wk = Wv in a Transformer? [Reported] [8]
90. How does Transformer inference work for decoder-only models? [Reported] [6]
91. How does KV caching work during autoregressive inference? [Reported] [6]
    - What is the memory cost? How do multi-query and grouped-query attention reduce it? [Likely]
92. What makes standard self-attention expensive for long contexts? [Reported] [6]
    - What is the time and memory complexity in sequence length?
    - How does FlashAttention speed attention up without changing the result? [Likely]
93. Explain Longformer and how its attention mechanism scales to long sequences. [Reported] [6]
    - What are the trade-offs between full attention, sparse or local attention, and long-context architectures? [Reported] [6]
94. What is the difference between a cross-encoder and a bi-encoder for retrieval? [Reported] [6]
    - When would you use each in a retrieve-then-rerank pipeline?
95. Explain LoRA. [Reported] [10]
    - Compare parameter-efficient fine-tuning with full fine-tuning. Which parameters and optimizer states are trained? [Likely] (see [22])
96. Which modern LLM training techniques do you know (pretraining, SFT, preference optimization)? [Likely] (see [34]: a 2026 Glassdoor review for an AGI-team role, not confirmed as an Applied Scientist report)
    - Compare RLHF and DPO. What does the DPO beta control? [Likely] (Alexa postings list SFT, DPO, RLHF and RLAIF [31])
97. How do decoding strategies differ (greedy, beam, top-k, top-p, temperature)? [Likely]
98. What is an encoder-decoder model such as T5, and how do you optimize it for inference? [Reported] [11]
99. What is a mixture-of-experts layer? What do you gain in compute per token, and what problems does routing introduce (load balancing, expert collapse, serving memory)? [Likely]
100. What do scaling laws say about splitting a fixed compute budget between model size and training tokens? When would you deliberately train a smaller model on more data? [Likely]

101. How would you reduce hallucination in an LLM-powered assistant? Where does retrieval-augmented generation (RAG) help, and where does it not? [Likely] (Amazon's shopping assistant uses LLMs including Amazon Nova, built on Bedrock [30])
102. How would you evaluate an LLM-based feature before launch? Compare reference-based metrics, LLM-as-a-judge and human evaluation, and name the biases of an LLM judge (position, length, self-preference). [Likely] (see [30])

K. Computer vision fundamentals

103. How does a CNN work, and how is it different from a feed-forward network? [Reported] [17]
    - What are parameter sharing and translation equivariance?
104. What is a receptive field, and how does it grow with depth, stride and dilation? [Reported] [17]
105. How do you reduce overfitting in CNNs (augmentation, dropout, weight decay, transfer learning)? [Likely] (see [17])
106. How does a Vision Transformer turn an image into tokens? When does a ViT beat a CNN, and why does it need more data or stronger augmentation? [Likely]
107. Describe how the Segment Anything Model (SAM) works. [Likely] (see [34]; the review could not be matched to the question or confirmed as an Applied Scientist role)
108. Compare two-stage and one-stage object detectors. What are IoU and non-maximum suppression? [Likely]
109. Explain contrastive multimodal learning such as CLIP, and diffusion models at a high level. [Likely] (see [22])
    - Write the InfoNCE loss. What do the temperature and the number of negatives control?

L. Applied breadth: production, ranking, forecasting and data issues

110. A model performs well offline but poorly online. What do you check? [Likely] (see [22])
    - How do you detect data drift or feature drift, and what would you monitor?
111. How would you handle noisy labels or non-stationary data? [Likely] (see [22])
112. Implement a rolling average when some dates are missing. What data-manipulation pitfalls arise? [Reported] [3][5] (this sometimes appears inside the breadth or screen round)
113. How does collaborative filtering work, and what are its cold-start problems? [Likely] (Amazon item-to-item collaborative filtering [37]; recommender and learning-to-rank skills are listed in [32]; a recommender design question is reported in [33])
    - Why does item-to-item collaborative filtering scale better than user-based collaborative filtering?
    - Compare two-tower retrieval with a ranking model. Why have separate stages?

114. When would you use a multi-armed bandit instead of a fixed A/B test? Compare epsilon-greedy, UCB and Thompson sampling, and explain how a contextual bandit changes the problem. [Likely]
115. How would you forecast demand for millions of products, including new products with no sales history? Compare per-series models (ARIMA, exponential smoothing) with a global deep model such as DeepAR or a pretrained time-series model such as Chronos. [Likely] (Amazon forecasting research [35][36])
    - Why would you produce a probabilistic forecast and train with a quantile (pinball) loss instead of MSE?
116. How would you measure whether a new ranking or recommendation model improved the customer experience, beyond offline metrics? [Likely] (see [24])
117. How would you choose between a simple interpretable model and a deep model for a business-critical launch? [Likely] (tests L5 judgment; the official prep page asks candidates to explain trade-offs [1])

Prioritization for an L5 loop (based on report frequency)

- **Highest frequency in reports:** bias-variance (21), bagging vs boosting (14), regularization L1/L2 (8, 24), overfitting (22), decision tree and random forest (12, 13), logistic regression (5-9), Transformer internals (87-94), loss functions (61, 62), and p-values or A/B tests (53, 55).
- **Rising in 2025-2026:** KV caching, long-context attention, cross-encoder vs bi-encoder, LoRA, and LLM training techniques (90-96).
- **Expect derivations:** loss functions, regularization as MAP, EM, LSTM vs Transformer in math, and RNN shapes in matrix form (10, 24, 40, 78, 80).
- **Team-dependent L5 extras (unverified, [Likely]):** calibration and counterfactual evaluation for Ads and Search (36, 38), forecasting for supply chain (115), and distillation or quantization for serving cost (75, 76).

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03. ML Depth and Science Application: Research and Project Deep Dives, and Applying Science to Amazon Problems

Format notes

- **Official frame.** Amazon's Applied Scientist interview-prep page describes one or two 60-minute technical phone screens with a senior leader, then a loop of four 55-minute interviews with members of the science community. Each interview covers technical competencies and Leadership Principles (LPs), and the behavioral portion focuses heavily on your past work [1].
- **What "Science Depth" means in practice.** Candidates consistently describe the depth round as a deep dive into one or two projects or papers from the resume, usually led by a Senior Applied Scientist.
    - An AS II (L5) Advertising Trust candidate described "Science Depth" as a resume deep dive plus GenAI fundamentals (Transformers) plus the business impact of past projects [3].
    - Other AS2 reports describe a "deep dive into the specifics of my resume and past projects" [22], "modeling details in my project" [11], and the questions "why went with a particular choice of architecture?" [12].
    - IGotAnOffer summarizes the round as walking through a research project, with follow-ups on technical decisions, the modeling trade-offs considered, and outcomes [2].
    - A 2026 aggregate of about 20 candidate reports (2026 Q1 to Q3) describes the depth round as a full hour on your most complex project or your publications. The interviewer reads the resume in advance and chooses the topics: problem framing, data cleaning, baselines, class imbalance, why that backbone, metrics, deployment, and what you would do with more compute. Interns had each publication taken in turn [41].
    - Some loops replace or supplement this round with a 30-40 minute research talk or job talk followed by Q&A [10][16][26]; senior loops more often include presentations [39]. Blind commenters say the recruiter or coordinator should send a brief naming the topics and LPs for each round, so ask for it [19]. The same 2026 aggregate warns that the recruiter's preview of the rounds is often inaccurate [41].
- **Phone screens.** 2026 reports describe screens of 30-60 minutes that combine a two-project resume dive (the data, the training problems and why the design was chosen), a theory block and a coding problem [41]. Machine Learning Engineer loops report a separate "project deep dive" round of 10-15 minutes per project followed by layered questions [42].
- **What "Science Application" means.** This is the second half of the chapter, and it is reported in two flavors:

    - (a) An open-ended Amazon business or product problem. You propose data sources, labeling, modeling, metrics and risks [2][5][6]. Examples include modeling view, click and purchase probabilities [7]; categorizing 10k products into 100 categories [9]; training retrieval from click logs [9]; trustworthy ad recommendation [3]; and Alexa ASR for accented speech [6]. 2026 prompts include agentic AI proposals, a multi-turn search agent, a chatbot architecture and an evaluation framework for brand names in generated images [41].
    - (b) Discussion "based on past project experience," per a Blind comment [18].
    - An L5 candidate (Nov 2024) listed "ML application" as a distinct onsite round alongside depth, breadth, DSA and Bar Raiser [4]. An Alexa guide lists "science application" as one of five loop rounds [5].
- **Interviewer type.** Usually Senior or Principal Applied Scientists, and sometimes the hiring manager (Applied Science Manager), who may run a "tech talk" style mix of design and behavioral questions [3][8].
    - LPs are probed in every round. Older reports say about 5-15 minutes of each interview goes to one or two LP questions [5][27]. 2026 reports describe much more: often about half the hour, so the technical part may not start until minute 25 or later [41][43].
    - The depth round is often paired with Dive Deep, Deliver Results or Earn Trust [41]; Ownership and Customer Obsession also fit it naturally [Likely].
- **L5-specific bar.** At L5 (Applied Scientist II), the depth interviewer is checking several things:
    - whether you were "the architect or just a passenger" [23][25], with your decisions separated from your collaborators' and your advisor's [41];
    - whether you can defend every choice (architecture, loss, metric, data split) against alternatives [23][24][41];
    - whether you understand leakage and "how models cheat" [23][24];
    - whether you can connect the science to measurable customer or business impact [3][41][42]. Level is judged here: one candidate passed every round on content but was told the projects lacked the impact expected for the target level. The 2026 aggregate's advice is to lead with the project that has the largest measured business effect [41][42].
    - A Blind commenter notes that a fresh PhD is typically placed at L5 [21]. That means the bar includes publication-level rigor plus evidence that you can ship [Likely].
    - Interviewers often want "plausible and scientific sounding ideas on the why," even when they do not know the answer themselves [20].
- **Recent trend (2025-2026).** Depth rounds increasingly drift from your project into the fundamentals behind it. Reported topics include:
    - transformer internals, KV caching, long-context attention (Longformer) and bi-encoder vs. cross-encoder retrieval [12];
    - BERT vs. GPT and DPO vs. RLHF [13];
    - beam search decoding [17];
    - knowledge distillation and learning with limited labels [8];
    - how Segment Anything works [12];
    - for generative-AI and post-training teams, LoRA, RLHF, GRPO, KV caching and distributed training, while ads and seller teams lean on statistics, experimentation and ranking [41].

    - AGI-team loops are reported to lean heavily on modern LLM training and to weight ML depth above behavioral questions [28].
- **Method note (2026-10-14 review).** The original pass came from about 38 web searches, using search-result extracts rather than full page reads.
    - In this review, the egress proxy refused every cited host except raw GitHub content (HTTP 403 on Glassdoor, Blind, LeetCode, Medium, Quora, amazon.jobs, amazon.science, IGotAnOffer, Interview Kickstart, Dataford, OA VO Service and others), and the web-search budget was already used up. Those references could not be re-checked. "Unreachable" does not mean dead, so they were kept.
    - Five GitHub-hosted sources were fetched and read in full: [28], [41], [42], [43] and [44]. [28] supports the claims it is cited for. [41]-[44] are new and back the added 2026 questions.
    - Retagging: [Reported] items that cite only a prep-vendor guide ([24], [27]) or a non-science Amazon loop ([38]) were retagged [Likely], in line with the policy below. Two generic "answer a question about our product" items and one near-duplicate were removed.
    - Several sources are prep-vendor guides (Interview Kickstart, Dataford, OA VO Service). When a question appears only in a vendor guide, it is tagged [Likely] with "see [n]". [41]-[44] are TrueInterview pages mirrored on GitHub. They aggregate candidate reports and name the round, but not individual candidates.

Tag legend: [Reported] means a candidate or source reports it for an Amazon Applied Scientist or closely related Amazon science loop (cited). Items reported at Amazon with no role stated are marked "(role not stated)", and items from Research Scientist or MLE loops are marked as such. [Likely] means a well-grounded inference from the role, level, team or Amazon products.

A. Opening the deep dive: framing, scope and ownership

1. Walk me through the project or paper on your resume that you are most proud of, end to end. [Reported] [2][3][22][41]
    - Describe your recent work and projects. Which would you choose for a 45-minute deep dive, and why that one? [Reported] [2]
    - What was the problem statement, and who was the customer? What was the business motivation? [Reported] [41][42]
    - Which parts did you personally design, implement and evaluate, and which were done by others? [Reported] [41][42]
    - If I asked your collaborator or manager, how would they describe your contribution?
2. Pick two projects from your resume. For each one, what was the data, what training problems did you hit, and why was the design chosen? [Reported] (phone screen, 2026) [41]
3. Describe one of your proudest projects. What's the most innovative thing you did in it? [Reported] [2]
    - What made it novel compared with the prior art or the existing baseline?
    - Was the novelty necessary, or would a simpler baseline have got most of the gain?

4. Tell me about a project where you were the single-threaded owner of the science. [Likely]
    - What decisions did you make without being told to?
    - What would have happened to the project without you?
5. What was the state of the art or the production baseline when you started, and how far did you move it? [Likely] (see [26])
    - How did you choose the baselines, and were they tuned as carefully as your method? [Reported] [41]
6. What was the hardest technical obstacle in this project, and how did you get past it? [Likely]
    - What did you try that failed?
    - How long did you spend on dead ends before pivoting? (Bias for Action and Dive Deep)

B. Problem formulation and data

7. How did you turn the business or research question into an ML problem (classification, ranking, regression, generation, RL)? [Likely] (see [41])
    - Could it have been framed differently, and why did you reject the alternative framings?
8. How large was the training data, and where did it come from? How did you clean it? [Reported] [41] (see also the Glassdoor depth-round reports summarized at [12])
    - How did you sample it, and is the sample representative of production traffic?
    - How did you down-sample or construct the dataset? [Reported] [7]
9. How did you obtain labels, and how noisy were they? [Likely] (see [2])
    - Did you measure inter-annotator agreement?
    - How did label noise affect your model, and did you correct for it?
10. How would you build this with very little labeled data? [Reported] [8]
    - Would you use weak supervision, semi-supervised learning, active learning, or LLM-generated labels?
    - How would you validate synthetic or LLM-generated labels before training on them?
11. How did you handle class imbalance in your project? [Reported] [27][41][42]
    - Why did you choose re-weighting, re-sampling or threshold tuning over the alternatives?
    - How did the imbalance change your choice of evaluation metric? [Reported] [27] [41]
12. Which features mattered most, and how do you know? [Likely] (see [24])
    - Would you trust impurity-based or permutation importance here, and why? [Reported] (phone-screen theory block) [41]
    - How did you guard against features that are not available at inference time?
13. How did you handle missing data, outliers and distribution differences between data sources? [Likely]
14. Would you standardize your inputs? How did you treat sparse numeric features, and what did you do about correlated features? [Reported] (role not stated) [44]

C. Modeling choices and alternatives

15. Why did you choose this particular architecture or backbone? [Reported] [12][41][42]
    - Why X over Y? What concrete evidence made you choose? [Likely] (see [23])
    - What would a simpler model (logistic regression or GBDT) have achieved?
16. What alternatives did you consider, and why did you rule them out? [Reported] [41] (see also the Glassdoor depth-round reports summarized at [12])
    - Did you run the alternatives, or reason about them?
17. Why did you use one gradient-boosting library or framework over another? [Likely] (see [23])
18. If you used a pretrained model, why that backbone, and how did you adapt it (full fine-tuning, adapters, LoRA, prompting)? [Likely] (see [28][41])
    - How did you choose the size of the model given latency and cost limits?
19. Derive or write out the core equations of your method on the whiteboard. [Likely] (see [25][41])
    - Which assumptions does the derivation rely on, and which of them does your data violate?
20. What are the main hyperparameters in your model, and how sensitive are the results to each? [Likely]
    - How did you tune them, and on which split?
21. If you used knowledge distillation, what exactly was distilled and why? [Reported] [8]
    - What were the teacher-student gap, the temperature and the loss mix?
    - Why distill instead of quantize or prune?

D. Loss, optimization and training dynamics

22. Why did you choose that specific loss function? [Likely] (see [23])
    - How does it align with the metric you actually care about?
    - What happens if you swap it for a ranking loss, focal loss or a calibrated probabilistic loss?
23. Tell me about loss functions and optimizers you used, and why. [Reported] (see the Glassdoor depth-round reports summarized at [12])
    - Why Adam or AdamW over SGD with momentum? [Reported] (phone-screen theory block) [41]
    - What learning-rate schedule did you use, and why?
24. Did you see overfitting or training instability, and how did you diagnose and fix it? [Likely]
    - Training accuracy is high but test accuracy is low: what do you do? [Reported] [2]
    - Your model is underperforming: diagnose and fix it. [Reported] (listed as a depth-round question) [41][42]
25. How was training distributed and how long did it take? What was the compute budget? [Likely]
    - Explain the parallelism strategies and collective operations your training used. [Reported] (role not stated) [44]
    - What would you do with more compute or resources? [Reported] [41][42]
    - What would you change if you had 10x less compute?

E. Evaluation and scientific rigor

26. How did you evaluate the model, and why did you pick those metrics? [Reported] [41] [42] (see also the Glassdoor depth-round reports at [12] and [25])
    - Choose between precision, recall, F1 and ROC-AUC for this problem, and justify your choice. [Reported] [27][41]
    - How does the offline metric relate to the online business metric? [Reported] [41] [42]
    - Can you write code that computes AUC from scores and labels? [Reported] [14]
27. How did you confirm the model wasn't just memorizing training data? [Likely] (see [23])
28. How did you handle data leakage in that specific pipeline? [Likely] (see [23][42])
    - How did you split train, validation and test (random, temporal, by user, by product)?
    - Give an example of leakage you found or almost missed.
29. Are your improvements statistically significant? [Likely]
    - How many seeds or runs did you use, and what is the variance?
    - What confidence interval or test did you use?
30. What ablations did you run, and which component contributed most? [Likely] (see [26])
31. Where does your model fail? Show me the error analysis. [Likely]
    - Which slices (new users, long-tail products, low-resource locales) are worst?
    - What would you do next to fix the worst slice?
32. Was the model A/B tested online? How did you design the test? [Reported] [41][42]
    - What was the treatment, the primary metric and the guardrails, and how long did the test run?
    - Your online A/B result fell short of the offline gain. How do you find out why? [Reported] [41][42]

F. Deployment, scale and post-launch

33. How was the model deployed, and what were the latency, throughput and cost constraints? [Likely] (see [24][41])
    - What did you trade away to meet latency?
34. How did you monitor the model after launch? [Likely] (see [23])
    - How would you detect data drift vs. concept drift? [Reported] [21]
    - What triggers retraining, and how do you roll back?
35. What happened when the data distribution shifted (seasonality, Prime Day, new marketplace)? [Likely]
36. If you had to scale this 100x (more traffic, more locales, more products), what breaks first? [Likely]

G. Impact, business value and reflection

37. What was the business impact of this project, in numbers? [Reported] [3][41][42]
    - How did you attribute the impact to your model rather than to other launches?

    - Who used the output, and how did it change their decisions?
38. What would you do differently if you started this project again today? [Likely] (see [25][42])
    - Which limitations did you not mention in the paper or launch doc? [Likely] (see [41])
    - What was a real mistake in this project, how was it caught, and what changed afterward? [Likely] (see [42])
39. Tell me about a time your results disagreed with a stakeholder's intuition. How did you resolve it? [Likely] (see [1][27])
40. How did you decide this project was worth doing over other options? [Likely]
41. Tell me about a time you simplified a scientific solution to ship faster. What did you give up? [Likely] (Invent and Simplify)

H. Research talk, job talk or paper round (when the loop includes one)

42. Present your PhD or research work in about 30 minutes, then take ML-focused Q&A. [Reported] [10]
    - For each publication, how did you choose the baselines, and are they the strongest available? [Reported] [41]
    - For each publication, what was your specific contribution vs. your co-authors'? [Reported] [41]
43. Present a project of your choice (a "tech talk"). [Reported] [16][12][28]
    - How would this research transfer to an Amazon product?
44. Why did you make this specific choice in your presentation? (Reported as an interviewer picking one detail and asking "why" repeatedly.) [Reported] [20]
    - What would you expect to happen if this assumption were removed?
    - Can you propose a plausible explanation for this unexpected result?
45. Paper-read round: discuss a paper's method, assumptions and weaknesses. [Reported] (listed among depth-round questions for FAR and "premium" loops; the exact prompt was not captured) [41]
46. What is the next paper you would write on this line of work? [Likely]
47. Which reviewer criticism of your paper was most valid, and how did you address it? [Likely]

I. Depth in your specialty: NLP and LLMs (common for 2025-2026 L5 loops)

48. Explain the transformer architecture in detail. [Reported] [15]
    - Explain the attention mechanism at "root level," including scaled dot-product, multi-head attention and positional encoding. [Reported] [13][15]
    - Walk me through the mathematical formulation of self-attention. [Likely] (see [23])
49. How do BERT and GPT differ (objectives, architecture, use cases)? [Reported] [13]
    - How is BERT trained? [Reported] [9]

    - What are the differences between BERT and RoBERTa? [Reported] (see the Glassdoor depth-round reports summarized at [12])
50. How does decoder-only inference work, and what is the KV cache? [Reported] [12]
    - Why does standard self-attention get expensive on long inputs? [Reported] [12]
    - How does Longformer, or sparse or local attention, handle long context? [Reported] [12]
51. Compare bi-encoders and cross-encoders for retrieval and re-ranking. [Reported] [12]
    - When would you distill a cross-encoder into a bi-encoder? [Likely] (see [34])
52. How does beam search decoding work for LLMs? [Reported] [17]
    - What are the trade-offs vs. sampling (temperature, top-k, top-p)?
53. What is the difference between DPO and RLHF? [Reported] [13]
    - Walk through an SFT training pipeline and its loss. [Likely] (see [27])
    - What does DPO's beta control? [Likely]
54. If you did post-training: why does GRPO remove the critic, what does that cost, and how would you design the reward? [Reported] (phone screen, generative-AI team) [41]
    - Which instabilities did you hit in GRPO-style training, and how did you fix them? [Likely] (see [41])
55. Explain the main parameter-efficient fine-tuning techniques, which parameters each trains, and when each beats full fine-tuning. [Reported] (online assessment, resume-based) [41][42]
    - How do you choose the LoRA rank and which layers to adapt? [Likely] (see [25] [28])
56. How does an inference engine such as vLLM raise serving throughput for large language models? [Reported] (online assessment, MLE loop) [42]
57. How do you evaluate an LLM-based system (hallucination, faithfulness, helpfulness)? [Likely] (see [28][32])
    - When would you trust an LLM-as-judge, and how would you calibrate it?
58. Explain an AI-agent project you built, and how it was adopted. [Reported] (MLE project deep dive) [42]
59. Why do LSTMs beat vanilla RNNs, and why did transformers replace both? [Reported] (see the Glassdoor reports summarized at [12])

J. Depth in your specialty: search, ranking and recommendations

60. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [9]
    - How do you correct for position bias in clicks? [Reported] [42] (see also [29] [40])
    - What other biases (query intent, presentation, trust) contaminate click labels? [Likely] (see [29])
61. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [40][42]
    - How would you estimate true relevance from clickstream logs with position bias? [Reported] [40]

62. Walk me through a search-ranking project you led: the pain point, why this model, the offline and online metrics, and how you diagnosed a disappointing A/B result. [Reported] [41][42]
63. Why did you choose pointwise, pairwise or listwise learning-to-rank in your project? [Likely]
    - How do NDCG and MRR relate to online engagement?
64. How do you handle cold start for new users or products in your recommender? [Likely]
65. How does matrix factorization for recommendation work, and when would you use it over a two-tower model? [Likely] (see [25])
66. Predict the click probability for music recommendations. [Reported] (role not stated; phone screen, Sep 2026) [44]

K. Depth in your specialty: vision and multimodal

67. Describe how SAM (Segment Anything Model) works. [Reported] [12]
    - How would you adapt it to a product-image task with few labels?
68. You have 10k product listings across 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [9]
    - How do you fuse text and image signals?
    - How do you handle a long tail of categories with very few examples?
    - How do you evaluate when the category taxonomy itself is noisy?
69. For a detection system you built, explain the backbone choice, how you handled class imbalance, and what more resources would change. [Reported] [42]
70. Compare approaches to building joint image-text embeddings (CLIP-style contrastive pretraining and others). Where would they help Amazon? [Reported] (online assessment) [41]

L. Depth in your specialty: forecasting, optimization and causal inference

71. Explain your forecasting model, and how you produce and evaluate uncertainty (quantiles, calibration). [Likely] (see [30][31])
    - Why probabilistic forecasts rather than point forecasts for inventory decisions?
72. What is the newsvendor problem, and what is its solution? [Likely] (see [38]; reported in Amazon supply-chain interviews, not a science loop)
    - How does a quantile forecast connect to the newsvendor critical ratio?
73. How would you model an optimization problem from your work: objective, decision variables and constraints? [Likely] (see [27])
    - When would you use a heuristic instead of an exact solver?
74. Build a package-allocation model for couriers. [Reported] (role not stated) [44]
75. Prove that a new allocation method outperforms the manual baseline. [Reported] (role not stated) [44]
76. Design an end-to-end regression model for energy demand. [Reported] (role not stated) [44]

77. Design a causal study to measure the impact of reminders on customer behavior. [Reported] (role not stated) [44]
78. How would you estimate the causal effect of a price or promotion change without a clean experiment? [Likely] (see [31])
    - When would you use double ML, instrumental variables, or difference-in-differences?
    - How do you validate a causal estimate you can never observe directly?
79. In your project, how did you separate correlation from causation in the impact claim? [Likely]

M. Science application: reported Amazon problems

80. A user moves from viewing to clicking to purchasing a product. Model the probability of each action given a user and a product. [Reported] [7]
    - Would you build three independent models, or model view, click and purchase jointly? [Reported] [7]
    - How would you construct the dataset, choose features, and down-sample (for example with reservoir sampling)? [Reported] [7]
    - How would you handle the selection bias that purchases are only observed after clicks?
81. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] [3]
    - Walk through the full data pipeline, from collection to deployment. [Reported] [3]
    - How do you trade off ad revenue against trust and safety errors?
82. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] [6]
    - Where would you get diverse accent training data, and how would you handle phonetic variation? [Reported] [6]
    - How would you measure improvement per accent group without hurting the majority?
83. You are given a high-level ML problem. Propose a solution, including data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Reported] [2]
84. Propose agentic AI solutions for a hypothetical project, and defend their viability. [Reported] (2026) [41]
85. Design a multi-turn search agent, and explain how it handles long conversation context. [Reported] (2026) [41]
86. Architect an AI chatbot: intent detection, dialogue management, generation and a knowledge base. [Reported] [41][42]
    - How do fallback behavior, evaluation and continuous learning work? [Reported] [41][42]
87. Design an evaluation framework that checks whether a text-to-image system spells a requested brand name correctly and places it naturally. [Reported] [41][42]
    - How would you combine OCR accuracy, embedding-based semantic checks and a human review protocol? [Reported] [41]

88. Design an LLM quality validation system. [Reported] [41][44]
89. Design a distributed training-data pipeline (reported for a FAR loop). [Reported] [42] [44]
90. How would you evaluate adding video ads? [Reported] (role not stated, Nov 2025) [44]
91. How would you measure product-market fit for Alexa Shopping? [Reported] (role not stated) [44]
92. Design an end-to-end spam detection system. [Reported] (role not stated) [44]
93. Discuss LLM decoding and training choices in the context of an Amazon use case (LLM team). [Reported] [17][28]
94. Design and evaluate an ML pipeline to improve personalized product recommendations on the Amazon homepage. [Likely] (see [25])
    - What is the north-star metric, and what are the guardrails?
    - How would you trade off customer engagement against sales? [Likely] (see [25])

N. Science application: likely Amazon problems by org (L5, 2025-2026)

95. Rufus (shopping assistant): how would you build retrieval-augmented answers to "What do I need for cold weather golf?" over the catalog and reviews? [Likely] (see [32][33])
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you evaluate it offline before an A/B test?
96. How would you summarize thousands of customer reviews into a short "customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?
97. Fraud and abuse: how would you detect fake reviews or abusive sellers with heavily imbalanced and adversarial labels? [Likely]
    - How do you handle adversaries adapting to your model?
98. Supply chain (SCOT): how would you forecast demand for millions of products, including new ones with no history? [Likely] (see [30][31][37])
    - How would forecast errors propagate into buying and placement decisions?
99. Pricing: how would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely] (see [31])
    - What about interference between treated and control products?
100. Last mile and geospatial: how would you predict delivery time or improve address geocoding accuracy? [Likely] (see [35])
    - What loss would you use if late deliveries cost more than early ones?
101. Ads: how would you predict click-through rate for sponsored products, and how does calibration interact with the auction? [Likely] (see [3][41])
    - Revenue is flat, CPC is up and conversions are down after a pacing change. How would you diagnose it?
102. Product categorization and catalog quality: how would you detect and fix wrong attributes (size, color, brand) on listings at catalog scale? [Likely]
103. Alexa or devices: how would you make an on-device model 10x smaller with minimal quality loss? [Likely] (see [34])

104. AWS AI services: how would you evaluate whether a new foundation-model fine-tuning method is worth exposing to customers? [Likely]
105. How would you build a query-understanding model (intent, product type, attributes) for a new marketplace and language with little data? [Likely] (see [36])

O. Applying your own research to Amazon

106. How would you apply your research or project to a problem at Amazon? [Reported] [18]
    - Which team would benefit most, and what metric would move?
    - What would have to change in your method to work at Amazon's scale and latency?
107. If you joined this team, what would you work on in your first six months, and how would you measure success? [Likely]
108. Take your method and tell me the first experiment you would run with our data. [Likely]
    - What result would make you abandon the approach?
109. How would you explain your project's value to a non-scientist product manager in two minutes? [Likely] (Earn Trust)
110. Write the one-paragraph "PR/FAQ" headline for a product built on your research. [Likely]

P. LP probes woven into the depth round

111. Tell me about a time you went deep into the data and found something that changed the direction of the project. [Likely] (Dive Deep; see [1][5])
112. Describe a decision you made with incomplete information, and how you later validated it. [Reported] [41] (Bias for Action)
113. Describe a calculated risk you took: what you knew at the time, and how it turned out. [Reported] [42]
114. Tell me about a time a client or decision maker proposed a plan that was far too aggressive. What did you do? [Reported] [41][42]
115. Tell me about a time a model you shipped failed or underperformed in production. What did you do? [Likely] (Ownership)
116. Tell me about a time you disagreed with a senior scientist's technical approach, for example about an evaluation metric, and then committed. [Likely] (see [41][42]; Have Backbone; Disagree and Commit)
117. Tell me about a time you raised the scientific bar on your team (reviews, standards, reproducibility). [Likely] (Insist on the Highest Standards)
118. Tell me about a time you learned a new area quickly to solve a problem. [Likely] (Learn and Be Curious)
119. Tell me about the most ambiguous problem you worked on, and how you scoped it. [Likely]
120. Tell me about a time you traded scientific purity for customer impact, or refused to. [Likely] (Customer Obsession)

121. How do you use AI tools in your work? Tell me about a time an AI tool gave you a wrong answer, and how you caught it. [Reported] (Amazon behavioral, 2025-2026) [43]

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04. Coding Round for Applied Scientists: DSA Questions Reported at Amazon

> Scope: the live-coding (data structures and algorithms) round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite or virtual loop, 2025-2026. Questions only, with no solutions. Tags: **[Reported]** means a candidate report or prep source ties the question to Amazon (citation [n]). Where the report comes from an SDE, intern or Data Scientist loop, or the source does not name the role, the line says so. **[Likely]** means a reasoned guess based on the role, team or Amazon's products. No source reported it.
>
> Research note: in this environment the egress proxy blocked Glassdoor, Blind, LeetCode, 1point3acres, Medium, amazon.jobs and the vendor guides, so those sources were read through search-result extracts and not as full pages. The GitHub-hosted sources (the TrueInterview mirrors and ombharatiya question lists, marked "fetched and read in full" in the References) were fetched and read in full. Treat the details of each report as what the candidate said happened, not as verified fact. ML-from-scratch coding (AUC, IoU, k-means, k-NN, attention, softmax, gradient descent and similar) is covered in Chapter 05 and is not repeated here.

Format notes for this round

- **Where coding appears.** Amazon's official AS prep page says a technical phone screen comes first, and some teams add a second one [1]. Candidate reports describe the L5 loop as 4 to 6 back-to-back 60-minute interviews: ML breadth, ML depth, live coding, science application or ML system design, and a Bar Raiser [2][3][4][5]. An L5 poster said their recruiter described **one coding interview, usually a LeetCode medium**, with no coding in the other rounds [4]. One Seattle 2026 report listed "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [6]. A 2026 aggregator of scientist reports says coding appears in one or two loop rounds, sometimes the hiring manager's, and shares the hour with Leadership Principle (LP) questions [7].
- **Difficulty.** Most reports say easy to medium, and often an "easier medium" [4][8][9][10]. Some reports say medium/hard, with "clean production level, optimal code" and edge cases expected [11]. Others got one medium and one hard, with brute force explained before the optimal solution [12], or a LeetCode Medium/Hard such as median of a data stream [13]. One April 2026 AS phone screen set an original dynamic program that was not on any standard list [7][14]. Some AS candidates say they got almost no LeetCode at all (one poster: "roughly 4/5 of the AS I asked") [4][9]. **The bar depends on the team and on who interviews you.** A Reddit commenter said an SDE interviewer usually gives medium DSA, while an AS interviewer often gives ML or data-science coding [2].

- **Time pressure.** LP questions often take the first 20 to 30 minutes of a technical hour [15][16][10]. One 2026 report says LP questions left about 15 minutes for the coding task [7]. Plan to finish a medium in about 25 minutes [10].
- **Environment.** Amazon's prep pages ask for syntactically correct code with no pseudo code, written in its Livecode shared editor with no IDE, and they say to check edge cases and bad input [1][17]. Reports say the editor does not run code, so you trace through it by hand [18][10]. Python is the norm for AS candidates, but any language is accepted [19]. The 2026 SDE online assessment added an AI-assisted repository-debugging section in some regions. Live coding rounds still ban AI, and no source reports the AI-assisted section for AS loops [20].
- **What is assessed at L5.** Problem framing and clarifying questions, choice of data structure, stating time and space complexity, moving from brute force to optimal, testing and edge cases, and readable, maintainable code [17][19][21][18]. Scientist interviewers also want edge cases and complexity stated without being asked, and they want you to narrate decisions as your own: one candidate who solved course schedule was told too many hints had been given [7][10].
- **Prep signals from candidates.** Candidates say Blind 75 or 150 at medium level is usually enough, and that you should know arrays, hash maps, trees, graphs, heaps and basic DP well [22][23][24]. L5 phone screens mix coding, statistics and LPs [25]. Alexa, AGI and LLM teams lean toward ML-implementation coding [26][27]. Amazon AGI phone screens were reported as "standard LC coding plus a backend task" [28]. Amazon SDE-tagged frequency lists (islands, LRU, top-k, intervals) are useful backup practice, because AS coding interviewers are often SDEs [29][30][31][32][20].
- **Reports too vague to list as questions.** Several AS reports name only the topic: "implement a data structure from scratch" (4 or 5 small prompts) [2], a sliding-window variant with "rigorous follow-ups" [9], a standard tree medium followed by Transformer questions [2], a DFS exercise and a binary-search question in Senior AS loops [33], a DP problem from the Blind 150 list asked by a hiring manager [2][34], two string DP and hash problems [35], and simple Python data wrangling or a medium SQL question [3][11].
- **Common follow-ups.** Scaling to data that does not fit on one machine (MapReduce) [36], streaming versions [14], and "now make it production-grade" [11].

A. Arrays, strings and hashing

1. [Reported] Given an array of integers and a target, return the indices of the two numbers that add up to the target. AS reports use it as a warm-up before a harder problem [34][7]. A 2026 onsite used an in-flight-movie framing: pick two movie lengths that fit the flight time [37].
    - What changes if the array is sorted, or if there are many queries against the same array?
    - How would you return all unique pairs without duplicates?
    - How would you solve it at scale, when the array does not fit in memory? [14]
2. [Reported] Find the maximum subarray sum in an integer array. [11][37]
    - How would you also return the start and end indices?
    - What if the array is circular?

3. [Reported] Compress a string in place by run-length encoding (for example "aabccc" becomes "a2bc3"), and analyze time and space complexity. [6][5]
    - How do you handle counts of 10 or more?
    - What if the compressed string would be longer than the original?
4. [Reported] For each position in a list, return the product of all other elements. A fraud-detection team screen asked this without allowing division. It was also reported in an April 2026 Amazon phone screen (role not stated). [38][37]
    - How do you handle zeros?
    - Can you do it in O(1) extra space, not counting the output?
5. [Reported] Find the longest substring without repeating characters. [5][19][37]
    - How would you return the substring itself, not just its length?
6. [Reported] Divide two integers without using multiplication, division or modulo. [19][14]
    - How do you handle overflow and negative numbers?
7. [Reported] In a sorted array where every element appears twice except one, find the single element. [19][14]
    - Can you beat O(n)?
8. [Reported] For each element, find the nearest smaller element to its left. This was reported in an Amazon loop, but the role is not specified. [39]
    - Which data structure gives you amortized O(n)?
9. [Reported] Merge two sorted arrays in place when the first array has spare capacity at the end. A third-party guide lists it as a sample AS question, and it was reported at Amazon in February 2026. [40][14]
    - What changes if you must merge three sorted arrays and remove duplicates? [14]
10. [Reported] A hash-map counting problem, followed by: what if the data is too big for one machine? [36]
    - How would you split it into map and reduce stages?
    - How do you handle a skewed key that overloads one reducer?
11. [Reported] Rotate an n x n matrix by 90 degrees in place. Reported in an AS loop (2024) and in Amazon screens and onsites through May 2026. [41][14][37]
    - How do you rotate counterclockwise, or by 180 degrees, with the same code?
12. [Reported] Count the subarrays whose sum equals k. This is on an Amazon-tagged list. [20]
    - Why does a sliding window fail when values can be negative?
13. [Reported] Group anagrams. Reported at Amazon in an August 2026 report and on Amazon-tagged lists (role not stated). [14][20][37]
    - How does your key choice change if the input is Unicode?
14. [Reported] Find all triplets that sum to zero (3Sum). This is on an Amazon-tagged list. [20]
    - How do you skip duplicate triplets without a set?
15. [Likely] Given customer reviews as strings, return the k most frequent words, ignoring stop words.
    - How do you break ties?
    - What if the vocabulary is far larger than k?
16. [Likely] Check whether two strings are one edit apart (insert, delete or replace).
    - How would you generalize this to a full edit-distance threshold k?

B. Intervals and sorting

17. [Reported] Merge overlapping intervals. Reported in AS loops (Sunnyvale 2020, a June 2026 AS technical screen) and as the coding question in a scientist online assessment. [2][42][43][7][37]
    - Should touching intervals such as [1,3] and [3,5] merge? How does your code change if they should not?
    - How would you insert one new interval into an already merged list?
18. [Reported] Merge overlapping time ranges for room reservations given as epoch-second pairs. This was tagged Amazon technical screen. [43]
    - How do you handle intervals where start equals end?
19. [Reported] Given meeting time intervals, find the minimum number of rooms needed. A scientist loop asked it with a min-heap after a two-sum warm-up. [7][37]
    - Can you do it with a sweep line and no heap?
    - How would you also return which meeting goes to which room?
20. [Reported] Given process start and end times, find the maximum number of processes running at once. Reported at Amazon in July 2026 (role not stated). [14]
    - How would you also return the time window when the peak occurred?
21. [Reported] Support adding non-overlapping intervals and querying whether a point is covered. This was reported at Amazon (role not stated). [14]
    - Which structure keeps both operations at O(log n)?
22. [Reported] Given a list of calendar events, find all pairs that conflict. Reported at Amazon in June 2026 (role not stated). [14]
23. [Likely] Sort a large file of (customer_id, timestamp) records that does not fit in memory.
    - How many passes does external merge sort need, given memory size M?

C. Two pointers and sliding window

24. [Reported] Design a counter that returns the number of clicks on an ad in the last k minutes. Reported in Amazon ML Engineer and scientist screens; the expected structure was a deque plus a hash map. [7][10][14]
    - How does memory grow with k and with the number of ads?
    - What changes if events can arrive out of order?
25. [Reported] Find the k closest elements to a target x in a sorted array. [44]
    - Can you do it in O(log n + k)?
26. [Reported] Return the maximum of every window of size k (sliding window maximum). Reported at Amazon in June 2026 and on Amazon-tagged lists. [14][20]
    - Why does a monotonic deque give O(n)?
27. [Reported] Find the shortest substring of a string that contains all characters of a pattern (minimum window substring). This is on an Amazon-tagged list. [20]
28. [Reported] Find the longest substring you can get by replacing at most k characters so that all characters are the same. This is on an Amazon-tagged list. [20]
29. [Reported] Trapping rain water. Reported in an April 2026 Amazon phone screen and on Amazon-tagged lists. [45][20][37]
    - Can you do it in O(1) extra space?

30. [Likely] Given daily sales, find the longest run of days whose total is at most a budget B, assuming all values are non-negative.
    - What breaks if negative values are allowed?

D. Linked lists and stacks

31. [Reported] Swap nodes in pairs (pairwise swap) in a linked list. A Research/Applied Scientist phone screen asked for this as a medium problem in about 25 to 30 minutes. [46][47]
    - Can you write both the recursive and iterative versions?
    - How would you generalize it to reverse every k nodes?
32. [Reported] Asteroid collision. Reported in an Applied Scientist intern DSA round (Bengaluru, 2026) and in an August 2026 Amazon report. [12][48][14]
    - Which edge cases involve equal sizes or asteroids all moving the same way?
33. [Reported] Check whether a string of brackets is valid. Reported in Amazon ML Engineer phone screens in 2026. [10][37]
    - How would you return the minimum number of insertions that make it valid?
34. [Reported] Evaluate an arithmetic expression. A July 2026 Amazon report used postfix notation. [14]
    - How would you extend it to infix with +, -, *, / and parentheses?
35. [Reported] Merge k sorted lists. This is on Amazon-tagged lists. [45][20]
    - What is the complexity with a heap compared with pairwise merging?
36. [Reported] Merge k sorted event streams behind an iterator with next() and hasNext(), without loading all events into memory. Reported in a September 2026 Amazon onsite. [14][20]
    - What happens when one stream is much slower than the others?
37. [Reported] Detect a cycle in a linked list, return where it starts, and break it. Reported at Amazon in June 2026. [14]
38. [Reported] Copy a linked list where each node has a next pointer and a random pointer. Reported at Amazon in March 2026. [14]
    - Can you do it in O(1) extra space?
39. [Reported] Implement a stack that pops the most frequent element, breaking ties by most recent (maximum frequency stack). Reported at Amazon in September 2026. [14]

E. Trees and BSTs

40. [Reported] Given an org tree with salaries, compute the average salary of each manager's direct reports. Reported in Amazon scientist and ML Engineer loops in 2026. [7][10][14]
    - How would you compute the average over all reports, not just direct ones, in one pass?
    - What if the input is a flat list of (employee, manager, salary) rows?
41. [Reported] Serialize and deserialize a binary tree. Reported at Amazon in February 2026 and on Amazon-tagged lists. [14][20]
    - How does your format handle null children compactly?

42. [Reported] Find the lowest common ancestor of two nodes in a binary tree. Reported in a February 2026 Amazon phone screen. [20][37]
    - What if one node might not be in the tree?
    - How does it get easier in a BST?
43. [Reported] Validate a binary search tree. Reported at Amazon in June 2026. [14]
44. [Reported] Find the kth smallest element in a BST. An August 2026 report included follow-ups. [14]
    - What if the tree is modified often and you need kth smallest many times?
45. [Reported] Return the right-side view of a binary tree. Reported at Amazon in February 2026. [14]
46. [Reported] Return the vertical order traversal of a binary tree. Reported at Amazon in June 2026. [14]
47. [Reported] Build a binary tree from a list of (parent, child, isLeft) descriptions and return its root. Reported at Amazon in August 2026. [14]
48. [Likely] Given a product category tree (Electronics > Audio > Headphones), return the path from the root to a category and the depth of the deepest leaf.

F. Graphs and grids

49. [Reported] Count the number of islands in a 2D grid. Reported in AS loops in 2020, 2025 and 2026, in a London 2026 screen, and in Amazon screens with a streaming follow-up in April 2026. [2][12][49][50][14][37]
    - Should diagonals count as connected?
    - What if the grid is streamed row by row and does not fit in memory?
    - Can you solve it with union-find?
50. [Reported] A BFS problem on a grid (for example, the minimum number of steps from start to target). [51]
    - How do you reconstruct the path?
51. [Reported] Shortest path with Dijkstra's algorithm where the grid or graph has obstacles. A 2026 source says weighted shortest-path problems are rising at Amazon. [52][20]
    - Why do negative edge weights break Dijkstra?
    - When would A* be the better choice?
52. [Reported] Explain how you would implement Dijkstra: which data structures, and what runtime? [53]
    - Which data structure gives O((V+E) log V), and which gives O(V^2)?
53. [Reported] Course schedule: given prerequisite pairs, return a valid order or report a cycle. Reported in 2026 Amazon ML Engineer screens, including a version framed around ads. [10][14][37]
    - How do you return all nodes that are part of a cycle?
54. [Reported] Given dependent tasks and a task that failed, order the tasks and find which tasks must be rerun. Reported in a September 2026 Amazon onsite. [14]
55. [Reported] Check whether adding an edge to a directed graph creates a cycle. Reported at Amazon in October 2025. [14]
56. [Reported] Word ladder (BFS on implicit graph). This is on Amazon-tagged lists. [18] [20]
    - How do you build neighbors efficiently using wildcard patterns?

57. [Reported] Rotting oranges and multi-source BFS: how many minutes until every reachable cell is rotten? Reported at Amazon in April 2026 and on Amazon-tagged lists. [14][20]
58. [Reported] Search a grid of letters for a word, then for a list of words with a trie (word search I and II). Reported at Amazon in January and February 2026. [14][20]
59. [Reported] Given equations such as a / b = 2.0, answer division queries (evaluate division). This is on an Amazon-tagged list. [20]
60. [Reported] Count connected components with union-find. Reported at Amazon in May 2026 (role not stated). [14]
    - How would you use it to group users who share a payment method or address for fraud detection?
61. [Likely] Clone a graph, for example a knowledge-graph subgraph used by Alexa.
62. [Likely] Given warehouse aisles as a grid with blocked cells, find the shortest route for a robot from a start to multiple pick locations. This is related to Amazon Robotics.
    - How does the problem change with 2 pick locations compared with k?

G. Heaps, top-k and streaming

63. [Reported] Find the median of a data stream. Reported as a LeetCode Medium/Hard in an AS loop (Luxembourg, 2023) and at Amazon in January 2026. [13][14]
    - What if most values are small integers in a known range?
    - How would you support a sliding-window median?
64. [Reported] Top K frequent elements. A September 2026 Amazon report added a streaming follow-up. [18][45][14]
    - Can you compare heap and bucket sort?
    - How would you approximate this with Count-Min Sketch when there are billions of items?
65. [Reported] K closest points to the origin. This is Amazon SDE-tagged. [45]
    - Can you use quickselect, and what is its worst case?
66. [Reported] Find the kth largest element in an unsorted array. This is on an Amazon-tagged list. [20]
67. [Reported] Rearrange a string so that no two adjacent characters are the same, or report that it is impossible. Reported at Amazon in May 2026. [14][20]
68. [Reported] Given tasks and a cooldown n between identical tasks, find the minimum time to run them all (task scheduler). This is on an Amazon-tagged list. [20]
69. [Likely] Return the top k products by click-through rate in a sliding one-hour window over a click stream.
    - What state must you keep per product?

H. Binary search

70. [Reported] Find the minimum of a rotated sorted array. Reported in Amazon scientist and ML Engineer phone screens. [7][10]
    - How do duplicates change the worst case?
    - How would you then search for a target in the same array?

71. [Reported] Search a 2D matrix whose rows and columns are both sorted, and find the optimal approach. [5]
    - Can you compare the staircase search with binary search on each row?
72. [Reported] Implement binary search and state its time complexity. [40]
    - Can you write it to find the first and the last occurrence of a value? [20]
73. [Reported] Koko eating bananas: find the minimum eating speed to finish within h hours. Reported in an April 2026 Amazon phone screen. [14][37]
    - How would you reframe it as "ship packages within D days"?
74. [Reported] Design a key-value store that keeps a time-stamped history of values and returns the value at a given time. Reported in a March 2026 Amazon onsite and a September 2026 phone screen. [14][37]
    - How would you use it to look up point-in-time feature values for training data?
75. [Reported] Find the kth smallest element in two sorted arrays. This was reported at Amazon (role not stated). [14]
76. [Likely] Given a monotonic model-score threshold function, find the smallest threshold that keeps precision at or above a target. This is binary search on the answer.

I. Dynamic programming

77. [Reported] Batch items into buckets so that total padding cost is as low as possible. This original DP was reported in an April 2026 Applied Scientist phone screen. [7][10][14]
    - What is your DP state, and what is the recurrence?
    - How does the answer change if the number of buckets is fixed?
78. [Reported] A minimum-coins variant solved with 1-D DP (AS intern, 2025), also reported at Amazon as "minimum bills and coins" in June 2026. [51][14][20]
    - Can you return the actual coins used, not just the count?
    - How do you count the number of ways instead?
79. [Reported] Edit distance between two strings, with a follow-up variant (July 2026). [14]
    - Can you reduce memory to O(min(m, n))?
80. [Reported] Word break: can a string be split into dictionary words? A July 2026 report gave a dictionary helper function. [14][20]
    - How does a trie help?
    - How would you return all valid segmentations?
81. [Reported] House robber and its variants (circular street, binary tree). Reported in Amazon screens and onsites in November 2025. [14][37]
82. [Reported] Best time to buy and sell stock, including the versions with unlimited trades and at most k trades. Reported at Amazon in March and July 2026. [14][20][37]
83. [Reported] Decode ways. This is on an Amazon-tagged list. [20]
84. [Reported] Find the longest palindromic substring. This is on an Amazon-tagged list. [20]
85. [Reported] Regular expression matching with . and *. This was reported at Amazon (role not stated). [14]
86. [Likely] Longest increasing subsequence.
    - Can you do it in O(n log n)?

87. [Likely] Viterbi-style DP: given emission and transition scores, find the highest-scoring tag sequence.

J. Randomization and sampling

88. [Reported] Random pick with weight: build a weighted random number generator from a probability array. Reported in AS and Senior AS screens in 2026. [33][5][54][7][10]
    - How do you handle all-zero weights or a single element?
    - What does the alias method give you, and what does it cost?
89. [Reported] Design a set with insert, delete and getRandom in O(1). This is on an Amazon-tagged list. [20]
    - How do you support duplicates?
90. [Likely] Reservoir sampling: choose k items uniformly from a stream of unknown length.
    - What changes for weighted reservoir sampling?
91. [Likely] Shuffle an array uniformly at random (Fisher-Yates), and explain why the naive swap-with-any-index version is biased.

K. Design-style data structures

92. [Reported] Implement an LRU cache. Reported in Amazon loops, including a basic-DS AS-adjacent report and an April 2026 phone screen. [49][18][37]
    - Can every operation be O(1)?
    - How would you make it thread-safe?
    - What about LFU, or TTL-based eviction? [20]
93. [Reported] Implement a trie that supports adding words and searching with a wildcard character. This was reported at Amazon, and autocomplete design is on an Amazon-tagged list. [14][20]
    - How would you return the top-3 completions by query frequency?
94. [Reported] Given a stream of (timestamp, message), decide whether each message may be printed, so that the same message prints at most once every 10 seconds. This was reported for Amazon new grads in August 2026. [20]
95. [Likely] Implement a min-stack or a max-queue that supports O(1) min or max.

L. Practical data-processing algorithms

96. [Reported] Given order events and ad-impression events, find for each user the pair with the smallest time difference. Reported at Amazon in February 2026. [14]
    - What is the complexity if each list is sorted by time?
97. [Reported] Given an Audible reading-event stream, flag books that receive abusive activity. Reported at Amazon in February 2026. [14]
98. [Reported] Parse log lines and aggregate them by a key (group-by). Reported at Amazon in March 2026. [14]
    - What if the logs do not fit in memory?

99. [Reported] Given send and receive CSV logs, compute each message's latency. Reported at Amazon in May 2026. [14]
    - How do you handle a receive with no matching send?
100. [Reported] Return the top k slowest requests from a set of logs. This was reported at Amazon (role not stated). [14]
101. [Likely] Given a log of (user_id, session_start, session_end), compute each user's total active time, merging overlapping sessions.

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05. ML Coding: Implementing Algorithms and Metrics from Scratch

> Scope: the "ML coding" variant of the coding round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite or virtual loop, 2025-2026. This round asks you to implement an ML algorithm, metric, loss or training loop in plain Python, NumPy or (on LLM teams) PyTorch, usually with no scikit-learn or high-level library calls. Questions only, with no solutions.
>
> Tags: **[Reported]** means a first-person candidate report, or an aggregator that compiles candidate reports, ties this coding task to an Amazon science or ML loop (citation [n]). Where the source is an aggregator or a vendor-published debrief, the reference list says so. **[Likely]** means an inference from the role, Amazon's teams and products, or a concept that was reported in a breadth or depth round, turned into a coding task. A [Likely] item was not reported as a coding question. Items listed only by a vendor prep guide, a prep blog or a practice site, and not tied to a candidate report, are tagged [Likely] with the source noted.
>
> Research note (read this): in the review pass for this chapter (October 2026), web search was unavailable and the egress proxy refused every non-GitHub host, so the 38 references carried over from the first draft could not be re-fetched. They were not removed, because a refused connection is not evidence that a page is dead; they reflect what the chapter 01-04 research recorded, mostly from search-result extracts. The new [Reported] items below come from GitHub-hosted aggregators that were fetched and read in full during this pass [39]-[43]. Two of those pages cover Amazon's Research Scientist and Machine Learning Engineer tracks, which share the science loop format with Applied Scientist; each item says which track it came from. Treat what each source describes as the candidate's account of what happened.

Format notes for this round

- **Where it sits in the loop.** Amazon's official AS prep page describes the onsite as four 55-minute interviews with members of the science community. Candidates for AS II usually report 5 rounds: ML breadth, ML depth, coding, science application or system design, and a Bar Raiser [1][3][11]. Most L5 loops have **one coding round**, sometimes two, and the hiring manager's round may add a short coding task [40][41]. Depending on the team and the interviewer, it can be LeetCode-style DSA, data manipulation in pandas or NumPy, or ML from scratch [11][12][13]. One Reddit-sourced comment, summarized on Glassdoor, says SDE interviewers tend to give medium DSA, while AS interviewers often give ML or data-science coding [6]. One aggregator says some applied-science loops had no algorithm problems at all, only hand-written sigmoid, softmax and attention during the technical discussion [41].

- **Which teams lean toward ML coding.** Reports from Alexa, AGI and LLM teams lean toward implementing ML from scratch [24][25]. Amazon AGI reports include "debug a broken Transformer implementation", "build/train a small classifier live" and "reinforcement learning fundamentals + coding" [43]. A Blind loop summary lists "implement a simple ML algorithm or training loop from scratch (e.g., logistic regression with gradient descent, k-means)" as one coding format [12]. Vendor guides say libraries are banned so the interviewer can test whether you truly understand the method [19][21]; an Amazon online-assessment softmax task explicitly says "do not rely on any third-party libraries" [42].
- **Environment.** You code in Amazon's shared editor (Livecode), with no IDE and usually no way to run the code. Amazon expects syntactically correct code, not pseudo code, and expects you to check edge cases [1][2][30]. Python is the default; NumPy and PyTorch are expected for numerical tasks [40].
- **Bar at L5.** You are expected to write correct, readable, vectorized code. Aggregated reports list what interviewers check: numerical stability (subtract the max before exponentiating, stable variance formulas), vectorization when asked, linking statistics to their maximum-likelihood estimators, and edge cases such as all-zero weights or a single element, stated without prompting [40][41]. Expect follow-ups that scale the solution up (batching, streaming, distributed) or tie it to an Amazon use case.
- **Time pressure from LPs.** Each technical round typically opens with Leadership Principle (LP) questions, often 20-30 minutes. Several reports describe only about 15 minutes left for the coding task after long LP answers, and a rejection that followed [1][21][40][41]. Keep each LP story near five minutes.
- **Typical pacing (inferred).** About 5 minutes to clarify the task and the input/output contract, 15-30 minutes to implement depending on how long LPs ran, 5 minutes on tests and complexity, then extensions. Not every report confirms this split.

A. Evaluation metrics from scratch

A1. Classification metrics

1. Write a program that computes ROC-AUC from a list of scores and binary labels. [Reported] [3][4][5][17]
    - How do you handle tied scores?
    - Can you do it in O(n log n) using ranks (Mann-Whitney U) instead of building the full ROC curve?
    - What do you return if all labels are the same class?
    - How would you compute AUC over a dataset too large for one machine, for example with a fixed-bin histogram of scores per class?
2. Compute precision, recall and F1 from predicted and true labels, without libraries. [Likely] (metric choice was reported in breadth and depth rounds [21]; precision, recall and AUC for imbalanced fraud data were reported in Amazon science screens [40])
    - How do you extend it to multi-class with macro, micro and weighted averaging?
    - What should the function return when there are no predicted positives?
3. Build a confusion matrix for K classes from two integer arrays, vectorized in NumPy. [Likely]

4. Given scores and labels, find the threshold that maximizes F1, or that maximizes recall while precision stays at or above a target. [Likely]
    - Can you do it in a single pass after sorting?
5. Compute the precision-recall curve and the average precision (PR-AUC). [Likely]
    - Why can ROC-AUC look optimistic under heavy class imbalance, for example fraud or abuse detection?
6. Implement binary and multi-class log loss (cross-entropy) with numerically safe clipping. [Likely] (the cross-entropy versus MSE question was reported in breadth [7])
7. Compute expected calibration error (ECE) with equal-width bins, and the data for a reliability diagram. [Likely] (ad click-through-rate (CTR) prediction and auction teams)
    - Equal-width or equal-mass bins: which would you use, and why?
8. Fit temperature scaling for a trained classifier: find the single scalar T that minimizes validation negative log-likelihood, by gradient descent or a 1-D search, and apply it to the logits. [Likely] (temperature scaling is named among the "numerical care" points interviewers look for [41])
    - Why does temperature scaling leave accuracy and AUC unchanged?
9. Implement isotonic regression with the pool-adjacent-violators algorithm to calibrate CTR scores. [Likely] (ads calibration)
    - When would you prefer it to Platt or temperature scaling, and what is its overfitting risk?

A2. Ranking and retrieval metrics

10. Compute precision@k, recall@k and NDCG@k for a ranked list of search results with graded relevance. [Likely] (Amazon search, ads and recommendations)
    - What is IDCG, and how do you handle queries with no relevant items?
    - How do you average across queries?
11. Compute mean reciprocal rank (MRR) and mean average precision (MAP) for a batch of queries. [Likely]
12. Given impressions with click logs, compute CTR per position, and suggest a simple correction for position bias. [Likely] (position bias in clickstream relevance was reported in a depth round)

A3. Vision, NLP and forecasting metrics

13. Write a function that computes Intersection over Union (IoU) for two bounding boxes. [Likely] (listed as an Amazon Senior AS coding question by a vendor guide [19], not by a candidate report)
    - What edge cases come up when boxes do not overlap or have zero area?
    - How would you vectorize it for N x M boxes?
14. Implement non-maximum suppression (NMS) using your IoU function. [Likely] (vision teams such as Lab126, Amazon Go and robotics)
    - What is the complexity, and how would you speed it up?
15. Compute mean average precision (mAP) at an IoU threshold for an object detector over a dataset. [Likely]
16. Implement BLEU (n-gram precision with brevity penalty) or ROUGE-L for a candidate and a reference. [Likely] (Alexa, translation and LLM teams)

17. Compute perplexity from per-token log-probabilities, handling padding masks. [Likely]
18. Compute word error rate (WER) with edit distance between a hypothesis and a reference transcript. [Likely] (Alexa ASR; accented-speech ASR was reported as a science-application problem)
    - How would you also return the counts of substitutions, insertions and deletions?
19. Implement forecasting metrics: MAPE, sMAPE, and weighted quantile (pinball) loss at P50 and P90. [Likely] (Amazon supply-chain forecasting teams)
    - What goes wrong with MAPE when actuals are zero?

B. Clustering, neighbors and similarity search

20. Implement k-means from scratch in Python or NumPy. [Reported] [12][16][17][39] (the aggregator listing records a December 2025 Amazon sighting of "Implement K-means and solve interval/frequency tasks" [39])
    - How do you initialize centroids? Implement k-means++ seeding as a separate function.
    - When should it stop, and what do you do with an empty cluster?
    - What is the per-iteration complexity? Can you vectorize the distance computation?
    - How would you choose k?
    - How would you run it on 1 billion product embeddings (mini-batch, distributed)?
21. Implement k-nearest neighbors classification from scratch. [Likely] (titled as an Amazon ML Scientist question by a prep blog [22], not a verified candidate report; kNN appeared in an AS II breadth report [38])
    - How do you choose k and break ties in voting?
    - Why must features be scaled, and how do you avoid leaking test statistics into the scaling?
    - What is the query cost, and how does it change in high dimensions?
    - Extend it to distance-weighted kNN regression.
22. Compute the Chamfer distance between two point clouds efficiently, using an exact nearest-neighbor structure such as a KD-tree instead of computing all pairwise distances. [Likely] (Amazon-framed practice question on a prep site [18], not a candidate report)
    - How does a KD-tree perform as dimensionality grows?
23. Build a KD-tree and run a nearest-neighbor query. [Likely] (follow-up to the Chamfer task)
24. Given a query embedding and a matrix of N item embeddings, return the top-k items by cosine similarity, using a heap rather than a full sort. [Likely] (semantic search, Rufus retrieval [33])
    - What is the complexity of the full sort versus the heap?
    - What changes for 100 million items (approximate nearest neighbor: IVF, HNSW, product quantization)?
25. Implement random-hyperplane locality-sensitive hashing for near-duplicate product detection. [Likely]
26. Compute pairwise Euclidean distances between two matrices without Python loops, using the expansion ||a||^2 - 2ab + ||b||^2. [Likely] (core subroutine of the reported k-means task)

    - What numerical issue can produce small negative values, and how do you fix it?
27. Implement greedy maximal marginal relevance (MMR) re-ranking: from N candidates with relevance scores and embeddings, pick k items that trade off relevance against similarity to items already picked. [Likely] (Amazon Science's MUSS work targets relevance-plus-diversity subset selection for recommendations [44])
    - What is the complexity of the greedy loop, and how would you scale it to millions of candidates?

C. Linear models, optimization and estimation

28. Implement a simple ML training loop from scratch, for example logistic regression with gradient descent. [Reported] [12]
    - Derive the gradient of the log loss before you code it.
    - How do you add L2 (and L1) regularization? Should the bias be regularized?
    - How do you avoid overflow in the sigmoid?
    - How do you check convergence?
29. Write gradient descent without libraries, and explain each line. [Reported] [21][39] (vendor-published debrief [21]; "Write and explain gradient descent pseudocode" in an Amazon aggregator listing, undated [39])
    - How would you change it to mini-batch SGD with momentum?
30. Code a traditional ML classifier from scratch, or build and train a small classifier live (the candidate chose the model). [Reported] [8][43] (the live-classifier task is from an Amazon AGI aggregator entry [43])
    - Why did you choose that model, and what are its assumptions?
    - How do you show the interviewer it is learning, with no plotting tools?
31. Implement linear regression with the closed-form normal equation and with gradient descent, and compare them. [Likely] (closed form versus GD and OLS were reported in breadth [13][23]; least-squares assumptions were reported in Amazon science screens [40])
    - When is the matrix X^T X singular, and what do you do about it (ridge, pseudo-inverse)?
32. Implement multinomial (softmax) regression for K classes with a vectorized gradient. [Likely]
33. Implement the Adam optimizer update for a parameter vector. [Likely] (Adam versus momentum SGD, and Adam versus AdamW, were reported in Amazon screens and depth rounds [40][41])
    - Why is bias correction needed in the early steps?
    - Change it to AdamW: where does decoupled weight decay enter the update, and why does it differ from L2 in the loss?
34. Implement a numerical gradient check to verify an analytic gradient. [Likely]
35. Implement L1-regularized regression with proximal gradient (soft-thresholding) or coordinate descent. [Likely] (L1 versus L2 sparsity was reported in breadth [10][16])
36. Implement a Newton-Raphson (IRLS) step for logistic regression. [Likely]
    - When does Newton's method beat gradient descent, and what does it cost per step?
37. Implement stratified k-fold cross-validation and use it to tune the regularization strength of your logistic regression. [Likely]

38. Write a NumPy function for the mean and variance of an array, then relate them to maximum-likelihood estimation. [Reported] [39][40][41] (Research Scientist and MLE loops; aggregator sighting dated September 2026 [39])
    - Why do the sample variance and the Gaussian MLE of the variance differ, and which is biased?
    - How do you compute variance in a numerically stable way, and along a chosen axis without loops?
39. Write the negative log-likelihood of a Bernoulli model and of a Gaussian model in code, and show that they reduce to log loss and squared error. [Likely] (squared loss as a Gaussian likelihood and log loss from the Bernoulli likelihood were reported in Amazon science screens [40])
40. Implement Welford's online algorithm for a running mean and variance over a stream, and merge the statistics from two shards. [Likely] (follow-up to item 38; "stable variance formulas" are named as an evaluation point [40])

D. Trees and ensembles

41. Implement the split search for a decision-tree node: find the feature and threshold that maximize information gain or Gini reduction. [Likely] (how splits are chosen was reported in breadth [36])
    - What is the complexity per node, and how does pre-sorting help?
    - How do you handle categorical features and missing values?
42. Implement a full decision-tree classifier with recursive building, max-depth and min-samples stopping rules, and predict. [Likely]
43. Implement prediction for a trained tree stored as arrays of nodes (feature, threshold, left, right, value). [Likely]
44. Implement a random forest on top of your tree: bootstrap sampling, feature subsampling and majority voting. [Likely] (bagging versus boosting is the most-reported breadth topic)
    - How would you compute out-of-bag error?
45. Implement gradient boosting for regression with depth-1 stumps and squared loss. [Likely]
    - What changes for log loss, and what are the "pseudo-residuals"?
46. Compute permutation feature importance for any trained model. [Likely] (impurity versus permutation importance was reported in Amazon science screens [40])

E. Neural networks and backpropagation

47. Implement forward and backward passes for a 2-layer MLP in NumPy, and train it on a toy dataset. [Likely] (backpropagation for an MLP was reported in breadth [35])
    - What are the shapes of every gradient?
    - How would you initialize the weights, and why?
48. Implement a numerically stable softmax with no third-party libraries, then softmax-plus-cross-entropy with its gradient with respect to the logits. [Reported] [39][41][42] (an Amazon online-assessment task requires the max-subtraction form for inputs in [-1000, 1000] [42]; hand-written softmax was also reported in AS loops [41])

    - Why does the combined gradient simplify to p - y?
49. Implement sigmoid by hand without overflow for large negative inputs, then ReLU, tanh and GELU with their derivatives. [Reported] (sigmoid [39][41]); the other activations are [Likely]
50. Implement dropout (inverted dropout) for training and inference. [Likely]
51. Implement the batch normalization forward pass with running statistics, and explain or implement the backward pass. [Likely] (batch norm was reported in breadth)
    - How does it behave differently at train and inference time?
52. Implement a 2D convolution (single channel, then multi-channel) with stride and padding, using loops and then im2col. [Likely] (vision teams; convolution and receptive fields were reported in breadth [41])
53. Implement a single LSTM or GRU cell forward step in matrix form. [Likely] (LSTM math and recurrent shapes in matrix form were reported in breadth [26][32])
    - How does backpropagation through time work, and where do vanishing gradients come from?
54. Build a minimal scalar autograd engine (a Value class with +, *, tanh and backward via topological sort). [Likely]
55. Implement a PyTorch training loop for a small model: data loader, forward, loss, backward, optimizer step, evaluation. [Reported] [39] (Amazon aggregator listing, undated; "practice a training loop from memory" is advised for the coding round [41])
    - Where do zero_grad, model.train() and model.eval() go, and what breaks if you forget each?
    - Add gradient accumulation and gradient clipping by global norm.
    - Add early stopping on validation loss and a warmup-plus-cosine learning-rate schedule.
56. Implement focal loss for binary classification and its gradient. [Likely] (class imbalance was reported in depth rounds [21])
    - How do the alpha and gamma parameters change the loss for easy and hard examples?
57. Implement cross-entropy with label smoothing. [Likely]
58. Implement symmetric per-channel int8 quantization of a weight matrix, dequantize it, and measure the error on a matrix multiply. [Likely] (Amazon Science publishes low-precision LLM training work [45])
    - Why does per-channel scaling beat per-tensor scaling when a few weights are outliers?

F. Attention, transformers and LLM components

59. Implement scaled dot-product attention by hand in NumPy or PyTorch. [Reported] [39][41] (hand-written attention reported in AS loops, aggregator sighting dated July 2026 [39]; attention "at root level" and the math of self-attention were also reported in depth and breadth rounds [27][29])
    - Why divide by sqrt(d_k)?
    - What changes if Wq = Wk = Wv? (reported as a breadth question [7])
60. Add a causal mask and a padding mask to your attention function. [Likely]
    - Where exactly is the mask applied, and why use -inf instead of 0?

61. Implement multi-head self-attention, including the reshape, split heads and output projection. [Likely]
    - What is the time and memory complexity in sequence length?
62. Implement sinusoidal positional encoding, or rotary embeddings (RoPE). [Likely] (writing the RoPE formula was reported in Amazon science depth rounds [40][41])
63. Implement layer normalization and RMSNorm. [Likely]
    - Pre-norm or post-norm: what is the difference in training stability?
64. Implement a decoder step with a KV cache: given cached K and V and a new token, return the output and the updated cache. [Likely] (KV caching and decoder-only inference were reported in a 2026 breadth round [6])
    - How much memory does the cache use for a given model size and context length?
    - How would multi-query or grouped-query attention change your code? (grouped-query attention was reported in screens [40])
65. Implement a sliding-window (local) attention mask, as in Longformer. [Likely] (Longformer and long-context attention were reported [6])
66. Implement beam search decoding over a toy next-token probability function. [Likely] (beam search decoding was reported as a conceptual question for an LLM-team AS II [28])
    - How do you handle end-of-sequence tokens and length normalization?
67. Implement temperature, top-k and top-p (nucleus) sampling from a logits vector. [Likely]
68. Implement byte-pair encoding (BPE): learn merges on a small corpus, then tokenize a new word. [Likely]
69. Implement a LoRA linear layer forward pass and explain which parameters are trained. [Likely] (LoRA was reported in AWS AS breadth [4])
    - How do you merge the adapter weights for inference?
70. Implement the SFT loss for a causal LM, masking prompt tokens so only response tokens count. [Likely] (SFT pipelines were reported in a depth round [21])
71. Implement the DPO loss given log-probabilities from the policy and reference models for chosen and rejected responses. [Likely] (Alexa and AGI job descriptions mention SFT, DPO and RLHF [37])
72. Implement a numerically stable, symmetric InfoNCE loss in PyTorch for a batch of paired image and text embeddings (image-to-text and text-to-image). [Reported] [39] [40][41] (Research Scientist and MLE loops; aggregator sighting dated February 2026 [39])
    - Where does the temperature go, and why use log-sum-exp in both directions?
    - Why are in-batch negatives a biased sample, and how does batch size change the loss?
73. You are handed a Transformer implementation that trains but performs badly. Find and fix the bugs. [Reported] [43] (Amazon AGI, aggregator; underlying reports not individually named)
    - Which bugs would you check first (mask direction, missing scaling, wrong softmax axis, residual or norm placement, positional encoding added twice)?
    - What small sanity tests (overfit one batch, compare with a reference shape) would you run?
74. Implement attention with an online (streaming) softmax over key blocks, keeping a running max and running sum, so the full N x N score matrix is never stored. [Likely]

(flash attention was reported in Amazon depth rounds [41])
    - Show that your blocked result matches the naive result.
75. Implement a mixture-of-experts layer with top-k gating and an auxiliary load-balancing loss. [Likely] (dense versus mixture-of-experts was reported in an Amazon "LLM depth" round; see chapter 07)
    - What happens to the experts if you drop the load-balancing term?
76. Implement the acceptance step of speculative decoding: given draft and target probabilities for k draft tokens, decide how many to accept and sample the correction token. [Likely] (inference-optimization questions on LLM teams)
77. Implement the PPO clipped surrogate loss given log-probabilities, old log-probabilities and advantages. [Likely] (PPO's clipped objective was reported in Amazon depth rounds [41])
78. Given G sampled responses per prompt with scalar rewards, compute GRPO group-relative advantages and the policy loss. [Likely] (why GRPO drops the critic, and how to design its reward, were reported in Amazon science screens [40][41])
    - What happens when every response in a group gets the same reward?
79. Code a reinforcement-learning fundamental from scratch, for example tabular Q-learning on a grid world or REINFORCE with a baseline. [Reported] (Amazon AGI, aggregator: "reinforcement learning fundamentals + coding" [43]); the specific algorithm is [Likely]
80. Fit a Bradley-Terry model from pairwise preference judgments (for example LLM-judge or human A/B comparisons between model outputs) to get a score per model. [Likely] (LLM evaluation was reported in 2025-2026 science-application rounds; see chapter 07)
81. Implement a triplet or margin ranking loss with in-batch hard-negative mining for a product-retrieval embedding model. [Likely]

G. Probability, sampling and statistics in code

82. Build a weighted random number generator from an array of weights (random pick with weight). [Reported] [9][39][40][41] (Senior AS screen, 2026 [9]; Research Scientist and MLE loops, aggregator sighting dated April 2026 [39])
    - What is the complexity per sample with prefix sums plus binary search? Can you do O(1) with the alias method?
    - What should happen if all weights are zero or there is a single element?
83. Implement reservoir sampling to pick k items uniformly from a stream of unknown length. [Likely]
    - How would you make it weighted?
84. Implement bootstrap sampling for the mean and its confidence interval, then remove the explicit Python loop. [Reported] [39][40] (Research Scientist and MLE screens; aggregator sighting dated August 2026 [39])
    - Which statistics does the bootstrap estimate poorly?
    - Extend it to a confidence interval for the AUC difference between two models scored on the same examples.
85. Implement a two-sample t-test (Welch) or a two-proportion z-test for an A/B test, from scratch. [Likely] (A/B testing, p-values and power were reported in breadth and screens [40])

    - How would you compute the required sample size for a given minimum detectable effect?
86. Implement a Bayesian (Naive Bayes) classifier from scratch, with Laplace smoothing and log-space computation, then apply it to scenarios the interviewer gives. [Reported] [40][41] (Research Scientist and MLE loops, sometimes in the hiring manager's round; Naive Bayes also appeared in an AS II breadth report [38])
87. Implement PCA via SVD, or via eigendecomposition of the covariance matrix, and project data onto the top k components. [Likely] (PCA and covariance were reported in breadth)
    - How would you implement power iteration for the top component only?
88. Implement EM for a 1-D Gaussian mixture model with two components. [Likely]
89. Implement Thompson sampling or UCB for a multi-armed bandit over ad creatives or widgets, and simulate regret. [Likely] (ads and personalization)

H. Recommendation, search and NLP building blocks

90. Implement matrix factorization for implicit or explicit ratings with SGD or ALS. [Likely] (Amazon recommender-systems roles list collaborative filtering [34])
    - How do you handle cold-start users and items?
91. Implement item-to-item collaborative filtering: compute cosine similarity between item vectors from a sparse user-item matrix, and recommend the top N. [Likely] (Amazon's well-known item-to-item recommendations)
    - How would you scale this computation to millions of items?
92. Implement TF-IDF and BM25 scoring for a query against a small document collection. [Likely] (product search)
93. Implement a pointwise or pairwise learning-to-rank loss (for example, RankNet) and its gradient. [Likely]
94. Implement n-gram language-model counting with add-k smoothing, and score a sentence. [Likely]
95. Implement a simple word2vec skip-gram negative-sampling update for one (center, context) pair. [Likely]
96. Deduplicate near-identical product titles using normalized string keys or character shingles plus Jaccard similarity. [Likely]

I. Data manipulation and ML pipeline coding

97. Compute a rolling average when some dates are missing. [Reported] [11]
    - Should missing days count as zero or be skipped? Defend your choice.
98. Load a dataset in pandas, clean it, engineer features, and fit a model, explaining each step. [Reported] [13][31]
    - Which steps can leak information from the test set?
99. Answer simple data-wrangling questions in Python (group-by, joins, aggregations) alongside standard DSA. [Reported] [3]
100. Return the top-K most frequent elements in an array, then in a stream. [Reported] [21][39] (aggregator sighting of a streaming follow-up variant dated September 2026 [39])

101. Given a company hierarchy as a tree with salaries, compute the average salary of each manager's direct reports. [Reported] [40][41] (Research Scientist and MLE loop coding)
    - What do you return for a manager with no direct reports?
102. Count an ad's clicks in the last k minutes from a stream of click events, using a deque and a hash map. [Reported] [40][41] (MLE and Research Scientist screens)
    - How does memory grow with the number of ads, and how would you expire idle ads?
103. Split an ordered list of sequences into contiguous batches, where each batch costs a fixed overhead plus padding up to its longest item, so total cost is minimized. Give the DP state, the recurrence and a worked example. [Reported] [39][40] (Applied Scientist phone screen, aggregator sighting dated April 2026)
    - What is the complexity, and can you bound the batch size to speed it up?
    - How would a greedy bucketing heuristic compare in practice for an inference server?
104. Implement a train, validation and test split that is stratified by label and grouped by user, so no user appears in two splits. [Likely]
105. Implement a time-based split and an expanding-window backtest for a demand-forecasting model. [Likely] (supply-chain forecasting)
106. Implement standard scaling and one-hot encoding as fit/transform classes that handle unseen categories. [Likely]
107. Implement mini-batch iteration over a dataset that does not fit in memory, with shuffling. [Likely]
108. Implement random oversampling, or class-weighted loss, for an imbalanced dataset, and show the effect on your metric function. [Likely] (class imbalance was reported in depth rounds [21])
109. Simulate data-parallel training: given gradient arrays from N workers, implement a ring all-reduce (reduce-scatter then all-gather) and check that it matches a plain average. [Likely] (inter-node communication and pipeline-parallel schedules were reported in the most specialized Amazon screens [41])
    - How much data does each worker send, as a function of N and the gradient size?

J. Round-level probes (how the interviewer pushes at L5)

110. Before writing code: what are the inputs, outputs, data types and sizes? What should happen on empty or degenerate input? [Likely] (Amazon's prep pages stress edge cases and bad input [1][2])
111. What is the time and space complexity of your implementation, and where is the bottleneck? [Likely]
112. You cannot run the code. Walk me through it on a tiny example by hand. [Likely] (third-party guides note that code is usually not executed [30]; aggregators say coding is "discussed rather than run" in screens [40])
113. How would you test this function? Which unit tests and invariants would you write? [Likely]
114. How would this change if the data were 1000x larger, or arrived as a stream? [Likely]

115. Now apply it to our team's problem. For example, where would k-means or kNN fit in product categorization or Rufus retrieval? [Likely]
116. Estimate the compute time and memory-transfer time of a residual MLP block at batch size 1 and batch size 256, given memory bandwidth, on-chip memory and peak throughput. Which case is memory-bound? [Reported] [39][40][41] (manager-level screen, aggregator sighting dated August 2026)
117. Tell me about a time you dove deep into data or code to find a root cause (an LP question opening or closing the coding round). [Reported] [21]

Prep priorities (derived from the tags above)

- **Most-reported ML coding tasks:** AUC from scores and labels (1), k-means (20), logistic regression or a gradient-descent training loop (28, 29, 55), a classifier from scratch or trained live (30), stable softmax, sigmoid and attention by hand (48, 49, 59), stable symmetric InfoNCE (72), NumPy mean and variance with MLE (38), bootstrap CI (84), a Bayesian classifier (86) and weighted random pick (82).
- **Team-specific:** AGI and LLM teams add debugging a broken Transformer (73) and RL coding (79); the [Likely] LLM items (60-71, 74-78) are the natural follow-ups.
- **Do not neglect:** data-manipulation and ML-flavored DSA coding (97-109), including the batching DP (103). It shows up in screens and loops where the interviewer is a scientist, not an SDE.
- **Time management:** practice each core primitive so it fits in 15 minutes, because long LP answers squeeze the coding time [40][41].

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06. ML and Science System Design: Recommendations, Search, Ads, Fraud, Forecasting, Alexa/Rufus LLMs, Pricing and Supply Chain

> Scope: the "Science Application" / "ML System Design (team-specific)" round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite loop, 2025-2026. This chapter is questions only, with no answers.
>
> Research note (please read): no web searches or page fetches worked for this chapter. The run's shared web-search budget was already used up, and direct fetches to Glassdoor, Blind, Medium, Dataford, IGotAnOffer, amazon.jobs and similar sites were refused by this environment's egress proxy. Every [Reported] tag below therefore comes from the sourced findings already gathered for chapters 01-04 of this guide. Those findings were themselves built from search-result extracts of the cited pages, not full reads. Each [Reported] question cites the original page. A question is tagged [Reported] only if one of those chapters attributes it to an Amazon science interview. Everything else is [Likely]. A follow-up session with a fresh search budget could add more first-person design-round reports, especially from Blind and 1point3acres.
>
> Review pass (2026-10-14): an adversarial review tried to re-verify the references and found none reachable. WebFetch could not resolve medium.com, leetcode.com, dataford.io or igotanoffer.com. The proxy returned 403 for amazon.jobs, amazon.science, spectrum.ieee.org, Glassdoor and Blind. The web-search budget was also used up. No reference could be confirmed live or confirmed dead, so none were removed. The three amazon.jobs job IDs [28][29][30] are especially unverified, because postings expire. The review removed duplicate and generic items and added a few [Likely] questions. It added no new [Reported] items, because none could be sourced.

Format notes for this round

- **Where it sits in the loop.** L5 AS loops are usually 4-6 interviews of 55-60 minutes each. One 2025-2026 Applied Scientist II report gave the loop as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Amazon's official prep page describes the loop as about four 55-minute interviews with members of the science community. It says interviewers are assigned competencies based on the Leadership Principles and typically ask 2-3 behavioral questions each [1]. Other candidates and guides list "ML application" or "science application" as a distinct onsite round [17][18].

- **L5 versus L4.** Blind commenters say L5 loops usually include the design or application round and L4 loops may not [5]. The L5 expectation is end-to-end ownership: framing a vague business problem, choosing data, labels and models, defining offline and online metrics, and launching and monitoring the system, with sound trade-off calls along the way [5][16].
- **Who runs it.** It is often the hiring manager or a senior scientist from the hiring team, and the problem is drawn from that team's domain [2][8]. Examples include Ads trust [9], Alexa speech [10], product search and recommendations [3][12][13], and LLM agents and RAG [3][14].
- **What is assessed.** IGotAnOffer describes the round as being given a high-level ML problem and proposing the data sources, annotation strategy, modeling technique, evaluation metrics, and trade-offs and risks [7]. In candidate reports, interviewers drill into the data pipeline "from collection to deployment" [9], dataset construction and sampling [11], and bias in implicit feedback such as click position bias [12][13]. In 2025-2026, GenAI design prompts appear more often: agentic recommenders, RAG and LLM-agent evaluation [3][6][14].
- **Leadership Principles.** Expect 1-2 LP questions inside this round, often Customer Obsession, Dive Deep, Invent and Simplify, or Think Big. Reports put LPs at roughly 5-15 minutes of each technical interview [1][16].
- **How it differs from SDE system design.** The weight is on the science (problem framing, labels, modeling, metrics, experimentation) more than on distributed-systems boxes. Candidates still report questions about scale, latency and deployment [9][22]. One campus loop had a separate "Scalable ML" round [21].

Tag legend: **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science interview (AS unless noted). **[Likely]** means a grounded inference from the team's products, job postings or Amazon Science publications. It has not been seen verbatim.

A. Generic framing (the questions every design round opens or pivots with)

1. Here is a high-level ML problem from our team. Propose a solution, covering data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Reported] [7]
    - Which parts of the problem actually need ML, and which could be rules or heuristics?
    - What is your v1 that could ship in one quarter, and what is v2?
2. Walk me through the full data pipeline for your design, from collection to deployment. [Reported] [9]
    - Where do labels come from, and how noisy are they?
    - How do you keep training and serving features consistent?
3. Pick an agentic product we could build. Who is the customer, and how do you measure success? (Seattle 2026 report) [Reported] [2]
    - What would the press release headline say (Working Backwards)?
    - Which single metric would you defend to a VP?

4. How would you take this from a prototype to Amazon-scale traffic? [Likely]
    - Where do latency and cost hit first: feature fetch, model inference, or candidate generation?
    - What do you precompute offline, and what must be computed in real time?
5. What offline metric and online metric would you pair, and what do you do when they disagree? [Likely]
    - How do you design the A/B test, including the unit of randomization, duration, power and guardrails?
6. What would make you kill or pivot this project after launch? [Likely]

B. Recommendations and personalization

7. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [6]
    - What is the north-star metric, and what are the guardrails?
    - How do you split the work into candidate generation, ranking and re-ranking?
    - How do you trade relevance against diversity and novelty on one page of widgets?
    - How do you avoid a feedback loop that only recommends already-popular products?
8. Design an agentic recommendation system for Amazon, covering component architecture, model choices, and quality and safety evaluation. [Reported] [6]
    - Which decisions should the LLM agent make, and which belong to a classical ranker?
    - How do you evaluate multi-turn recommendation quality before an A/B test?
    - How do you stop the agent from recommending unsafe, unavailable or irrelevant products?
9. Design a recommender system (reported as the design round in an AS loop). [Reported] [15]
    - Two-tower retrieval, matrix factorization, or sequence models: which, and why?
    - How do you handle cold start for new users and new ASINs?
10. A user goes from viewing to clicking to purchasing a product. Model the probability of each action given a user and a product. [Reported] [11]
    - Would you build three independent models, or model view, click and purchase jointly? [Reported] [11]
    - How would you build the dataset, choose features, and down-sample (for example with reservoir sampling)? [Reported] [11]
    - How do you handle the selection bias that purchases are only observed after clicks?
11. Design "Frequently bought together" / "Customers who bought this also bought". [Likely]
    - How do you separate substitutes from complements?
    - How do you deal with spurious co-purchases from promotions or bundles?
12. Design recommendations for Prime Video or Amazon Music, where consumption is sequential and the feedback is implicit. [Likely]
    - How do you define a positive label for a show that was watched for only 3 minutes?

13. How would you personalize email or push notifications without fatiguing customers? [Likely]
    - How would you frame this as uplift modeling rather than response modeling?
14. Design a size and fit recommendation system for apparel and shoes, so customers order the right size the first time. [Likely]
    - How do you learn from return reasons such as "too small" or "too large" as weak labels?
    - How do you handle brands whose sizing differs from the size printed on the label?
15. A product is out of stock. Design a system that recommends substitutes on the detail page and in the cart. [Likely]
    - How do you tell a true substitute from a merely similar item, and how do you evaluate substitutability offline?
16. How would you detect and fix popularity bias or exposure unfairness toward small sellers in recommendations? [Likely]

C. Product search and ranking

17. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [12]
    - How do you correct for position bias in clicks? [Likely] (see [23])
    - What other biases (presentation, trust, query intent) contaminate click labels?
18. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [13]
    - How would you estimate true relevance from clickstream logs with position bias? [Reported] [13]
    - Pointwise, pairwise or listwise learning-to-rank: which would you choose, and why?
    - How do you define "relevant" for a query like "running shoes" versus "nike pegasus 40 size 10"? (search-relevance application questions are reported in [3])
19. Compare bi-encoders and cross-encoders for retrieval and re-ranking. Where would each sit in a product-search stack? [Reported] [3]
    - What is your latency budget per stage, and how do you distill the cross-encoder?
20. Design query understanding for Amazon search (intent, product type and attribute extraction). [Likely] (see [27])
    - How would you launch it in a new marketplace and language with little labeled data?
21. Design a system to detect and reduce zero-result or low-quality-result searches. [Likely]
    - When should you relax the query, and when should you show a "did you mean"?
22. How would you combine relevance with business objectives (conversion, margin, sponsored placement) in the final ranking? [Likely]
    - How do you stop business boosts from eroding relevance over time?
23. Design an evaluation framework for search relevance using human annotation and LLM judges. [Likely]

    - How do you measure annotator agreement and calibrate an LLM judge against humans?
24. How would you build semantic or vector search over a catalog of hundreds of millions of products? [Likely]
    - How do you handle index freshness for new listings and price or stock changes?

D. Advertising (Sponsored Products, DSP, Ads Trust)

25. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] [9]
    - Walk through the full data pipeline from collection to deployment. [Reported] [9]
    - How do you trade ad revenue against trust and safety errors?
    - How do you get labels for rare policy violations?
26. Design click-through-rate prediction for Sponsored Products ads. [Likely] (see [20])
    - Why does calibration matter in a second-price or first-price auction?
    - How do you handle delayed conversions in conversion-rate modeling?
27. Revenue is flat, cost-per-click is up and conversions are down after a budget-pacing change. How would you diagnose it? [Likely]
28. Design an ad-relevance or ad-moderation classifier for images and text in sponsored creatives. [Likely] (see [9])
    - How do you handle adversarial advertisers who iterate on rejected creatives?
29. How would you measure the incremental impact (lift) of an ad campaign for an advertiser? [Likely]
    - Ghost ads, geo experiments or observational methods: what are the trade-offs?
30. Design a bidding or budget-pacing algorithm for advertisers on Amazon DSP. [Likely]
    - How would you frame it as constrained optimization, or as a bandit problem?
31. Design keyword or audience targeting recommendations for advertisers. [Likely]
32. Design the ad-load and ad-placement policy for streaming video ads (Prime Video). How do you trade ad revenue against viewer engagement and churn? [Likely]
    - Which long-term metric would you use as a guardrail, and how do you measure it within an experiment's duration?

E. Fraud, abuse, trust and payments

33. Design a fraud detection system for Amazon payments transactions. [Likely] (see [28])
    - How do you deal with extreme class imbalance and delayed chargeback labels?
    - How do you pick the operating threshold given the costs of false positives and false negatives?
    - How do you handle adversaries adapting to your model?
34. How would you detect fake reviews or abusive sellers with heavily imbalanced and adversarial labels? [Likely]
    - Would you use graph-based signals across accounts, devices and payment instruments?
    - How do you evaluate when your labels come from your own past enforcement?
35. Design a system to detect counterfeit products or listings. [Likely]
    - How do you combine image, text, seller history and brand-registry signals?

36. Design account-takeover detection at login time under a strict latency budget. [Likely]
    - What do you do with the "uncertain" band: friction, step-up auth, or human review?
37. Predict, at checkout time, the probability that an order will be returned. How would this be used, and what are the risks? [Likely]
    - How do you keep the model from penalizing protected groups or legitimate high-return categories?
38. Design returns-abuse detection. How do you avoid punishing legitimate customers? [Likely]
39. How would you prioritize a human-investigator queue when you can review only 1% of flagged cases? [Likely]
    - How do the investigators' decisions feed back into training without biasing it?
40. Your fraud model's precision drops sharply over two weeks. Walk me through the investigation. [Likely]

F. Computer vision and catalog quality

41. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [8]
    - How do you collect and label defect images when defects are rare?
    - Edge or cloud inference, and what is the throughput requirement per station?
42. You have 10k product listings across 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [12]
    - How do you fuse text and image signals?
    - How do you handle a long tail of categories with very few examples?
    - How do you evaluate when the taxonomy itself is noisy?
43. How would you detect and fix wrong attributes (size, color, brand) on listings at catalog scale? [Likely]
    - How would you use an LLM for attribute extraction, and how do you verify its outputs?
44. Design duplicate-listing detection (the same product listed under multiple ASINs). [Likely]
    - How do you scale pairwise matching to hundreds of millions of items?

G. Forecasting, supply chain and inventory (SCOT)

45. How would you forecast demand for millions of products, including new ones with no history? [Likely] (see [24][25][29])
    - Why use probabilistic or quantile forecasts rather than point forecasts for buying decisions?
    - Would you use a global model or per-item models, and why?
46. What is the newsvendor problem, and how would you use it to set inventory levels? (Amazon supply chain interviews) [Reported] [26]
    - How does a quantile forecast map to the newsvendor critical ratio?
47. How would you model an optimization problem from your work, including objective, decision variables and constraints? [Reported] [16]

    - When would you use a heuristic instead of an exact solver?
48. Design an inventory placement system that decides which fulfillment centers should stock each product. [Likely] (see [25])
    - How do forecast errors propagate into placement and transportation costs?
49. How would you forecast demand during Prime Day or other promotions, when history is sparse and distorted? [Likely]
    - How do you correct for censored demand when items were out of stock?
50. How do you evaluate a forecasting system against a business outcome (in-stock rate, holding cost) rather than MAPE alone? [Likely]
    - How would you run an experiment on a buying policy when products share warehouses and trucks?
51. Design a system to predict delivery time (the promise date shown on the detail page). [Likely] (see [19])
    - What loss would you use if a late delivery costs more than an early one?
    - How do you calibrate promises across regions and carriers?
52. How would you improve address geocoding or delivery-location accuracy for last-mile delivery? [Likely] (see [19])
53. Design a capacity or labor-planning forecast for fulfillment centers. [Likely]
54. For grocery delivery (Amazon Fresh or Whole Foods), design the system that picks a replacement when an ordered item is unavailable at picking time. [Likely]
    - How do you learn each customer's tolerance for substitutions, and what is the success metric?

H. Pricing, causal inference and experimentation

55. Design a system that recommends competitive prices for millions of products. [Likely] (see [29])
    - How do you estimate price elasticity when prices are set by your own algorithm (endogeneity)?
    - What guardrails stop the system from causing price spirals or customer-trust problems?
56. How would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely]
    - How do you handle interference between treated and control products, for example with switchback or cluster designs?
57. How would you estimate the causal effect of a promotion without a clean experiment? [Likely]
    - When would you use double ML, instrumental variables, or difference-in-differences?
58. How would you design an A/B test for a model launch? [Reported] [4]
    - What do you do if the test is underpowered or the metric is heavy-tailed?
    - How would you use variance reduction such as CUPED?
59. A new ranking model wins offline but is neutral online. What could explain it, and what do you do next? [Likely]
60. Design the system that chooses which seller's offer wins the featured offer (Buy Box) on a detail page. [Likely]

    - Which signals (price, delivery speed, seller performance) do you use, and how do you avoid being gamed by sellers?
    - How do you evaluate the effect on customers versus on seller fairness?
61. How would you decide whether to discount a product, and by how much, for a deal event? [Likely]
    - How do you account for cannibalization and pull-forward demand?

I. Alexa, Rufus and LLM-powered products

62. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] [10]
    - Where would you get diverse accent training data, and how would you handle phonetic variation? [Reported] [10]
    - How would you measure improvement per accent group without regressing the majority?
63. Design an enterprise RAG (retrieval-augmented generation) service. [Reported] [14]
    - How do you chunk, embed and index heterogeneous documents?
    - How do you enforce document-level access control in retrieval?
    - How do you evaluate retrieval quality separately from generation quality?
64. Evaluate an LLM agent with metrics beyond accuracy, such as tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you build an offline evaluation set for multi-step agent tasks?
    - When would you trust an LLM-as-judge?
65. Design a GitHub Copilot-style coding agent (reported in a Senior AS loop, Boston). [Reported] [3]
    - What context does the agent retrieve, and how do you rank it?
    - How do you measure whether it helps developers (acceptance rate versus downstream bugs)?
66. On an LLM team, discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] [22]
    - Greedy, beam or sampling decoding for a customer-facing assistant: which would you use, and why?
67. Design Rufus: answer "What do I need for cold weather golf?" using the catalog, reviews and Q&A. [Likely] (see [27][31][32])
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you evaluate it offline before an A/B test, and what is the online success metric?
    - How do you mix organic and sponsored products in a generative answer?
68. How would you summarize thousands of customer reviews into a short "Customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?
    - How do you stop manipulated reviews from steering the summary?
69. Design an Alexa skill or intent router that maps an utterance to the right skill or API (natural-language understanding). [Likely]
    - How do you handle new skills with no training data?
    - What is your latency budget end to end, from wake word to response?

70. How would you serve an LLM-based assistant to millions of users at acceptable cost and latency? [Likely]
    - Would you use distillation, quantization, caching, speculative decoding, or routing between small and large models?
71. How would you fine-tune and align an assistant for a new language or marketplace (SFT, DPO, RLHF or RLAIF)? [Likely] (see [27])
    - How do you get preference data, and how do you detect reward hacking?
72. How would you make an on-device model 10x smaller with minimal quality loss (Echo or Kindle devices)? [Likely]
73. How would you detect and mitigate unsafe, biased or policy-violating responses from a shopping assistant? [Likely]
    - Which red-teaming process and which guardrail models would you use?
74. Design proactive suggestions for Alexa ("hunches" or follow-up prompts) that are helpful without being intrusive. [Likely]
    - How do you set the interruption threshold, and how do you measure negative feedback that customers never state explicitly?
75. Design a conversational customer-service agent that can issue refunds or replacements. [Likely]
    - Which actions require human-in-the-loop, and how do you decide?
    - How do you evaluate resolution rate without incentivizing over-refunding?

J. Scaling, MLOps and lifecycle (probes across any design)

76. How do you monitor this model in production, and how do you detect data or concept drift? [Likely]
    - What triggers a retrain, and how do you validate it before rollout?
77. How would you run inference at scale, in real time versus batch, and on what infrastructure (for example, SageMaker endpoints)? [Likely]
78. How do you handle feature-store consistency and training/serving skew? [Likely]
79. How would you roll back safely if the new model misbehaves on a subset of traffic? [Likely]
80. Your labels arrive weeks late (returns, chargebacks, long-term retention). How do you train and evaluate under label delay? [Likely]
81. How would you reduce the cost of this system by 50% with minimal quality loss? [Likely] (Frugality)
82. Which parts of your design would you build, and which would you reuse from existing Amazon services or other teams? [Likely]
83. How would you handle personally identifiable information and privacy constraints in the training data? [Likely]

K. L5-bar probes and LP crossovers inside the design round

84. Of everything you proposed, what is the single riskiest assumption, and how would you de-risk it in two weeks? [Likely]

85. A product manager wants to launch in a month, and your offline evaluation is incomplete. What do you do? [Likely] (Bias for Action versus Insist on the Highest Standards)
86. Your design improves conversion but increases returns. How do you decide whether to ship it? [Likely]
87. Which metric would you refuse to optimize directly, and why? [Likely]
88. How would you explain the trade-offs in your design to a non-scientist stakeholder? [Likely] (Earn Trust)
89. If you had 10x the data, or one-tenth of it, how would the design change? [Likely]
90. How would you hand this system over to an engineering team for long-term ownership? [Likely]

L. Team-specific "warm-up" prompts to rehearse (inferred from postings and Amazon Science)

91. (SCOT Buying Science) Design a system that decides how much of each product to buy from which vendor and when. [Likely] (see [29])
92. (Payments) Design a model to predict payment-method failure and pick the best fallback for the customer. [Likely] (see [28])
93. (Last mile) Design a model that predicts whether a package will be delivered successfully on the first attempt. [Likely] (see [19])
94. (Alexa International) Design a multilingual natural-language-understanding model for a marketplace with code-switching (for example Hinglish). [Likely] (see [10][17])
95. (Ads Trust) Design a policy-violation detection system for ads with an annotation budget of 5,000 labels per month. [Likely] (see [9])

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07. LLM and GenAI Questions: Transformers, Fine-Tuning, RLHF/DPO, RAG, Evaluation, Hallucination, Efficiency, and Amazon Product Context (Bedrock, Nova, Rufus, Alexa+)

Scope: the LLM and GenAI questions an L5 Applied Scientist (Applied Scientist II) candidate should expect across the 2025-2026 onsite loop. This chapter is questions only, with no answers.

Research note (please read): this chapter draws on three kinds of evidence:

- **Candidate-report findings gathered for chapters 01-06 of this guide.** They came from search-result extracts of the cited pages, not full reads. Each [Reported] tag cites the original page.
- **GitHub-hosted interview aggregators, fetched and read directly** [22][23][24][25][26][35][36]. They compile Amazon candidate reports. Most cite trueinterview.io (CC BY 4.0); some do not name their sources. They are labeled as aggregators wherever used.
- **Amazon job postings and product pages,** used only to ground [Likely] questions.

Verification status (October 2026 review pass). The shared web-search budget was used up, and the egress proxy refused connections (HTTP 403) to every non-GitHub host this pass tried: Glassdoor, Blind, LeetCode, 1point3acres, Medium, amazon.jobs, amazon.science, aboutamazon.com, IEEE Spectrum, IGotAnOffer, Dataford, Interview Kickstart, Enginebogie and OA VO Service.

- **Re-fetched and checked line by line:** the seven GitHub aggregator pages [22][23][24][25][26][35][36].
- **Tags fixed after that check:** questions whose tags those pages did not support were retagged [Likely] or relabeled. That covers transformers replacing RNNs, mixture-of-experts specifics, pretraining vs fine-tuning, LLM production trade-offs, the content-moderation attribution, and several round labels.
- **Not re-checked:** the non-GitHub references [1]-[21] and [27]-[34] could not be re-opened. They are kept as captured from earlier search extracts, not confirmed dead. Re-verify them before relying on any single item.

Format notes for this round

- **There is usually no separate "LLM round."** LLM and GenAI questions come up inside the standard loop seats: ML/Science Breadth, Science Depth (your own project), Science Application / ML System Design, and sometimes ML Coding.
    - Amazon's official page describes a loop of about four 55-minute interviews with members of the science community. Each interviewer is assigned technical competencies and Leadership Principles and typically asks 2-3 behavioral questions [1].
    - Candidate reports for AS II more often describe five rounds of 55-60 minutes, for example "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2].
    - An aggregator built from about 20 Amazon science reports (mostly 2026) says generative-AI teams ask about LoRA, RLHF, GRPO, KV caching and distributed training. It also describes an optional online assessment covering PEFT, vLLM serving and image-text embeddings [22].
    - On generative-AI teams, "LLM depth is the most common technical reason for rejection" [23].
- **Transformers are now baseline breadth, even off-LLM teams.** An L5 virtual-onsite thread on Blind treats transformers as core breadth [13]. An Amazon Ads candidate reported being rejected after not preparing for attention, transformers and LLMs [14]. 2025-2026 reports cover:
    - decoder-only inference and KV caching;
    - Longformer-style long-context attention;
    - bi-encoders vs cross-encoders;
    - LoRA;
    - BERT vs GPT;
    - DPO vs RLHF;
    - beam search;
    - "modern LLM training techniques" [3][4][7][8][10][11].
- **Team-dependent depth.**
    - AGI/Nova-team loops are reported to lean hard on modern LLM training and ML coding. One aggregator reports a "debug a broken Transformer implementation" coding problem, and candidates compare the ML-coding style to OpenAI's [10][15][17][25]. Another aggregator says AGI candidate reports cover training techniques, evaluation (including perplexity and preference-based evals) and data quality [26].
    - LLM-team and Rufus/Alexa loops tie decoding, training and evaluation to an Amazon use case [8][27][28].
    - Science-application prompts in 2025-2026 include agentic products, agentic recommenders, enterprise RAG, LLM-agent evaluation and LLM quality validation [2][3][16][21][22].
- **Interviewer type.** Usually Applied Scientists, Senior Applied Scientists or Principal Applied Scientists. The application round is often run by the hiring manager [1][2].
- **The L5 bar.** L5 answers are expected to hold up through two or three layers of follow-up, against one layer at L4 [19]. Blind commenters say L5 candidates are vetted harder on whether they "raise the bar" [34]. For LLM topics that means:

    - deriving the mechanics (attention, the DPO loss, KV-cache memory);
    - naming failure modes and trade-offs;
    - choosing between prompting, RAG, fine-tuning and preference tuning for a concrete Amazon problem;
    - defining how you would evaluate it offline and online, including the cost and latency budget.
- **Leadership Principles still apply.** Expect 1-2 LP questions inside each technical round [1][19]. A GenAI-flavored behavioral prompt is now reported in Amazon screens: how you use AI tools, and a time one gave you a wrong answer [24].

Tag legend.

- **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science interview. That is Applied Scientist unless noted. Items from the Research Scientist or MLE aggregators are marked "(Amazon RS aggregate)" or "(Amazon MLE aggregate)".
- **[Likely]** means a grounded inference from the role, Amazon's products or job postings, or adjacent reported questions. It has not been seen verbatim.

A. Transformer fundamentals

A1. Architecture and attention

1. Explain the Transformer architecture in detail: encoder, decoder, and what each component does. [Reported] [4][9][11]
    - Why scale the dot product by sqrt(d_k)? What happens to the softmax if you don't?
    - Why multiple heads instead of one wide head?
    - Where are the residual connections and layer norms, and why does pre-norm vs post-norm matter for training stability?
2. Explain the attention mechanism "at root level," including scaled dot-product attention, multi-head attention and positional encoding. [Reported] [5][7]
    - Write the attention equation with shapes for Q, K and V.
    - How does the causal mask enter the equation?
3. What happens if you set Wq = Wk = Wv in a Transformer? [Reported] [6]
    - What does that do to the attention matrix's symmetry, and what expressiveness do you lose?
4. Explain the difference between LSTMs and Transformers mathematically. [Reported] [12]
    - How does each handle long-range dependencies, and what are the parallelism and compute trade-offs?
    - Why did transformers replace RNNs for language modeling? [Likely] (an Amazon-focused question list that says it is synthesized, not reported [26])
5. Why do Transformers need positional information? Compare sinusoidal, learned, RoPE and ALiBi encodings. [Reported] (Amazon RS aggregate: RoPE and ALiBi in a generative-AI phone screen) [22]

    - Write the RoPE formula and show how it makes attention depend only on relative position. [Reported] (Amazon RS and MLE aggregates) [22][23]
    - How does RoPE behave when you extrapolate past the training context length? [Likely]
6. Compare multi-head and grouped-query attention, and extend the comparison to multi-query attention. [Reported] (Amazon MLE aggregate, Ads-team phone screen) [23]
    - How does each change KV-cache memory and quality?
7. Transformer internals: how does the feed-forward block contribute compared with attention, and where do most of the parameters and FLOPs go? [Reported] (Amazon RS aggregate, generative-AI breadth hour) [22]
    - Where do the parameter count and FLOPs come from as d_model, the number of layers and the sequence length change?
8. Contrast Adam and AdamW. Why does AdamW decouple weight decay, and why does that matter for training transformers? [Reported] (Amazon RS and MLE aggregates) [22][23]
9. You are handed a Transformer implementation that trains but performs badly. Debug it. [Reported] (Amazon AGI, aggregator) [25]
    - What bugs would you check for first (mask direction, softmax axis, scaling, positional-encoding offsets, layer-norm placement)?
10. What is latent attention (low-rank compression of keys and values), and how does it cut KV-cache memory compared with GQA? [Reported] (Amazon MLE aggregate: "latent attention" listed among the most specialized training-at-scale screens) [23]
    - What does it cost in compute or quality, and how does it interact with RoPE? [Likely]

A2. Model families

11. How do BERT and GPT differ in objectives, architecture and use cases? [Reported] [7]
    - How is BERT trained (MLM and NSP), and why did RoBERTa drop NSP?
    - When would an encoder-only model still beat a decoder-only LLM for an Amazon task, such as product classification?
12. What is an encoder-decoder model such as T5, and how would you optimize it for inference? [Reported] [5]
13. Compare LLM architectures: encoder-only vs decoder-only vs encoder-decoder, and dense vs mixture-of-experts. [Reported for the architecture discussion] (Amazon MLE aggregate: a rejected candidate described "a full round on architectures, scaling laws and RLHF trade-offs") [23]; MoE specifics [Likely]
    - How do you load-balance experts in MoE, and what goes wrong without it? [Likely]
14. What do scaling laws say about how to split a fixed compute budget between parameters and tokens? [Reported] (Amazon MLE aggregate, LLM depth round) [23]
    - How would you use a scaling-law fit to decide whether to train a larger Nova-sized model or serve a smaller one? [Likely]
15. Compare a small and a large model from the same text-embedding family. What do you gain, and what do you pay? [Reported] (Amazon RS aggregate, breadth: a weak answer here was felt to hurt the round) [22]

- Trace the history of embedding models (word2vec, then BERT pooling, then contrastive sentence encoders, then LLM-based embedders). What changed at each step? [Likely]

A3. LLM components written by hand

16. Hand-write sigmoid, softmax and scaled dot-product attention, numerically stable, in NumPy or PyTorch. [Reported] (Amazon RS and MLE aggregates; sighting dated July 2026 in the Amazon question list) [22][23][36]
    - Where do you subtract the max, and how do you apply a causal mask without producing NaNs?

B. Inference, decoding and long context

17. How does inference work for decoder-only models? [Reported] [3]
    - Walk through prefill vs decode. Which phase is compute-bound and which is memory-bandwidth-bound?
18. What is the KV cache, and how does it speed up autoregressive generation? [Reported] [3][22][23]
    - Estimate the KV-cache memory for a given model (layers, heads, head dim, sequence length, batch). [Reported] (Amazon MLE aggregate: "memory arithmetic behind LoRA, quantization and KV caching") [23]
    - How do GQA/MQA, paged attention and cache quantization reduce it? [Likely]
19. What makes standard self-attention expensive for long contexts? [Reported] [3]
    - What are the time and memory complexity in sequence length, and what does FlashAttention change (and not change)? [Reported] (Amazon MLE aggregate: flash attention in the depth round) [23]
20. Explain Longformer and how its attention scales to long sequences. [Reported] [3]
    - Compare full attention, sparse or local attention, and other long-context approaches. [Reported] [3]
    - For a multi-turn shopping or search agent, when would you summarize or retrieve rather than extend the context window? [Reported] (Amazon RS aggregate: multi-turn search agent with long-context handling) [22]
21. How does beam search decoding work for LLMs? [Reported] [8]
    - When does beam search hurt open-ended generation, and why?
    - Compare greedy, beam, temperature, top-k and top-p sampling for a customer-facing assistant. [Reported] (LLM team: decoding choices for an Amazon use case) [8]
22. How would you add constrained or structured decoding so an assistant always returns valid JSON for a tool call? [Likely]
    - What does it cost in latency and quality?
23. What is speculative decoding, and when does it give a real speed-up? [Likely]
    - How do you pick the draft model, and how does the acceptance rate depend on the domain?
24. How does prefix (prompt) caching relate to the KV cache? When does it cut cost and latency for a Bedrock-style workload? [Likely] (follow-up in an Amazon-focused synthesized list [26])

    - What cross-tenant leakage risk does a shared prompt cache create, and how do you prevent it?
25. When is extra test-time compute (reasoning traces, self-consistency, best-of-n with a verifier) worth its latency and cost for a shopping or voice assistant? [Likely]
    - How would you decide per request whether to "think longer"?
26. How do you evaluate whether a model actually uses a long context, beyond needle-in-a-haystack tests? [Likely]
    - How would you test for "lost in the middle" effects on real multi-document tasks?

C. Pretraining, fine-tuning and PEFT

27. Which modern LLM training techniques do you know (pretraining, SFT, preference optimization)? [Reported] [10]
    - For an AGI-team role, expect several follow-ups in a row on each stage. [Reported] [10]
28. Pretraining vs fine-tuning: what changes in data, objective and learning rate? [Likely]
    - What is catastrophic forgetting, and how do you detect and limit it?
29. Walk me through an SFT training pipeline and its loss. [Reported] [19]
    - Why mask the prompt tokens in the loss?
    - How do you build and clean an instruction dataset for a domain like shopping or Alexa intents?
    - How do you mix general and domain data, and how would you choose the ratio?
30. Explain the main parameter-efficient fine-tuning techniques, which parameters each trains, and when each beats full fine-tuning. [Reported] (Amazon RS and MLE aggregates, online assessment) [22][23]; LoRA specifically [Reported] [4]
    - Explain LoRA. Which matrices do you adapt, and how do you choose the rank and alpha?
    - What is the memory cost of an adapter compared with full fine-tuning, counting optimizer states? [Reported] (Amazon MLE aggregate) [23]
    - Compare LoRA variants (QLoRA, DoRA, adapters, prefix tuning). [Reported] (Amazon RS aggregate) [22]
31. You need a domain-specialized model with limited compute. Do you use prompting, RAG, LoRA, full fine-tuning or continued pretraining? [Likely]
    - What evidence would make you move from one option to the next?
    - How does the answer change if the base model is only reachable as a hosted endpoint, for example through Bedrock? [Likely]
32. If you used knowledge distillation from an LLM, what exactly was distilled, and why? [Reported] [20]
    - Why distill instead of quantizing or pruning?
    - How would you distill an LLM into a small classifier using unlabeled data? [Likely] (see Amazon Science work on LLM distillation [31])
33. How would you generate synthetic training data with an LLM without collapsing diversity or leaking evaluation data? [Likely]
34. Your fine-tuning run's loss spikes at step 40k. How do you debug it? [Likely]
    - Which checks would you run on the data, the learning-rate schedule, mixed precision and gradient norms?

35. How does tokenizer choice (BPE vocabulary size, byte fallback) affect cost, latency and quality across Alexa's locales and scripts? [Likely] [27][32]
    - How would you extend a tokenizer for a new language without retraining from scratch?

D. Alignment: RLHF, DPO and beyond

36. What is the difference between DPO and RLHF? [Reported] [7]
    - Derive how DPO removes the explicit reward model. What role does the reference model play?
    - What does DPO's beta control, and what happens at very small or very large beta?
37. Explain RLHF and PPO, including the clipped objective. [Reported] (Amazon RS and MLE aggregates) [22][23]
    - Why include a KL penalty to the SFT policy, and what failure mode does it prevent?
38. Compare PPO, GRPO and GSPO. Why does GRPO drop the critic? [Reported] (Amazon RS and MLE aggregates) [22][23]
    - What does removing the critic cost, and how would you parallelize GRPO training? [Reported] (generative-AI phone screen; specialized screens) [22][23]
    - When would you prefer verifiable rewards (RLVR) over a learned reward model? [Likely]
39. If you have run GRPO-style training yourself: which instabilities did you hit (entropy collapse, reward spikes, length blow-up), and how did you fix them? [Reported] (Amazon RS aggregate: post-training teams look for "evidence you have run GRPO-style training and fixed its instabilities") [22]
40. RLHF in practice: what goes wrong at scale? [Reported] (Amazon MLE aggregate: "RLHF in practice" and "RLHF trade-offs" in depth rounds) [23]
    - Walk through reward hacking, length bias, sycophancy and mode collapse. [Likely]
    - How would you detect reward hacking from logs? [Likely]
41. How do you design a reward signal or preference dataset for a shopping assistant? [Reported] (Amazon RS aggregate: "how would you design the reward") [22]
    - Who labels, how do you measure agreement, and how do you handle annotator disagreement?
    - Where could RLAIF replace human labels, and where can it not? [Likely] (Alexa postings list SFT, DPO, RLHF and RLAIF [27])
42. Your DPO-tuned model gives longer but less accurate answers. Diagnose it. [Likely]
    - How do you control for length in the preference data and in evaluation?
43. How would you align a multilingual Alexa model when preference data is scarce in some locales? [Likely] (Alexa International Tech posting [27]; the Alexa International loop guide [32])

E. Retrieval-augmented generation (RAG)

44. Design an enterprise RAG service. [Reported] [16]

    - How do you chunk, embed and index heterogeneous documents?
    - How do you enforce per-user, document-level access control at query time, when permissions change between crawls?
    - How do you evaluate retrieval quality separately from generation quality?
45. What is the difference between a cross-encoder and a bi-encoder for retrieval? [Reported] [3]
    - When would you use each in a retrieve-then-rerank pipeline, and with what latency budget?
    - When would you distill a cross-encoder into a bi-encoder? [Likely] [31]
46. Explain RAG, and where it helps and does not help with hallucination. [Reported] (Amazon RS aggregate: LoRA, RAG and LLM agents in breadth) [22]
    - What failure modes remain once the right passage is retrieved (ignoring it, misreading it, or mixing it with parametric knowledge)?
47. Compare dense, sparse (BM25) and hybrid retrieval for product search queries. [Likely]
    - How do you handle exact attributes such as model numbers, sizes and brand names?
48. Contrastive embedding training: write a numerically stable, symmetric InfoNCE loss for paired image and text embeddings. [Reported] (Amazon RS and MLE aggregates, coding round) [22][23]
    - How do hard negatives change what the encoder learns, and how do you mine them without false negatives?
49. Rufus-style question: build retrieval-augmented answers to "What do I need for cold weather golf?" over the catalog, reviews and community Q&A. [Likely] [28][29][30]
    - How do you ground claims in product facts and avoid made-up specifications?
    - How do you handle stale or conflicting sources (listing vs reviews)?
    - How do you show citations, and how do you check that a citation supports the claim?
50. How would you decide between a longer context window and better retrieval for an assistant? [Likely]
51. How would you evaluate a RAG system offline (context recall and precision, faithfulness, answer relevance)? [Likely]
    - How do you build the golden set, and how often do you refresh it?
52. Much of the catalog is structured (attributes, prices, availability). When would you answer through a structured lookup or text-to-query tool instead of text retrieval, and how would the LLM decide? [Likely]
53. Use an LLM to rewrite or expand product-search queries. How do you keep it inside a search latency budget (offline caching for head queries, a distilled model for the tail)? How do you measure the effect on relevance? [Likely]
54. Would generative retrieval or semantic IDs (an LLM that decodes item identifiers) work for Amazon-scale product recommendation? What breaks as the catalog changes daily? [Likely]

F. Agents and tool use

55. Pick an agentic product we could build. Who is the customer, and how do you measure success? [Reported] [2]

56. Propose agentic AI solutions for a hypothetical project, and defend their viability. [Reported] (Amazon RS aggregate, science application) [22]
57. Design an agentic recommendation system for Amazon, covering the component architecture, model choices, and quality and safety evaluation. [Reported] [21]
58. Design a GitHub Copilot-style coding agent. [Reported] (Senior AS loop) [3]
    - What context does the agent retrieve, and how do you rank it?
    - How do you measure whether it actually helps (acceptance rate vs downstream defects)?
59. Evaluate an LLM agent with metrics beyond accuracy: tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you build an offline evaluation set for multi-step agent tasks?
    - How do you assign credit when a five-step trajectory fails at step four?
60. Design a multi-turn search agent that handles long conversation context. [Reported] (Amazon RS and MLE aggregates) [22][23]
61. Architect an AI chatbot with intent detection, dialogue state, retrieval, generation and fallback. [Reported] (Amazon RS aggregate, science application; Amazon MLE aggregate, ML system design) [22][23]
    - Which layers would you keep as classic NLU or rules, and which would you hand to the LLM?
    - How does the system learn continuously from production conversations? [Reported] [23]
62. Alexa+-style question: an LLM-based assistant has to call real device and service APIs (timers, smart home, ordering). How do you make tool calls reliable and safe? [Likely]
    - How do you handle ambiguous requests: ask a clarifying question, or act?
    - Which actions require confirmation, and how do you decide?
63. Design a browser-operating agent (in the spirit of Nova Act) that completes multi-step web tasks. How do you make it reliable enough to ship? [Likely] (Nova Act is named as an active Amazon AGI product [25]; the prompt also appears in a synthesized Amazon list [26])
64. How do you defend an agent against prompt injection from retrieved web pages, product listings or reviews? [Likely]
65. Design long-term memory for an assistant like Alexa+: what to store about a household's preferences, how to retrieve it, and how to let users see and delete it. [Likely]
    - How do you stop a wrong memory from repeatedly degrading answers?
66. When would you split an assistant into several specialized agents or "experts" behind an orchestrator, rather than one model with many tools? [Likely]
    - How do you evaluate and debug the routing between them?

G. Evaluation

67. Design an LLM quality validation system. [Reported] (Amazon RS and MLE aggregates; the MLE aggregate says strong answers combine automated checks with human review for generated output) [22][23][36]
    - What do you measure offline, and which online guardrails do you set? [Reported] [23]

    - How do you measure hallucination, faithfulness and helpfulness separately? [Likely] (follows from [3][16])
    - When would you trust an LLM-as-judge, and how would you calibrate it against human labels? [Likely]
    - What biases do LLM judges have (position, length, self-preference), and how do you control for them? [Likely]
68. Design an evaluation framework that checks whether a text-to-image system spells a brand name correctly and places it naturally (OCR accuracy, embedding checks, human review protocol). [Reported] (Amazon RS and MLE aggregates) [22][23]
69. How would you decide an LLM-powered assistant is ready to launch to millions of customers? [Likely] (synthesized Amazon list [26])
    - Which launch gates would you use for quality, safety, latency and cost?
70. Offline metrics improved but the A/B test fell short. How do you find out why? [Reported] (Amazon RS and MLE aggregates, depth round, general ML) [22][23]
    - How do you design an online experiment for a conversational assistant, where sessions interact and the outcome (purchase) is delayed? [Likely]
71. How do you detect benchmark contamination when evaluating a new foundation model? [Likely]
72. How would you measure an assistant's calibration and teach it to say "I don't know"? [Likely] (calibration reported as an agent-evaluation metric [3])
73. How would you run a red-teaming and safety evaluation for a shopping or voice assistant (toxicity, unsafe advice, PII leakage, jailbreaks)? [Likely]
74. How would you evaluate a multilingual assistant fairly across locales when you have far fewer human raters in some languages? [Likely] [27][32]
75. When is perplexity a misleading metric for comparing foundation-model checkpoints? How do preference-based evaluations (pairwise win rate, Elo or Bradley-Terry) complement it? [Reported for the topic] (Amazon AGI candidate reports summarized as covering "evaluation (including perplexity and preference-based evals)" [26]); exact wording [Likely]
76. How would you measure product-market fit for a GenAI shopping experience on Alexa? [Reported] (question title in an Amazon aggregator list, "Measure PMF for Alexa Shopping") [36]
    - Which engagement, retention and purchase metrics would you trust, and how would you separate novelty effects from lasting value?
77. How would you use implicit feedback (thumbs, reformulations, abandonment, purchases) from an assistant to improve it? How do you avoid feedback loops and selection bias? [Likely]

H. Hallucination and grounding

78. How would you reduce hallucination in an LLM-powered assistant? [Likely] (Amazon's shopping assistant runs on LLMs including Amazon Nova, built on Bedrock [29])
    - Which causes are data-side, which are decoding-side, and which are retrieval-side?
    - How do you trade off abstaining against helpfulness?
79. An assistant states a wrong battery life for a product. Walk through your root-cause analysis. [Likely] [28][30]

    - How do you tell whether the cause was retrieval, a stale catalog attribute, or the model?
80. How would you build an automatic, claim-level factuality checker for generated answers? [Likely]
    - How do you validate the checker itself?
81. How would you fine-tune an LLM to be a shopping assistant that cites its sources? [Likely]
    - How would you reward grounded claims and penalize unsupported ones in preference tuning?
82. How would you summarize thousands of customer reviews into a short "Customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?
83. Design a guardrail layer that sits before and after the model (input filters, output filters, a grounding check, PII redaction). How do you set its thresholds and measure the false-block rate? [Likely] (guardrails as a composable layer in a synthesized Bedrock-shaped design [26])

I. Efficiency, serving and cost

84. What are the trade-offs of putting LLMs into production (cost per token, latency, reliability, model right-sizing)? [Likely] (Frugality applied to GPUs is listed as an Amazon design axis [26])
85. Explain the memory arithmetic of quantization. Compare int8 and int4, and weight-only vs activation quantization (GPTQ, AWQ). What is the quality cost? [Reported for the memory arithmetic] (Amazon MLE aggregate) [23]; method comparison [Likely]
    - Which layers or tasks are most sensitive?
86. A customer's Bedrock-hosted workload costs too much. How would you cut inference cost sharply without unacceptable quality loss? [Likely] (Bedrock-shaped prompt, synthesized in an Amazon prep list [26])
    - Compare model routing, caching, distillation, quantization, batching and prompt compression.
    - How do you prove the quality loss is acceptable?
87. Your LLM endpoint's p99 latency doubled after a deploy, but the model weights did not change. Walk me through debugging it. [Likely] [26]
88. How does an inference engine such as vLLM raise serving throughput for large language models? [Reported] (Amazon RS and MLE aggregates, online assessment) [22][23]
    - Explain paged KV memory and continuous batching, and what each buys you.
89. Compare data, tensor and pipeline parallelism for training a large model across hundreds of accelerators. [Reported for pipeline-parallel schedules] (Amazon MLE aggregate, staff-level screens) [23]; the full comparison also appears in a synthesized list [26]
    - Where do the communication bottlenecks sit for each, and how do pipeline schedules reduce the bubble? [Reported] [23]
    - How does ZeRO/FSDP sharding change memory per device? [Likely]
90. Explain the collective operations used in distributed training (all-reduce, all-gather, reduce-scatter) and which parallelism strategy relies on each. [Reported] (question

title in an Amazon aggregator list, "Explain parallelism and collectives in training") [36]
    - How does inter-node bandwidth change your choice of strategy? [Reported] [23]
91. For a residual MLP of width 4096 in BF16, which of batch size 1 and batch size 256 is compute-bound? Do the analysis from bandwidth and peak throughput. [Reported] (Amazon RS and MLE aggregates, manager-level screen) [22][23]
92. You have one latency budget for Rufus answers (say, first token in under 1 second). How do you split it across retrieval, reranking, generation and safety filters? [Likely] [28]
93. When would you route a request to a small model and when to a large one, and how would you learn the router? [Likely]
94. Design a distributed pipeline that ingests, tokenizes, deduplicates and filters LLM pretraining data. [Reported] (Amazon MLE aggregate, staff loop) [23]
    - How do you detect near-duplicates at web scale?
    - How do you measure the downstream effect of a quality filter?
95. Split an ordered list of sequences into contiguous batches. Each batch costs a fixed overhead plus padding up to its longest item. Minimize the total cost: give the state, the recurrence and a worked example. [Reported] (Amazon RS and MLE aggregates, Applied Scientist phone screen) [22][23][36]
    - How does this relate to length bucketing when serving or training LLMs?
96. Design a multi-tenant inference platform that serves many foundation models to thousands of customers (Bedrock-shaped). [Likely] (synthesized Amazon list [26])
    - How do you handle noisy neighbors, quotas and a 10x traffic spike on one model?
97. You must serve hundreds of customer-specific LoRA adapters on one shared base model. How do you batch requests across adapters, and what does it cost in latency compared with merged weights? [Likely]

J. Multimodal and speech (team-dependent)

98. Compare a dual-encoder (CLIP-style) model with a vision-language model that uses a resampler or adapter. [Reported] (Amazon RS aggregate) [22]
    - Which would you use for product image search, and which for visual question answering on a listing?
99. How does the Segment Anything Model work? [Reported] [3][10]
100. How do diffusion noise schedules affect sample quality and training? [Reported] (Amazon RS aggregate) [22]
101. What causes codebook collapse in VQ-VAE, and how do you fix it? [Reported] (Amazon RS aggregate) [22]
102. Compare approaches to building joint image-text embeddings. [Reported] (Amazon RS aggregate, online assessment) [22] - How is a dual-encoder embedding trained, and what does batch size do to it?
103. For a speech-to-speech assistant (in the spirit of Nova Sonic or Alexa+), what changes when you move from an ASR, then LLM, then TTS cascade to an end-to-end speech model? [Likely] [25] - How do you evaluate turn-taking, barge-in and latency?
104. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents, and could an LLM rescoring stage help? [Reported for

the base question] [32]; LLM follow-up [Likely]

K. Applying GenAI to Amazon problems (science-application flavor)

105. Here is a high-level problem from our team. Would you solve it with an LLM at all? Propose a solution: data sources, annotation strategy, modeling, evaluation, trade-offs and risks. [Reported for the frame] [33]; the LLM-or-not framing is [Likely]
    - What is the cheapest baseline that might be good enough?
106. Use an LLM to categorize millions of products into a large taxonomy. How do you combine LLM labeling, a distilled classifier and human audit? [Likely] (product categorization is reported in classical form in earlier chapters; LLM distillation context [31])
107. Use an LLM to generate or enrich product listings (titles, bullet points, attributes) from seller input. How do you prevent made-up attributes? [Likely]
108. Use LLMs as relevance judges for product search. How do you validate them against human judges and behavioral signals? [Likely]
109. How would you personalize Rufus answers with a customer's purchase history while respecting privacy? [Likely] [29]
    - How would you measure whether personalization helps rather than creeps people out?
110. Sponsored products in a generative answer: how do you blend ads into an LLM response while keeping trust, and how do you measure the impact? [Likely]
111. Design an LLM-based system to detect policy-violating content or fraud in reviews or ads. [Likely] (Advertising Trust context [18])
    - How do you deal with adversaries who adapt to the classifier?
112. Design a system that turns customer-support transcripts into structured issue reports. How do you evaluate extraction quality? [Likely]
113. A team wants to replace an existing BERT classifier with an LLM prompt. What do you need to see before you agree? [Likely]
    - How do you compare cost per 1,000 requests, latency and accuracy on the long tail?
114. How would you detect LLM-generated fake reviews at Amazon scale, and what false-positive rate on genuine reviewers is acceptable? [Likely]

L. Depth probes on your own LLM project (depth-round style)

115. Walk me through your most significant LLM or NLP project, including the GenAI fundamentals behind it and its business impact. [Reported] [18]
116. Why did you choose this base model and size? What would you do with more compute? [Reported] (Amazon RS aggregate: backbone choice, use of extra resources) [22]
117. How did you build the evaluation set, and how do you know it reflects real customer traffic? [Likely]
118. What did your error analysis show, and which fix had the biggest effect? [Likely]

    - What did you try that did not work?
119. If you fine-tuned, how did you rule out leakage between training data and evaluation prompts? [Likely]
120. Where would your approach break if deployed to a different marketplace or language? [Likely]
121. How did you decide the project was "done" and ready to ship (Deliver Results, Insist on the Highest Standards)? [Likely]
122. Explain an AI-agent project you built, and how it was adopted. [Reported] (Amazon MLE aggregate, project deep dive) [23]
    - What share of the target users actually used it, and what stopped the rest?
123. Paper-read round: discuss a recent LLM paper's method, assumptions and weaknesses. [Reported for the format] (Amazon RS aggregate: paper-read round on FAR and "premium" loops) [22]; the choice of an LLM paper is [Likely]

M. GenAI behavioral and LP crossovers

124. How do you use AI tools in your work, and when did one give you a wrong answer? How did you catch it? [Reported] (Amazon general aggregate, phone screen and Bar Raiser) [24]
125. What hands-on experience do you have with post-training or agents? [Reported] (Amazon recruiter screens for generative-AI teams) [22][24]
126. Walk me through a concrete AI-assisted engineering workflow: what you delegated, what context you protected, and how you verified the output. [Reported] (Amazon new-grad screens, August 2026, SDE track; aggregator) [35]
127. You get a small repository, some failing tests and a guard-railed AI assistant. How do you work, and how do you know the fix is right? [Reported] (Amazon online assessment, AI-assisted debugging task, SDE and MLE pipelines) [23][24]
128. Tell me about a time you pushed back on using an LLM where a simpler model was better (Frugality, Invent and Simplify). [Likely]
129. Tell me about a time an LLM feature you shipped behaved badly in production. What did you do in the first 24 hours (Ownership, Dive Deep)? [Likely]
130. Tell me about a time you had to explain LLM risks (hallucination, safety) to a non-scientist stakeholder (Earn Trust). [Likely]

Coverage notes for prep

- **Highest-yield set across reports:**
    - transformer and attention mechanics (questions 1-2, 16);
    - BERT vs GPT (11);
    - KV cache, decoder inference and long context (17-21);
    - training, SFT and PEFT/LoRA (27, 29-30);
    - DPO vs RLHF, and PPO/GRPO (36-38);
    - bi-encoder vs cross-encoder, and RAG design (44-45);
    - agents and LLM evaluation (55-61, 67);
    - vLLM serving (88).

Sources: [3][4][7][8][10][16][19][22][23].

- **Team-specific tilt:**
  - AGI/Nova teams: training, RL, parallelism, evaluation and ML coding [15][17][22][25][26].
  - Rufus/Stores: RAG, grounding, ranking and evaluation [28][29][30].
  - Alexa: multilingual alignment, tool use and speech [27][32][36].
  - AWS/Bedrock-adjacent science: PEFT, serving cost and customization [4][26].
- **Aggregator caveat:** questions marked "(Amazon RS aggregate)" or "(Amazon MLE aggregate)" come from compilations of Amazon Research Scientist and ML Engineer reports [22][23]. Those loops overlap heavily with AS loops but are not identical. The aggregator pages regenerate often, so a quoted item may later move or disappear.

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08. Statistics, Causal Inference and A/B Testing: Hypothesis Tests, Power, Variance Reduction, Weblab, Bandits and Forecasting

> Scope: the statistics and experimentation content an L5 Applied Scientist (Applied Scientist II) candidate meets in the Amazon onsite loop. This chapter covers hypothesis testing, power and sample size, metric design, variance reduction, experiment pitfalls at Amazon scale, causal inference without clean randomization, bandits, and forecasting. Questions only; no answers.

> Sourcing note: references [1]-[21] come from findings that chapters 01-06 of this bank gathered from search-result extracts. In this run, none of those pages could be re-fetched: WebFetch could not resolve amazon.jobs, igotanoffer.com, interviewkickstart.com, medium.com or teamblind.com, the proxy refused Glassdoor, and the shared web-search budget was used up. So no reference could be confirmed live or dead, and none was removed. Each [Reported] tag was checked against how chapters 01-06 attribute that question, and the attributions matched. References [22]-[24] are new. They are TrueInterview candidate-report pages mirrored on GitHub (sync of 2026-10-14). They were read in full from a local clone, and every question that cites them was checked against that text. Questions without a direct candidate report are tagged [Likely], with the reasoning given.

Format notes for this round

- **There is no dedicated "stats round" in most L5 AS loops.** Amazon's official prep page describes the onsite as four 55-minute interviews with members of the science community. Each interviewer is assigned competencies and typically asks 2-3 Leadership Principle questions [1]. Candidates more often report 5 rounds of 55-60 minutes for AS II: ML breadth, science depth, science application or ML design, coding, and a bar raiser [2][3][15].
- **Where statistics shows up:**
  - **ML/Science Breadth round:** p-values, statistical tests, CLT, and "design an A/B test" are reported as part of the breadth survey [4][8][9].
  - **Phone screen:** an L5 phone screen mixed coding, statistics and LPs [12], and a 2025 first-round AS interview covered ML and statistics alongside LLM theory [21]. In 2026 reports, statistics-heavy teams such as Ads asked about least-squares assumptions, losses as likelihoods, A/B test power, anomaly detection and fraud metrics. The coding part was a bootstrap confidence interval with a loop-free follow-up [22][23].
  - **Coding round:** numerical code from scratch is reported, such as a NumPy mean and variance followed by maximum-likelihood questions, and a Bayesian

classifier [22].
    - **Science Application / ML design round** (often the hiring manager, and usually present at L5 [20]): candidates are asked how they would A/B test the launch, choose metrics and handle conflicting metrics [5][6][10]. Ranking prompts include online experiments and position-bias correction [23].
    - **Science depth round:** interviewers ask how the project was A/B tested and how you debugged an online result that missed the offline gain [22][23].
    - **Bar raiser:** a Principal TPM bar raiser asked a few stats checks, such as interpreting a p-value or a confidence interval [7].
    - **Team-specific depth:** supply-chain, pricing and forecasting teams (SCOT) ask forecasting and inventory questions such as the newsvendor problem [16][17][18]. A 2026 aggregator says Ads and seller teams lean on classical statistics and experimentation, while generative-AI teams lean on fine-tuning and alignment [22].
- **The L5 bar.** L5 candidates are expected to own a problem end to end. That includes framing a vague business goal, defining offline and online metrics, designing the launch experiment and making the ship or no-ship call when metrics disagree [6][20][22]. In stats questions, interviewers probe past definitions into judgement. Expect follow-ups like "the test is underpowered, now what?", "the primary metric is up but a guardrail is down", and "you can't randomize, now what?" [5][6][10].
- **Experimentation at Amazon (context).** Amazon's internal A/B testing platform is widely known as Weblab. Teams treat online experiments as the launch gate for ranking, recommendation, ads, pricing and Alexa/Rufus changes. Interviewers outside retail may phrase this generically as "how would you A/B test this?" Questions that mention Weblab are tagged [Likely], because no source in this bank reports a candidate being asked about Weblab by name.

A. Probability and statistical foundations

1. State the Central Limit Theorem. Why does it matter for A/B tests? [Reported] [8]
    - When does the CLT approximation break down for a revenue-per-visitor metric with a heavy right tail?
    - How large must n be before you trust a normal approximation for a skewed metric?
    - How does the law of large numbers differ from the CLT? [Likely]
2. What is Bayes' rule? Explain the prior, likelihood and posterior. [Reported] [13]
    - Walk through a base-rate example: a fraud model flags 1% of orders, fraud prevalence is 0.1%, and recall is 90%. What is the precision?
3. Work through a coin-flip probability question on the whiteboard. [Reported] [14] (the exact wording was not captured)
    - Derive the MLE for p after k heads in n tosses. What is its variance?
    - How would you build a confidence interval for p when k = 0?
4. State Hoeffding's inequality and use it. [Reported] [11]
    - How many samples do you need to estimate a click-through rate to within plus or minus 0.1% with 95% confidence?
    - How does a Hoeffding bound compare with a CLT-based interval in practice?

5. Write a NumPy function for the mean and variance of an array, then relate it to maximum-likelihood estimation. [Reported] [22][24] (coding round; aggregator sighting September 2026)
    - Why is the Gaussian MLE of the variance biased? When do you divide by n, and when by n - 1?
    - How do you compute a variance in a numerically stable way on a stream (Welford)?
    - What are the MLE, the Fisher information and the asymptotic variance for a Poisson rate? [Likely]
6. Show that squared loss corresponds to a Gaussian likelihood and log loss to a Bernoulli likelihood. [Reported] [22] (statistics-heavy team phone screens, 2026)
7. State the assumptions of ordinary least squares, and say what breaks when each one fails. [Reported] [22][23]
    - With heteroscedastic or clustered errors, how do you get valid standard errors? [Likely]
8. What is the difference between a frequentist confidence interval and a Bayesian credible interval? [Likely] (follows from the reported CI interpretation probe [7])
9. Explain the delta method. When do you need it for an A/B test metric? [Likely] (ratio metrics like CTR or revenue per session are standard in Amazon retail and ads)
    - Derive the variance of a ratio metric such as clicks / impressions when the randomization unit is the customer.
10. Implement bootstrap sampling for a mean and its confidence interval. Say which statistics it estimates poorly, then remove the explicit loop. [Reported] [22][23][24] (Ads-team phone screen; aggregator sighting August 2026)
    - When would you use the bootstrap instead of an analytic standard error?
    - How do you bootstrap correctly when observations are clustered by customer?
11. Compare the Poisson, negative binomial and zero-inflated distributions for modeling orders per customer. [Likely] (retail count data)
12. Implement a Bayesian classifier from scratch and apply it to a few scenarios. [Reported] [22] (coding round)
    - How do you handle zero counts? When does the naive conditional-independence assumption hurt calibration? [Likely]

B. Hypothesis testing and p-values

13. What is a p-value? Interpret it in the context of an A/B test. [Reported] [4][9] (also asked as "interpret a p-value or a confidence interval" by a Principal TPM bar raiser [7])
    - What does a p-value of 0.03 NOT tell you?
    - What is a common misinterpretation of a 95% confidence interval?
    - Explain it to a product manager with no statistics background.
14. Walk through hypothesis testing end to end: which statistical tests would you use, and when? [Reported] [5][8][9]
    - When would you use a t-test, a z-test, a chi-squared test or a non-parametric test?
    - Welch's t-test or Student's t-test: which is the safer default, and why?

15. Define Type I and Type II errors. Which is more costly for a checkout-flow change compared with a recommendation widget? [Likely]
16. Compare one-sided and two-sided tests. When, if ever, is a one-sided test justified for a launch decision? [Likely]
17. How would you test whether conversion rates differ between two arms? Compare a two-proportion z-test, a chi-squared test and Fisher's exact test. [Likely] (conversion is the core retail metric)
18. When would you use Mann-Whitney U or a permutation test instead of a t-test? [Likely]
    - Does Mann-Whitney test a difference in means? What does it actually test?
19. You observe a statistically significant 0.05% lift on a site with hundreds of millions of sessions. How do you think about statistical versus practical significance? [Likely] (Amazon traffic scale)
20. What is a paired test? When could an Amazon experiment use a paired design? [Likely]
21. How do you test for equivalence or non-inferiority, for example to show that a cheaper model is "no worse"? [Likely] (cost-reduction launches such as model distillation)
    - How do you choose the non-inferiority margin?
22. How would you run a Bayesian A/B test that reports the probability that treatment beats control and the expected loss of shipping it? [Likely]
    - When can it give a different launch call from a frequentist test?
    - Is a Bayesian test immune to peeking? Defend your answer.
23. How do you build a confidence interval for the difference in median or p90 latency between two arms? [Likely] (latency is a standard guardrail for model launches)

C. A/B test design, power and sample size

24. Explain hypothesis testing, then design an A/B test for a model launch. [Reported] [5] [6][9][10]
    - How do you choose the randomization unit, primary and guardrail metrics, and sample size or power?
    - What do you do if the test is underpowered or the metric is heavy-tailed?
25. What determines the power of an A/B test? Compute the power for a given effect size, variance and sample size. [Reported] [22][23] (Ads-team phone screen, 2026)
    - How does power change if you split traffic 90/10 instead of 50/50? [Likely]
26. Treatment improves the target metric, but a population-level metric drops. What do you do? [Reported] [6]
    - How would you decide whether the drop is real, a segment effect, or noise?
    - Who do you bring into the decision, and what do you write in the launch doc?
27. Derive the sample-size formula for a two-sample test of means. Which inputs drive it the most? [Likely]
    - How does required sample size scale with the minimum detectable effect (MDE)?
28. How do you choose the MDE for an experiment? [Likely]
    - Who should own that number, science or the business?
29. Your power analysis says you need 8 weeks, but the team wants an answer in 2. What are your options? [Likely]

30. How do you choose the randomization unit (customer, session, page view, product, seller, marketplace)? [Likely] (the Amazon marketplace has customers, sellers and products as distinct units)
    - What goes wrong if the analysis unit differs from the randomization unit?
31. How long should an experiment run? How do you account for day-of-week effects and paydays? [Likely]
32. What is an A/A test, and what would you learn from one? [Likely]
    - Your A/A test shows a significant difference. Walk through your debugging.
33. What is a sample ratio mismatch (SRM)? How do you detect and diagnose it? [Likely]
    - The split is 50.4/49.6 on 20 million users. Is that a problem?
34. How do you handle triggering and dilution, for example when only 3% of users ever see the changed widget? [Likely] (retail page widgets)
    - How do you translate a triggered-population effect into a site-wide impact estimate?
35. How would you run an experiment for a change that affects only Prime members, or only a single marketplace? [Likely]
    - A small marketplace has too little traffic for its own test. How could a hierarchical model borrow strength from larger marketplaces, and what does that risk?

D. Metrics: primary, guardrail and long-term

36. Choose an overall evaluation criterion for a change to product search ranking. [Likely] (search is a core AS team)
    - Why might clicks be a poor primary metric? What about revenue?
37. What guardrail metrics would you add for a recommendation change? [Likely]
    - Consider latency, returns, customer-service contacts, and seller-side metrics.
38. How do you define a metric that captures long-term customer value when the experiment runs for only two weeks? [Likely]
    - How would you build and validate a surrogate or proxy metric?
39. Your model's offline metric improved, but the online A/B result fell short of the offline gain. How do you find out why? [Reported] [22][23] (science depth round)
    - How does your offline metric (NDCG or AUC) relate to the online business metric?
40. How do you evaluate a metric's sensitivity, or directionality, before using it as a launch criterion? [Likely]
41. How would you measure a change whose effect shows up as fewer returns 30 days later? [Likely] (delayed outcomes in retail)
42. What is Simpson's paradox? Give an Amazon-flavoured example where it could reverse an experiment readout. [Likely]
43. A metric moved by +2% in the US and -1% in the UK. How do you decide whether to launch globally? [Likely]
44. Design an evaluation framework that checks whether generated images spell a brand name correctly and place it naturally. [Reported] [22] (science application round)
    - How do you reconcile automatic metrics, such as OCR accuracy and embedding similarity, with a human review protocol? [Likely]

    - How do you measure agreement between raters, and how many labels do you need to estimate the defect rate within plus or minus 2%? [Likely]
45. You use an LLM judge to compare two model versions on paired prompts. How do you validate the judge against human labels and put a confidence interval on the win rate? [Likely] (LLM evaluation is reported in application rounds; see [22] and chapter 07)
    - How many paired comparisons do you need to detect a 3-point win-rate difference, and how do you handle ties and position bias in the judge?

E. Variance reduction and sensitivity

46. What is CUPED, and why does it reduce variance? [Likely] (CUPED appears as a probe on the reported "design an A/B test" question in this bank's design chapter)
    - Which pre-period covariate would you choose, and what happens for new customers with no pre-period data?
    - How does CUPED relate to regression adjustment (ANCOVA)?
47. Compare stratified sampling, post-stratification and regression adjustment as variance reduction methods. [Likely]
48. How do you handle heavy-tailed metrics like revenue per customer? [Likely] (the heavy-tail probe is in [10])
    - Compare winsorization, capping, log transforms and switching to a quantile metric. What do you trade off with each?
    - Does capping introduce bias, and when is that acceptable?
49. When would you use interleaving instead of an A/B test for a ranking change? [Likely] (search and ads ranking teams)
    - What can interleaving tell you, and what can it not?
50. How can you use ML predictions as control variates for variance reduction? [Likely]
51. Your experiment is underpowered and you cannot get more traffic. List five ways to increase sensitivity. [Likely]

F. Multiple testing, peeking and sequential testing

52. Your dashboard tracks 40 metrics. Three are significant at 0.05. What do you conclude? [Likely]
    - Compare Bonferroni, Holm and Benjamini-Hochberg. When is each appropriate?
    - How does your answer change when you test many variants against one control at the same time? (follow-up to a reported A/B design question; see [6])
53. A PM checks results every day and wants to stop when $p < 0.05$. What is wrong with that? [Likely]
    - How would you design a sequential test (alpha spending, group sequential, mSPRT or always-valid p-values) to allow safe early stopping?
54. You slice results by 20 segments and one shows a large effect. How do you decide whether to believe it? [Likely]
    - How would you estimate heterogeneous treatment effects in a principled way?
55. Many teams run experiments on the same pages at the same time. How do you handle interactions between concurrent experiments? [Likely] (Amazon runs many concurrent experiments across teams)

G. Pitfalls at Amazon scale: interference, novelty and marketplaces

56. How do you measure a recommender when network or interference effects break user-level randomization? [Likely] (follow-up to a reported design question; see [6])
57. How would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely] (pricing science teams; see [18])
    - When would you use switchback, cluster or marketplace-level randomization?
58. A new seller-facing feature changes how sellers price, which affects buyers in control. How do you get an unbiased estimate? [Likely] (two-sided marketplace)
59. How would you experiment on a fulfillment or inventory-placement policy when products share warehouses and trucks? [Likely] (SCOT; see [18][19])
60. What are novelty and primacy effects? How do you detect them in a two-week test? [Likely]
    - The test wins in week 1 and is flat in week 3. What do you report?
61. How do you A/B test an ads model when advertiser budgets are shared between treatment and control? [Likely] (ads teams; see [8])
    - What is budget cannibalization, and how do budget-split designs address it?
62. How would you A/B test a change to Alexa or Rufus, where the conversation spans many turns and outcomes are sparse? [Likely] (LLM product teams)
    - What would your primary metric be, and how would you combine it with human or LLM-judge evaluation?
63. When is an A/B test the wrong tool, and what would you do instead? [Likely]
64. Design a search retrieval and ranking pipeline, including online experiments and position-bias correction. [Reported] [23] (ML design round; the estimation follow-ups below are [Likely])
    - How do you estimate position bias from logged clicks: result randomization, click models or intervention harvesting?
    - How do you use the estimated propensities in training and in offline evaluation, and what happens to variance when a propensity is tiny?

H. Weblab, launch decisions and post-launch monitoring

65. Walk me through how you would set up, monitor and close out an experiment on Amazon's internal experimentation platform (Weblab) for a model change. [Likely] (Weblab is Amazon's internal A/B platform; no source here reports it being named in an interview)
    - What would you check on day 1, and what before the launch decision?
66. How would you ramp a risky change (1%, 5%, 50%), and what triggers a rollback? [Likely]
67. What should go in a launch recommendation, a narrative-style write-up for leadership, after an experiment? [Likely] (Amazon's written narrative culture)
    - How do you present a result with a wide confidence interval that includes zero?
68. The result is significant on the primary metric, but the effect is smaller than the cost of serving the new model. How do you decide? [Likely]

69. Leadership wants to launch despite a non-significant result. How do you respond? [Likely] (also a Leadership Principle probe for "Have Backbone; Disagree and Commit")
70. After launch, how do you check that the measured lift persists? [Likely]
    - What is a holdback (long-term holdout) group, and how big should it be?
71. How would you estimate the cumulative impact of 30 launched experiments in a year? Why might the sum of individual lifts overstate it? [Likely]
    - What are the winner's curse and regression to the mean, and how do you correct for them?
72. Your model is live. How would you set alert thresholds on its online metrics so that real regressions are caught without paging on noise, and what triggers a rollback? [Reported] [23] (a candidate reported spending much of an ML design hour on alert thresholds and rollback)
73. How would you detect a shift between training and serving feature distributions? [Likely] (monitoring is graded in design rounds [23])
    - Compare the population stability index, the KS test and a domain classifier. With millions of rows every test is significant, so how do you set a practical threshold?

I. Causal inference without clean randomization

74. Design a causal study to measure the impact of reminders on customer behavior. [Reported] [24] (role not stated)
    - The reminders were already sent to every customer who met a rule-based threshold. What can you still estimate, and under what assumptions? [Likely]
75. Prove that a new allocation method outperforms the manual baseline. [Reported] [24] (role not stated; paired with "build a package-allocation model for couriers")
    - You cannot randomize couriers or regions. Compare switchback, matched-pair and difference-in-differences designs for this case. [Likely]
76. How would you estimate the causal effect of a promotion or price change without a clean experiment? [Likely] (pricing and promotion teams; see [18])
    - When would you use difference-in-differences, synthetic control, instrumental variables or double ML?
77. Explain difference-in-differences. What is the parallel-trends assumption, and how would you check it? [Likely]
    - What goes wrong with staggered adoption, for example a feature rolled out marketplace by marketplace?
78. Explain synthetic control. When would you use it to measure a launch in a single country? [Likely]
79. What is an instrumental variable? Give a plausible instrument for estimating price elasticity on Amazon. [Likely]
    - How do you defend the exclusion restriction?
80. How do you estimate price elasticity when prices are set by your own algorithm (endogeneity)? [Likely] (pricing system design; see [18])
81. Explain propensity score matching and inverse propensity weighting. What assumptions do they need? [Likely]
    - How do you check overlap (positivity)?

    - What does doubly robust estimation buy you?
82. What is regression discontinuity? Give an example from a free-shipping threshold. [Likely] (free-shipping thresholds are a familiar Amazon retail mechanic)
83. How would you measure the effect of joining Prime on customer spending? [Likely]
    - Why is a naive comparison of Prime and non-Prime members biased, and in which direction?
84. What is double or debiased machine learning, and why use cross-fitting? [Likely]
85. Draw a causal DAG for "recommendation exposure leads to purchase". Name the confounders, colliders and mediators. [Likely]
86. How would you measure the incremental impact (lift) of an ad campaign for an advertiser? [Likely] (ads teams; see [8])
    - What are the trade-offs between ghost ads, geo experiments and observational methods?
87. What is uplift modeling? How does it differ from response modeling? [Likely]
    - How would you evaluate an uplift model (Qini or uplift curves)?
88. How do you validate a causal estimate when you can never observe the counterfactual? [Likely]

J. Bandits and adaptive experimentation

89. When would you use a multi-armed bandit instead of an A/B test? [Likely] (ads creative and widget selection)
    - What do you give up in inference when you use a bandit?
90. Explain epsilon-greedy, UCB and Thompson sampling. How do their exploration strategies differ? [Likely]
    - Write the Beta-Bernoulli posterior update for Thompson sampling.
91. Define regret. What regret bounds do UCB and Thompson sampling achieve? [Likely] (Hoeffding bounds are reported [11] and underpin UCB)
92. How would you design a contextual bandit for recommending widgets on the homepage? [Likely]
    - How do you handle delayed rewards, such as purchases that happen hours later?
    - How do you handle non-stationarity, for example during Prime Day?
93. How do you evaluate a new policy offline from logged bandit data? [Likely]
    - Compare inverse propensity scoring, self-normalized IPS and doubly robust estimators. When does each have high variance?
    - What must the logging policy record for off-policy evaluation to be possible?
94. A bandit is optimizing CTR on deal-page creatives and converges to clickbait. How do you fix the reward? [Likely]
95. How would you frame budget pacing or bidding as a bandit problem or as constrained optimization? [Likely] (ads teams; see [8])

K. Forecasting and time series

96. What is the newsvendor problem, and what is its solution? [Reported] [16] (reported in Amazon supply-chain interviews, not in a science loop)

    - How does a quantile forecast map to the newsvendor critical ratio?
97. How would you forecast demand for millions of products, including new ones with no history? [Likely] (SCOT forecasting teams; see [17][18][19])
    - What does a global model trained across products buy you over per-product models?
98. Why use probabilistic or quantile forecasts rather than point forecasts for inventory decisions? [Likely] (see [17])
    - What is quantile (pinball) loss, and how do you evaluate calibration of P50 and P90 forecasts?
99. Compare MAPE, sMAPE, WAPE, RMSE and quantile loss. Which would you use for intermittent demand? [Likely]
    - Why does MAPE behave badly for low-volume products?
100. How do you backtest a forecasting model correctly? [Likely]
    - What is an expanding-window or rolling-origin evaluation, and how does leakage creep in?
101. How do you forecast demand during Prime Day or other promotions, when history is sparse and distorted? [Likely] (see [17])
    - How do you separate a promotion lift from the baseline?
102. Demand is censored when products are out of stock. How do you correct for it? [Likely] (in-stock is a core SCOT metric)
103. Compare ARIMA, exponential smoothing, and neural forecasters such as DeepAR- or transformer-style models. When does each win? [Likely] (Amazon publishes neural forecasting research; see [17])
104. Explain stationarity, autocorrelation and seasonality. How do you test for them? [Likely]
105. How would you forecast a hierarchy (product to category to total) so the forecasts add up? [Likely]
106. How do you evaluate a forecasting system against a business outcome (in-stock rate, holding cost) rather than accuracy alone? [Likely] (see [18])
    - How would you A/B test a new forecast when buying decisions in treatment and control share inventory?
107. How would you detect anomalies or structural breaks in a demand time series, such as the start of a pandemic? [Likely]
108. Design an end-to-end regression model for energy demand. [Reported] [24] (role not stated)
    - Weather is an input, but at prediction time you only have a weather forecast. How does that error propagate into your prediction interval? [Likely]
109. How would you build prediction intervals with guaranteed coverage around any forecaster (conformal prediction)? What happens to coverage under distribution shift? [Likely]
110. How would you model time to repurchase or time to churn when many customers have not yet repurchased (right-censoring)? [Likely]
    - Compare Kaplan-Meier, Cox proportional hazards and a discrete-time classifier. How do you check the proportional-hazards assumption?

L. Applied stats case questions (mixed, interview-style)

111. Orders dropped 5% week over week. Walk me through how you would diagnose it. [Likely]
112. A product page change increased add-to-cart but not purchases. What hypotheses would you test, and how? [Likely]
113. You are asked to estimate how many customers would buy a new Echo device at a given price. What data and statistical approach would you use? [Likely]
114. Customer reviews have an average rating of 4.6 from 5 reviews and 4.3 from 2,000 reviews. How should ranking treat them? [Likely] (product reviews ranking)
    - How would you apply Bayesian shrinkage, or a lower confidence bound, to the rating?
115. How would you estimate the false-positive rate of a fraud model when you only see labels for orders you blocked? [Likely] (selective labels in fraud and trust teams)
116. Two models have AUCs of 0.812 and 0.815 on the same test set. Is the difference meaningful? [Likely]
    - How would you get a confidence interval for the AUC difference (DeLong or bootstrap)?
117. How would you design a survey or human-labelling study to estimate the defect rate of search results, with a confidence interval? [Likely]
    - How many labelled samples do you need, and how do you handle disagreement between labelers?
118. Evaluate a fraud model on heavily imbalanced data with precision, recall and AUC. [Reported] [22] (statistics-heavy team phone screen)
    - Why can ROC AUC look good when the model is useless in practice? When do you prefer PR AUC? [Likely]
    - Fraud prevalence doubles next quarter. What happens to precision at a fixed threshold? [Likely]
119. Predict the click probability for music recommendations. [Reported] [24] (phone screen; aggregator sighting September 2026; the statistics follow-ups below are [Likely])
    - How do you check whether the predicted probabilities are calibrated (reliability diagram, expected calibration error)?
    - You downsampled negatives for training. How do you correct the predicted probabilities?
120. Anomaly detection, reported as a phone-screen topic on an Ads team [23]: how would you detect anomalies in a stream of ad clicks, such as click fraud or a broken tracking pixel? [Reported] [23] (the scenario is [Likely])
    - How do you set the alert threshold to control the false-alarm rate when you have very few labelled anomalies?
121. Interpret a logistic-regression coefficient as an odds ratio, and give its confidence interval. [Likely] (logistic regression and its loss are reported in breadth and screens [22])
    - How do collinearity and separation change what the coefficients mean?

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09. Leadership Principles Bank: Behavioral Questions for the L5 Applied Scientist Loop, with L5 Probes and Bar Raiser Favorites

> Scope: behavioral (Leadership Principle, "LP") questions as asked of Amazon Applied Scientists, grouped by all 16 LPs, plus the follow-up probes used to test the L5 (Applied Scientist II) bar, and the bar raiser's favorite questions and pressure tactics. Questions only, no model answers.
>
> Research note (read this first): this chapter was built under tight limits. The session's web-search budget had already been used by earlier chapters, and this environment's egress policy blocked direct fetches of Glassdoor, Blind, LeetCode, Medium, IGotAnOffer, interviewing.io and amazon.jobs. Two kinds of evidence were used: (a) Sourced claims and reported questions already recorded, with citations, in chapters 01-08 of this bank. Those were gathered from search-engine excerpts of the cited pages and are carried over here with the same sources. (b) Pages read in full for this chapter: GitHub-hosted material, including a candidate's notes that repost 1point3acres Amazon AS interview reports [21][22] and a curated Amazon AI-role question guide [20]. [Reported] means a source attributes the question, or a close paraphrase, to Amazon interviews (AS where possible). [Likely] means an inference from the LP text, the AS role and Amazon's products. It has not been seen verbatim. Treat vendor guides as weaker than first-person reports.

Format notes for the behavioral portion

- **No single "LP round"; LPs are in every round.** Amazon's AS prep page says interviewers are assigned competencies "based off of our Leadership Principles" and each typically asks 2-3 behavioral questions [1]. Candidates report 1-2 LP questions opening or closing every technical round. Estimates of LP time per round range from 5-15 minutes [15] to "at least 30 minutes" of each hour-long interview [21]. One L5 candidate put behavioral content at over 60% of the loop [23].
- **The bar raiser (BR) round is mostly or entirely LP.** One 1point3acres AS report said the BR round was "pure LPs" [21]. The BR comes from outside the hiring org, has been trained and shadowed into the role, chairs the debrief, and holds a de facto veto [3][19]. Glassdoor AS and AS II reports describe the BR round as LP-heavy [4][5]. AS candidates have reported BRs who were non-science, including a Principal TPM who also asked about ETAs, dependencies and a few stats checks [7].
- **Assigned LPs.** Each interviewer usually owns 2-3 LPs, so the loop covers the set without overlap [1][16]. Some interviewers name the LP before asking [21]. Candidates recommend asking the recruiter in the prep call which LPs each

interviewer is assigned [21]. A candidate's compiled notes say an Amazon recruiter advised against reusing one story with more than two interviewers [22].

- **Fresh-PhD adaptation.** Interviewers prompt candidates without industry projects to draw on research experience. Reported examples: "what did you do when you didn't like a project your advisor assigned?" and "have you worked in a research area you weren't familiar with?" [21]
- **What the L5 bar looks like in LP answers.** L5 means owning a project end to end, making the technical calls, and influencing peers, with some spill-over to sister teams [8]. A vendor debrief says L5 stories face 2-3 layers of follow-up, against about 1 layer at L4 [15]. Blind commenters say LP narratives, not coding, are what separate L5 from L6, and that L6 needs multiple strong examples and influence beyond the team [8][9]. A BR can pull the level down even when the hiring manager is positive. One AS candidate's BR thought they were "at the cusp of L4 and L5" [6].
- **Grading habits reported for BRs.** BRs push on metrics ("how was that measured?"), on individual contribution ("I", not "we"), and sometimes say outright that your approach was wrong to see how you hold your position [17][18][20]. Weak or rehearsed stories "collapse" under several "why" layers [20].
- **The 16 LPs** are published on amazon.jobs [2]. The last two, Strive to be Earth's Best Employer and Success and Scale Bring Broad Responsibility, were added in 2021 and are asked less often in reports, but they are in scope.

Questions

Tag legend: [Reported] = a source attributes it to Amazon interviews (cite). [Likely] = a grounded inference. Indented bullets are follow-up probes, phrased as interviewers ask them.

1. Customer Obsession

1. Tell me about a time you started from the customer and worked backwards to define the science problem. [Likely]
    - Who exactly was the customer: the end shopper, an internal team, or a seller?
    - How did you get the customer signal: data, anecdotes, or a PR/FAQ?
    - What metric did you choose as the customer proxy, and where did it diverge from what customers actually felt?
2. Tell me about a time you used data to anticipate a customer need nobody had asked for. [Likely]
    - How did you validate that the need was real before you built it?
3. Tell me about a time you traded scientific elegance or novelty for customer impact, or refused to. [Likely]
    - Looking back, was that the right call for the customer?
4. Tell me about a time a model's offline metric improved but the customer experience got worse. [Likely]
    - How did you find out, and what did you change in how you evaluate?
5. Tell me about a time you dealt with a difficult or demanding internal customer (a product, ops or business team consuming your model). [Reported] [22]
    - What did they want that you wouldn't give them, and why?

6. Tell me about a time you had to say no to a stakeholder request because it would hurt end customers. [Likely]
    - Did the stakeholder agree in the end? If not, what happened?

2. Ownership

7. Tell me about a time you went above and beyond your job responsibilities. [Reported] [21]
    - Why was it your job to do it, and who else could have?
8. Tell me about a time you went well beyond the scope of a project and still delivered. [Reported] [21]
    - What did you stop doing to make room for it?
9. Tell me about a time you took on something significant outside your area of responsibility. [Reported] [22]
    - How did the owning team react?
10. Tell me about a time you worked on a project with unclear responsibilities. [Reported] [22]
    - How did you decide what you would own?
11. Tell me about a time a model you shipped failed or underperformed in production. What did you do? [Reported] [12] (model underperformance reported in BR-style questioning; production framing [Likely])
    - Who found it first, you or someone else?
    - What did you put in place so it couldn't happen silently again?
12. Tell me about a time you sacrificed a short-term result for long-term value, for example paying down technical or science debt instead of shipping another model. [Likely]
    - How did you justify it to your manager?
13. Tell me about a time you took initiative without waiting for permission. [Likely] (chapter 01 notes a rejection reportedly explained as "seemed to wait for permission")
    - What would have happened if you had waited?
14. Tell me about something you own today that will keep working after you leave the team. [Likely]
    - What mechanism, such as a runbook, monitor or review, makes it independent of you?

3. Invent and Simplify

15. Tell me about a time you gave a simple solution to a complex problem, or found a new way to do something. [Reported] [22]
    - What did the complex alternative look like, and why did you reject it?
16. Tell me about the most innovative project you have worked on. [Reported] [20] (Exponent-style "invented a solution to a problem that didn't have an obvious answer"; quoted in a prep source, so weak)
    - What, specifically, was your invention versus prior art?
17. Tell me about a time you simplified a complex ML approach into something that could ship. [Likely]
    - What did you give up in accuracy, and how did you decide it was worth it?

18. Tell me about a time you started from a heuristic or simple baseline and it beat the sophisticated model. [Likely]
19. How do you solve a problem when the capability you need isn't in your programming tools or libraries? [Reported] [21] (AS NLP phone screen)
    - Give me a specific time you built that missing piece yourself.
20. Tell me about a time you reused an idea from outside your field or from another team rather than inventing from scratch. [Likely]
    - How did you know it would transfer?

4. Are Right, A Lot

21. Tell me about a time you made a critical scientific or technical decision without full data. [Reported] [14]
    - What did you learn, and how did you pivot? [Reported] [14]
22. Tell me about a time you realized your initial modeling hypothesis was wrong. [Likely]
    - What evidence made you change your mind, and how fast did you act on it?
23. Tell me about a time you were wrong about a technical call. How did you find out? [Likely]
24. Tell me about a time your judgment went against the data, or the data went against your intuition. Which did you trust? [Likely]
25. Tell me about a time you sought out a view that disagreed with yours before making a decision. [Likely]
    - Whose view did you seek, and did it change the decision?
26. Tell me about a time you had to choose between two model architectures or approaches with no clear winner. [Likely]
    - What was the decision rule, and in hindsight was it right?

5. Learn and Be Curious

27. Tell me about a time you learned a new area quickly to solve a problem. [Likely] (fresh-PhD variant reported: "worked in a research area you weren't familiar with" [21])
    - How did you know you had learned enough to make decisions?
28. Tell me about a time you worked in a research or technical area you were not familiar with. [Reported] [21]
29. What is something you taught yourself in the last year (for example LLM fine-tuning, RL from human feedback, or a new framework), and how have you applied it? [Likely]
    - Where did you get stuck?
30. Tell me about a time curiosity about an anomaly in the data led to a bigger finding. [Likely]
31. How do you stay current with the literature, and tell me about a paper that changed how you work. [Likely]

6. Hire and Develop the Best

32. Tell me about a time you mentored a junior scientist, engineer or intern. [Likely]
    - How did you measure their growth?
    - What did you do when they struggled?

33. Tell me about a time you raised the science bar on your team through reviews, standards or reproducibility. [Likely]
34. Tell me about a time you gave someone difficult feedback. [Likely]
35. Tell me about a time you were involved in hiring or interviewing. What bar did you hold? [Likely]
    - Did you ever vote no on someone others liked?
36. Tell me about a time you helped a teammate move to a role or team where they could grow, even though your team lost them. [Likely]

7. Insist on the Highest Standards

37. Tell me about a time you raised the quality bar on a deliverable even though it delayed you. [Likely]
    - Who disagreed with the delay, and how did you handle it?
38. Tell me about a time you had to deliver a model under a tight deadline. What corners did you refuse to cut? [Likely]
39. Tell me about a time you found a bug or leak in someone else's experiment, such as data leakage or a bad A/B setup. [Likely]
    - How did you tell them?
40. Tell me about a time you were not satisfied with the quality of your own shipped work. What did you do about it? [Likely]
41. Tell me about a time insisting on high standards turned out to be the wrong call. [Likely]
    - How do you now decide when "good enough" is good enough?

8. Think Big

42. Tell me about the most ambitious science bet you have proposed. [Likely]
    - How did you sell it, and how far did it get?
43. Tell me about a time you turned a one-off model into a platform or capability other teams use. [Likely]
44. Where do you think your field will be in 3-5 years, and what would you build at Amazon toward that? [Likely] (prep sources describe a BR "bold vision" prompt; not seen verbatim in first-person reports)
45. Tell me about a time you challenged a team to aim for a much bigger goal than planned. [Likely]
    - What was the inspiring direction, in one sentence?
46. Tell me about a time your vision for a project was larger than your manager's. What happened? [Likely]

9. Bias for Action

47. Tell me about a time you had to make a decision with incomplete information. [Reported] [14] [20]
    - Was it a one-way or a two-way door?
    - What was your backup plan if you were wrong?
48. Tell me about a time you had to take a risk, or had very little time to decide. [Reported] [22]
49. Tell me about a time you took a calculated risk and it failed. [Reported] [22]

    - What did you learn, and would you take the same risk again?
50. Tell me about a time you did something without asking for your manager's approval. [Reported] [22]
    - How did your manager react afterwards?
51. Tell me about a time you shipped a quick baseline or experiment to unblock a decision instead of waiting for the full model. [Likely]
52. Tell me about a time you moved too fast. What broke? [Likely]

10. Frugality

53. Tell me about a time you did more with less: compute, labels or headcount. [Likely]
    - What specific constraint forced you to be creative?
54. Tell me about a time you reduced the training or inference cost of a model significantly. [Likely]
    - What did it cost in quality?
55. Tell me about a time you dealt with limited or no labeled data. [Likely]
    - Why not just buy more annotation?
56. Tell me about a time you chose not to use a large or expensive model (for example an LLM) because a cheaper one was sufficient. [Likely]

11. Earn Trust

57. Tell me about a time you failed, or the biggest mistake you made. [Reported] [20][22]
    - What signal existed before the failure that you missed? [Reported] [20]
    - How did you tell your customers or leadership? [Reported] [20]
58. Tell me about a time you received negative feedback. [Reported] [22]
    - What did you change, and how would the person who gave it know?
59. Tell me about a time you earned the trust of an engineering or product partner who doubted your science. [Likely]
60. Tell me about a time you had to deliver bad news, such as a model not working or a launch slipping. [Likely]
    - When did you know, and when did you tell them?
61. Tell me about a time you explained a complex model trade-off to a non-scientist stakeholder. [Likely]
62. Tell me about a time you were vocally self-critical about your own team's work. [Likely]

12. Dive Deep

63. Tell me about a time you had to dive deep into data or code to find a root cause. [Reported] [15] (asked opening or closing an AS coding round)
    - What was the first hypothesis, and how did you rule it out?
    - At what level of detail did you personally look at the raw data?
64. Tell me about a time your model underperformed and you had to diagnose the issue. [Reported] [12]
65. Tell me about a time a metric and anecdotes disagreed. Which did you believe? [Likely] (taken from the LP text)
66. Tell me about a time you found a data quality or pipeline problem that was silently degrading a model. [Likely]

    - How long had it been broken, and why didn't monitoring catch it?
67. Tell me about a time an A/B test result looked too good to be true. [Likely]
68. Walk me through one number in your resume. How exactly was it computed? [Likely] (BR "how was that measured?" probe reported [17][18])

13. Have Backbone; Disagree and Commit

69. Tell me about a time you had a conflict with a coworker or manager, and how you approached it. [Reported] [10][17][18]
    - What did you do once the decision went against you?
70. Tell me about a time you had a major conflict with a Principal Engineer (or senior scientist). How did you resolve it? [Reported] [13]
71. Tell me about a time you disagreed with your team's technical direction. What did you do? [Reported] [20]
    - What would you have done if you had been overruled and then proven right? [Reported] [20]
    - Tell me about a time you were the one who was wrong. [Reported] [20]
72. Tell me about a time you disagreed with a teammate or manager. [Reported] [22]
73. Tell me about a time you pushed back on a stakeholder's metric or launch decision using data. [Likely]
    - Did you commit after being overruled? How would we know you really committed?
74. Tell me about a time you disagreed with a senior scientist's technical approach. [Likely]
75. Tell me about a time you committed to a decision you disagreed with. How did it turn out? [Likely]
76. Tell me about a time you stopped, or tried to stop, a model launch. [Likely]

14. Deliver Results

77. Tell me about a time you had a tight deadline. [Reported] [21] (AWS AS phone screen)
    - If you did it again, what would you do better? [Reported] [21]
78. Tell me about a time you missed a deadline. [Reported] [22]
    - When did you realize it would slip, and who did you tell?
79. Tell me about your most challenging or proudest project, or a time you solved a complex problem. [Reported] [22]
    - What was the business result, in numbers?
80. Tell me about a time you delivered a model under a hard deadline. What did you descope? [Likely]
    - What did that descoping cost?
81. Tell me about a project failure and how you managed the aftermath. [Reported] [13]
    - What would you do differently? [Reported] [11]
82. Tell me about a time you hit a major setback mid-project, such as data being unavailable or a dependency slipping, and still shipped. [Likely]
83. Tell me about a time you had to choose between shipping something imperfect on time and slipping for quality. [Likely]

15. Strive to be Earth's Best Employer

84. Tell me about a time you made your team's work environment better. [Likely]
85. Tell me about a time you noticed a teammate struggling (workload, morale, inclusion) and acted. [Likely]
86. A senior manager assigned you work when you already had a full load. How did you handle it? [Likely] (seen in consolidated MLE behavioral banks; not verified as Amazon-specific)
87. Tell me about a time you made a process or on-call rotation less painful for others. [Likely]

16. Success and Scale Bring Broad Responsibility

88. Tell me about a time you considered the downstream harm or fairness impact of a model. [Likely]
    - What did you measure, and what did you change?
89. Tell me about a time you identified a secondary effect of your model on sellers, customers or another team that others had missed. [Likely]
90. Tell me about a time you handled privacy, safety or responsible-AI constraints on a model, such as hallucination in a customer-facing LLM feature. [Likely]
91. Tell me about a time you decided not to launch because the risk to customers outweighed the gain. [Likely]

Bar raiser favorites and pressure probes

These are the questions and tactics most often attributed to the BR seat. The BR may come from any org, so expect plain-language questions, not science ones.

92. Tell me about your most significant failure. What happened, and what did you change afterwards? [Reported] [20] (described as "a Bar Raiser staple")
93. Tell me about a time you had a conflict with a coworker or manager (Have Backbone; Disagree and Commit). [Reported] [10][17][18]
94. (Pushback) "I think your approach was wrong. Convince me otherwise." [Reported] [17][18]
    - What data would make you concede?
95. How was that measured? [Reported] [17][18]
96. What was your exact individual contribution? Say it without "we." [Reported] [17][18]
97. (TPM or non-science BR) How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [7]
    - When did a science timeline slip, and how did you tell stakeholders?
98. (TPM BR) Interpret this p-value or confidence interval for me. [Reported] [7]
99. Give me a second, different example where you showed the same principle. [Likely] (L6 rejections often cite a lack of multiple strong examples [8][9])
100. Why are you leaving your current role, and why Amazon now? [Likely]
101. What is the hardest feedback you have received from a manager, and what did you do with it? [Likely]
102. Tell me about a time you were not proud of how you behaved in a disagreement. [Likely]

L5 calibration probes (any interviewer, layered on any story)

These follow-ups test whether a story is L5 scope rather than L4 execution. Expect 2-3 layers per story [15].

103. What was the largest scope you have owned end to end, from problem framing to production? [Likely]
    - Who else depended on it, beyond your own team?
    - What were you accountable for that your manager was not?
104. Tell me about a time your work changed a sister team's roadmap or technical direction. [Likely]
105. Tell me about a time you set the technical direction for other scientists or engineers, not just carried it out. [Likely]
    - How did you get buy-in without authority?
106. Where did you make a judgment call that your manager did not make for you? [Likely]
107. What mechanism did you build (a review, monitor or tool) that kept working without you in the room? [Likely]
108. If I called your skip-level manager, how would they describe your contribution to this project? [Likely]
109. What were the alternatives you rejected, and what data did you use to reject them? [Likely]
110. What would you do differently if you ran this project again today? [Reported] [11][21]

Generic deepening probes interviewers attach to any LP story

These are not standalone questions. Expect a few of them after each story. [Likely, except where cited]

- What was the situation in one sentence, and what was your specific task?
- What did you personally do, as opposed to your team? [Reported] [17][18]
- What was the measurable result, and how was it measured? [Reported] [17][18]
- What was the hardest moment, and what options did you consider?
- Who disagreed with you, and what was their best argument?
- What did you learn, and where have you applied it since? [Reported] [14]
- How long ago was this? Do you have a more recent example?
- What would have happened if you had not acted?

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10. Bar Raiser Round: Reported Questions, Mixed Technical and Behavioral Probing, and Follow-up Patterns

> Scope: the one interview in the L5 Applied Scientist (Applied Scientist II) onsite run by the Bar Raiser (BR). It covers who the BR is, what they assess at L5, the questions candidates report, the questions the format makes likely, and the follow-up patterns BRs use to test whether a story holds up.
>
> Research note (please read): in this review pass, the web-search budget was already used up, and page fetches to the cited sites (4dayweek.io, igotanoffer.com, interviewkickstart.com, dataford.io, barraiserai.com, amazon.jobs) failed with DNS errors or proxy refusals. The one source family that could be read in full was the GitHub question-bank aggregator [28]-[32]. It was cloned and read directly; it says its content is based on candidate reports, mostly from Q1-Q3 2026. Every question tagged [Reported] that cites [28]-[32] was checked against that text. References [1]-[27] carry over from chapters 01-08 of this guide and were **not re-fetched in this pass**. Their [Reported] tags rest on search-result extracts gathered earlier, so treat them as unverified until someone re-opens the pages.

Format notes

Aspect	What is known	Sources
Length and slot	One of the onsite loop interviews. Amazon's official AS page describes the loop as four 55-minute interviews with members of the science community. Candidates more often report 5 rounds of 55-60 minutes, one of them labeled "Bar Raiser". The BR round is sometimes held on a separate day, from the day before the loop to weeks after it.	[1][8] [9] [13] [28]
Who the BR is	A trained interviewer from outside the hiring team. BRs are nominated, trained, and shadow veteran BRs before they run debriefs. Science candidates report BRs from non-tech orgs, and one AS II candidate was told the final interview would be a Principal TPM.	[2][3] [5][6] [10] [29]
Power	The BR chairs the debrief and, with the hiring manager, makes the hire/no-hire call. The BR holds an effective veto that former Amazonians say is rarely used explicitly.	[2][3] [7] [12] [29]
What is assessed	For scientists the round is usually entirely behavioral: 3-4 LP questions, each followed many layers down. Customer Obsession, Learn and Be Curious and Insist on the Highest Standards are the LPs named for science BRs. In one L6 loop the BR also asked for coding (a priority queue from scratch). The "better than 50% of current employees at that level" line is BR folklore, not a published Amazon standard.	[4][5] [6] [28] [29] [30]
ML content	Science loops assign each round 2-3 LPs; one five-round loop covered ten different LPs. Interviewers expect stories to contain an ML decision about models, data or experiments, and	[28] [29]

Aspect	What is known	Sources
in stories	one team required ML work in every story. "A disagreement about an evaluation metric beats one about meeting schedules."	[30]
L5-specific bar	L5 means owning a project end to end, making the technical calls, and influencing peers and sometimes sister teams. A vendor-published debrief says L5 LP answers face 2-3 layers of follow-up, against 1 layer at L4. Blind commenters say LP narratives, not coding, separate L5 from L6. Scope is judged against the target level, so solid work can still be marked down for insufficient impact.	[14] [15] [18] [29] [30]
Leveling	A BR can push the level down even when the HM is positive. One L5 candidate's BR saw them "at the cusp of L4 and L5". Downleveling from L6 to L5 and from L5 to L4 is commonly reported.	[11] [15]
Behavioral weight overall	Amazon says each interviewer typically asks 2-3 behavioral questions. One candidate counted 17 LP questions in a four-hour loop. In one five-round science loop, LP questions left only about 15 minutes for coding in the fourth hour.	[1] [16] [18] [28] [29]
Split decisions and cooldowns	When the debrief splits, candidates report an extra round, for example one more coding round for an AGI-team AS II. Recent science and ML rejections carried cooldowns of 6-12 months.	[17] [29] [30]

Observed probing patterns (what makes the BR round different)

- **Layered "why" and "how do you know":** 3-4 layers on one story: why that path, which alternatives, what data, what you would change. [18][19][21][22][28]
- **"I" not "we":** BRs separate your individual contribution from the team's, down to dates, data and your exact role. [21][22][28][30]
- **Deliberate pushback:** candidates report BRs saying the approach was wrong, or repeatedly saying "that is not what I'm looking for", to see whether the candidate defends with data or switches to a better story. [21][22][29][30]
- **After-commit focus:** on disagree-and-commit stories, the grade turns on what you did after committing, for example quietly reducing the risk you had raised. [28][29] [31]
- **Metric interrogation:** "How was that measured?", including baselines, attribution and counterfactuals. [21][22]
- **Lay-audience translation:** a non-science BR (for example a Principal TPM, or a backend engineer BR in another track) asks for project-management mechanics, basic stats interpretation, or technical trade-offs in plain language. [10][32]
- **Closing sweep:** some BRs end with an open invitation to share anything earlier interviewers missed. [32]

Questions

Tag legend:

- **[Reported]**: a candidate or source attributes this question, or a close paraphrase, to an Amazon BR round or to Amazon AS/science LP questioning. When the report

comes from another round or track, the item says so in parentheses.
- **[Likely]**: an inference grounded in the BR format, the 16 LPs, the L5 bar and Amazon's science products. It has not been seen verbatim.

Indented bullets are follow-up probes. In a BR round, expect at least two of them per story.

A. Opening and scope calibration (sets the level the BR will argue in the debrief)

1. Walk me through your career and the one project that best shows the level you are operating at today. [Likely]
    - What part of that project would not have happened without you?
    - Who else could have done what you did on that project?
    - If I called your manager, what would they say was your scope?
2. Tell me about the largest-scope science project you have owned end to end, from problem framing to production. [Likely]
    - Where did the problem come from: did you find it or were you handed it?
    - What was the decision you made that most changed the outcome?
    - How many people or teams depended on your output?
3. Describe your exact individual contribution to that project, without saying "we". [Reported] [21][22][28]
    - Which design document or analysis did you write yourself?
    - What did you delegate, and why?
    - That sounds like a team result. What specifically did you do? [Reported] [21][22]
4. Tell me about a time your work influenced a team other than your own. [Likely]
    - How did they adopt it: mechanism, code, paper or persuasion?
    - What would have happened if they had said no?
5. What is the hardest technical problem you have solved in the last two years? [Likely]
    - Why was it hard for you specifically, and not just for the field?
    - Explain it so someone outside ML understands why it mattered.
6. Give me your flagship project in two minutes. Now take ten minutes and go deeper on the part you are proudest of. [Likely] [29]
    - What did you leave out of the two-minute version that a skeptic would ask about?

B. Conflict, backbone and disagree-and-commit

7. Tell me about a time you had a conflict with a coworker or manager, and how you handled it. [Reported] [19][21][22][29][30]
    - What did you do once the decision went against you?
    - Looking back, who was right?
    - How is your relationship with that person now?
8. Tell me about a time you had a major conflict with a Principal Engineer (or a more senior scientist). How did you resolve it? [Reported] [24]
    - What data did you bring to the conversation?
    - Did you escalate? Why or why not?
9. Tell me about a time a client or decision maker proposed a plan that was far too aggressive. What did you do? [Reported] [29][30]

    - What evidence showed you the plan was unrealistic?
    - What did you counter-propose, and what did they accept?
10. Tell me about a time you disagreed with a product manager's launch decision based on your model's results. [Likely]
    - What metric were they looking at that you thought was wrong?
    - What happened after launch, and were you proven right?
    - What would have convinced you to commit to their plan?
11. Tell me about a time you disagreed with a manager or challenged a team decision, committed anyway, and what happened next. [Reported] [28][30][31]
    - After you committed, how did you reduce the specific risk you had raised? [Reported] [28][31]
    - How did you make sure your doubts did not slow the team down?
    - Did you ever revisit the decision? What triggered that?
12. Tell me about a disagreement with a colleague over which evaluation metric or baseline to use. [Likely] [29]
    - How did you settle it: argument, experiment or escalation?
    - Which metric does the team use today?
13. Tell me about a time an engineer said your model could not ship in its current form. [Likely]
    - What did you give up to get it into production?
    - Who made the final call?
14. Tell me about a time you had to tell a senior leader that their idea would not work scientifically. [Likely]
    - How did you frame it?
    - What did they do next?
15. Describe a time you saw your team heading in the wrong direction. What did you do? [Reported] [32] (front-end BR round, same format)
    - How did you know before the results showed it?
    - Whose decision was it to change course?
16. Tell me about a time you raised a risk, were overruled, and the risk then happened. [Likely]
    - What did you say afterwards, and to whom?

C. Failure, being wrong and learning

17. Tell me about a project failure and how you managed the aftermath. [Reported] [24]
    - What would you do differently? [Reported] [20][28]
    - What mechanism did you put in place so it does not happen again?
18. Tell me about a time your model underperformed in production and you had to diagnose the issue. [Reported] [23][30]
    - What did you learn, and how did you pivot? [Reported] [25]
    - How long did it take you to notice, and why that long?
19. Tell me about a time you were wrong about a technical call. How did you find out? [Likely]
    - Who told you, or did you find it yourself?
    - What did it cost the team?
    - Did you have to admit it in front of the team or stakeholders?

20. Tell me about a time you got critical feedback that surprised you. [Reported] [28] (listed as a recurring BR theme)
    - What did you change afterwards, and how would someone see that change?
21. What do you most need to improve, and what have you changed so far? [Reported] [31] (hiring-manager round)
    - What evidence would your manager point to that you have improved?
22. Tell me about an experiment or research direction you killed. [Likely]
    - How did you decide it was dead rather than under-tuned?
    - How much effort had gone in, and how did you tell the team?
23. Describe a deadline you were going to miss: when you realized it, who you told, and what you proposed. [Reported] [28]
    - Why didn't you see it earlier?
24. Tell me about a time AI gave you a wrong answer. How did you catch it? [Reported] [28][31]
    - What do you now check by hand that you used to trust?
    - How did you realize the output of a generative AI tool was not good enough, and what did you do? [Reported] [31] (hiring-manager round)
25. How would you test code or analysis that an AI assistant generated for one of your experiments? [Reported] [31] (hiring-manager round; adapted from "AI-generated code")
    - What failure would slip past your unit tests but break the experiment?

D. Decisions under ambiguity (Bias for Action, Are Right a Lot)

26. Describe a decision you made with incomplete information and how you later validated it. [Reported] [25][29]
    - What data would you have wanted, and why couldn't you wait for it?
    - How did you limit the downside if you were wrong?
    - Was it a one-way or two-way door? What would you have done if it had turned out wrong?
27. Describe a calculated risk you took: what you knew at the time and how it turned out. [Reported] [28][29][30]
    - What was the worst case you planned for?
    - Who else knew about the risk before you took it?
28. Tell me about a time you took initiative without waiting for permission. [Likely]
    - Whose permission would you normally have needed?
    - What did your manager say afterwards?
29. Tell me about a time you had to choose between a quick heuristic and a principled model. [Likely]
    - What did you ship first, and did the principled model ever replace it?
30. Tell me about a time two offline metrics disagreed and you had to make a ship call. [Likely]
    - Which metric did you trust, and why?
31. Tell me about a time the online A/B result fell short of, or contradicted, your offline gain. How did you find out why? [Reported] [29][30] (science-depth and project-dive rounds)
    - What was the root cause?
    - Did you change your offline evaluation afterwards?

32. Tell me about a time you settled a technical debate with a quick experiment instead of more argument. [Likely]
    - How did you make sure both sides would accept the result?

E. Ownership and Deliver Results (the core L5 signal)

33. Tell me about a time you took ownership of something outside your formal scope. [Reported] [28] (listed as a recurring BR theme)
    - Why didn't the actual owner handle it?
    - What did you stop doing to make room?
34. Tell me about a time you delivered a model under a hard deadline. [Likely]
    - What did you descope, and what did that cost?
    - What corners did you refuse to cut?
35. Tell me about the most significant business result your science has driven. [Likely]
    - How was that measured? [Reported] [21][22]
    - How do you know it was the model and not seasonality, a UI change or a marketing push? What was the counterfactual?
    - What was the baseline?
36. Tell me about a time a project was about to fail and you turned it around. [Likely]
    - What was the first thing you did differently?
37. Tell me about a time you had to keep a model working long after launch (monitoring, retraining, on-call). [Likely]
    - Who owned it when it broke at 2am?
38. Tell me about a time you were blocked by another team. [Likely]
    - What did you do in the first week?
    - Did you find a way around them, or did you wait?
39. Tell me about a time two partner teams wanted your model to optimize different metrics. [Likely]
    - Whose metric won, and how was that decided?

F. Customer Obsession and Working Backwards, for a scientist

40. Tell me about a time you went above and beyond for a customer. [Reported] [29][30]
    - Was the customer internal or external?
    - What did going beyond cost you or the team?
41. Tell me about a time you started from the customer and worked backwards to the science problem. [Likely]
    - Who was the customer, and how did you get their signal?
42. Tell me about a time you chose a simpler model because it was better for the customer. [Likely]
    - What accuracy did you give up, and how did you justify it?
43. Tell me about a time a model you built hurt some customers even though the average metric improved. [Likely]
    - How did you find that segment?
    - What did you do about it?
44. Tell me about a time you used customer anecdotes to overturn what the data said. [Likely]
    - How many anecdotes was enough?

45. If you wrote a press release for your best project, what would the headline and customer quote be? [Likely]
    - What would the FAQ's hardest question be?

G. Dive Deep and scientific rigor (the technical side of the BR round)

46. Describe a time you found an issue nobody else had noticed: how you isolated it, and why your first hypothesis was wrong. [Reported] [18][27][31][32]
    - What was the first anomaly you noticed?
    - How did you rule out the other explanations?
47. Tell me about a time you found a data-quality or label-leakage problem in your own work. [Likely]
    - How long had it been in production?
    - Which numbers did you have to retract?
48. Walk through a major technical decision from your resume, the alternatives you rejected, and whether you would make the same choice again. [Reported] [28][31]
    - What experiment told you the alternative was worse?
    - Who did you have to convince?
49. For your main project or publication, what was your specific contribution, and how did you choose the baselines? [Reported] [29] (science-depth round)
    - Which strong baseline did you leave out, and why?
50. Walk me through how you validated that your evaluation set reflected production traffic. [Likely]
    - What distribution shift did you see after launch?
51. Tell me about a time a result looked too good to be true. [Likely]
    - What did you check first?
52. Tell me about a time you had to defend a statistical result to a skeptical audience. [Likely]
    - What was the effect size, and was it practically significant?
53. You mentioned an improvement of X%. What was the confidence interval? [Likely]
    - How many samples, and how long did the test run?
54. Tell me about a time you reproduced a paper's results and could not match them. [Likely]
    - What did you conclude, and what did you ship instead?
55. What is the biggest weakness of your flagship model that I have not asked about yet? [Likely] [29]
    - What would it take to fix it?

H. Technical checks from a non-science Bar Raiser (TPM, SDE or business BR)

56. How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [10]
    - When did a science timeline slip, and how did you tell stakeholders?
    - How do you estimate a project when you do not know if the idea will work?
57. Interpret a p-value or a confidence interval for me. [Reported] [10]
    - What would you tell a business leader who reads p = 0.06 as "no effect"?

58. Explain your most recent model to me as if I were a business leader with no ML background. [Likely]
    - What decision does it let someone make that they could not make before?
59. Tell me about a technique you learned on your own and applied. How did it hold up across the conditions it had to handle? [Likely] [32] (asked in a front-end BR about a front-end technology across devices; adapted here to ML)
    - Explain the trade-off to me as a non-specialist.
60. How do you decide when a research prototype is ready to hand to engineering? [Likely]
    - What does the handoff document contain?
61. Your science project needs six months but the business wants results in six weeks. What do you do? [Likely]
    - What would a six-week version look like?
62. How do you know a model is still working a year after launch? [Likely]
    - Who gets paged, and what is the rollback plan?
63. Tell me about a time you had to explain a model's mistake to a customer-facing or operations team. [Likely]

I. Invent and Simplify, Think Big, Frugality, Learn and Be Curious

64. Tell me about a time you simplified a complex scientific approach so it could ship. [Likely]
    - What did the complex version get right that the simple one missed?
65. Tell me about the most ambitious science bet you have proposed. [Likely]
    - Who funded it, and what did you have to prove first?
    - If it had been fully funded, what would the three-year version be?
66. Tell me about a time you did more with less: compute, labels or headcount. [Likely]
    - What was the cost of the cheaper approach in model quality?
67. Tell me about something you invented, such as a method, tool or mechanism, that others now use. [Likely]
    - How did you get people to adopt it?
68. Tell me about a time you used an LLM or foundation model where a custom model would have been the default. [Likely]
    - How did you weigh inference cost against development time?
69. Tell me about a time curiosity about an odd pattern in data led you to a new direction for the project. [Likely] [29] (Learn and Be Curious, named for science BRs)
    - How much time did you spend before you told anyone?
70. Tell me about a time you learned a new area fast to unblock a project. [Likely]
    - How did you know you had learned enough?

J. Earn Trust, Hire and Develop the Best, influence without authority

71. Tell me about a time you earned the trust of an engineering or product partner who doubted your science. [Likely]
    - What was the turning point?
72. Tell me about a time a teammate's work did not meet the bar, and what you did. [Reported] [29]
    - Did you fix it yourself or help them fix it? Why?

    - How did they react, and what changed afterwards?
73. Tell me about a time you mentored someone or raised the bar of your team's science. [Likely]
    - How do you know they improved?
74. Tell me about a time you influenced a decision without having authority over the people involved. [Likely]
    - Who was the hardest person to convince?
75. Tell me about a time you reviewed someone else's experiment and found a serious flaw. [Likely]
    - How did you raise it?

K. Insist on the Highest Standards, Broad Responsibility

76. Tell me about a time you raised the quality bar on a deliverable even though it delayed you. [Likely] [29]
    - Who disagreed with the delay?
77. Tell me about a time you refused to launch something that was not ready. [Likely]
    - What was the minimum bar, and who set it?
78. Tell me about a time you insisted on better labels or data quality when the team wanted to train on what they had. [Likely] [29]
    - How did you show the cost of the noisy data?
79. Tell me about a time you considered the downstream harm or fairness impact of a model. [Likely]
    - How did you measure it?
    - What trade-off did you accept?
80. Tell me about a time you handled sensitive customer data in a modeling project. [Likely]
    - What would you not do even if it improved the model?

L. Deliberate pushback and stress probes (BR-specific)

81. I think your approach was wrong. Convince me otherwise. [Reported] [21][22]
    - What data would make you concede?
    - What would the alternative have cost?
82. That is not what I'm looking for. Give me a different example for the same principle. [Reported] [29][30]
83. That example is from three years ago. Do you have a more recent one? [Likely]
84. Your manager made that call. Where was your judgment in that? [Likely]
85. If you had not been on the project, what would have been different? [Likely]
86. What was the worst decision you made on that project? [Likely]
87. What would your harshest critic on that team say about how you handled it? [Likely]
88. You said the result was a success. Who would disagree? [Likely]
89. Stop the story there. What exactly did you say in that meeting? [Likely]

M. Leveling probes: L4 vs L5 vs L6

90. Tell me about a time you set the technical direction for a project, not just executed it. [Likely]
    - Who was in the room when the direction was decided?

91. Tell me about a time you identified a science problem nobody had asked you to solve. [Likely]
    - How did you get time and resources for it?
92. Tell me about a mechanism (process, tool, review) you created that outlived your involvement. [Likely]
    - Who runs it now?
93. Tell me about a time you had to coordinate science work across several people or teams. [Likely]
    - How did you divide the work, and what broke?
94. What is the biggest ambiguity you have resolved, from vague goal to concrete plan? [Likely]
    - What was the first artifact you produced?
95. How would your current manager describe the gap between you and the next level? [Likely]
    - What are you doing about it?
96. What was the largest measured business effect of your work? [Likely] [29][30] (scope is judged against the target level)
    - If it were twice as big, what would you have done differently?

N. Mixed technical-behavioral prompts (BR who is a scientist)

97. Walk through a ranking (or similar) project: the pain point, why this model, the metrics, and how you diagnosed a disappointing A/B result. [Reported] [30] (project deep-dive round)
    - What would you change in the experiment design today?
98. For a model you built, why did you choose that backbone, how did you handle class imbalance, and what would you do with more resources? [Reported] [29][30] (science-depth and project-dive rounds)
    - Which of those choices would you defend to a skeptical Principal Scientist?
99. Explain an AI-agent or LLM project you built and how it got adopted. [Reported] [30] (project deep-dive round)
    - Who used it in the first month, and who stopped using it?
100. Tell me about a time a technical trade-off (latency, cost, accuracy) forced a business compromise. [Likely]
    - Who signed off on the compromise?
101. Tell me about a time you moved a model from research to production at scale. [Likely]
    - What failed first in production that never failed offline?
102. Tell me about a time you had to choose between publishing and shipping. [Likely]
    - What did the business get out of the choice?
103. Tell me about a GenAI or LLM feature you built or evaluated. How did you decide it was safe to launch? [Likely]
    - How did you measure hallucination, and what threshold did you accept?
104. Tell me about a time an evaluation metric you chose turned out to be the wrong proxy. [Likely]
    - How did you detect the mismatch?
105. (Senior BR adding coding) Implement a priority queue from scratch. [Reported] [28] (reported in an L6 loop; possible but less likely at L5)
    - What is the cost of each operation, and how would you support decrease-key?

O. Post-loop, split-debrief and closing scenarios

106. (Extra round after a split debrief) Let's revisit coding with a follow-up problem. [Reported] [17]
107. (Downlevel conversation) We think your scope maps to L4. Would you consider that level? [Likely] (BR-driven downleveling reported in [11][15])
    - What evidence from your past work would change our assessment?
108. Why Amazon, and why this team? [Reported] [28]
    - Which LP do you find hardest to live by?
109. Is there anything the earlier interviewers did not ask that you want me to know? [Reported] [32]
    - Which LP does that story show?
110. What questions do you have for me about how Amazon hires and develops scientists? [Likely]

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11. Amazon Science Team Context: What Each Hiring Org Asks, and the Recent Amazon Science Work to Know

> Scope: the team-specific part of the L5 Applied Scientist (Applied Scientist II) onsite. That means the hiring-manager / Science Application round, the "why this team" probes, and the "have you seen our work?" questions. It covers the orgs that most often hire L5 AS: Search (Stores Foundational AI, formerly M5), Ads, Alexa AI / AGI (Nova), AWS AI, Rufus, Personalization and recommendations, SCOT (supply chain), Pricing, and Prime Video. It also lists recent Amazon Science publications and open-source releases, framed as questions an interviewer might ask. This chapter is questions only, with no answers.
>
> Research note (please read): in both the first pass and the October 2026 review pass, the shared web-search budget was used up and only github.com could be reached. Glassdoor, Blind, Medium, LeetCode, 1point3acres, amazon.jobs, amazon.science, aboutamazon.com, IEEE Spectrum and the prep-vendor sites did not resolve from this environment. The evidence comes from two places:
>
> - Candidate-report findings already gathered and cited in chapters 01-09 of this guide. Those were built from search-result extracts of the cited pages, not full reads. Each [Reported] tag here cites the original page, and I tag a question [Reported] only when one of those chapters attributes it to an Amazon science interview. These non-GitHub citations could not be re-fetched in the review pass, so treat them as unverified.
> - Pages fetched and read directly: the amazon-science GitHub organization and more than 20 of its repositories [33]-[49], [60]-[69], and four pages from two GitHub interview aggregators [19][20][21][59]. The review pass re-read the aggregators. As a result, it retagged two items that the aggregators list only as round topics, not as asked questions (RoPE/ALiBi/KV caching, and the fraud-metrics item), and it fixed the DeepMTL2R method count.
>
> **Team-specific first-person reports for Prime Video and Pricing were not found, so every question in those sections is [Likely].** Before your loop, ask your recruiter for the team's name and re-check its newest postings and papers. A follow-up session with a fresh search budget could add Blind and 1point3acres threads for each org.

Format notes for this round

- **Where team context shows up.** Every L5 loop includes a team-tailored round, usually called "Science Application" or "ML System Design (team-specific)", and it is often run by the hiring manager or a senior scientist from the hiring team. One 2025-

2026 AS II loop was reported as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Amazon's own prep page describes about four 55-minute interviews with members of the science community. It says each interviewer is assigned competencies and asks 2-3 behavioral (Leadership Principle) questions [1]. Blind commenters say L5 loops usually include the application/design round and L4 loops may not [53].

- **Duration and interviewer.** Rounds run 55-60 minutes. Expect about 40 minutes on the team problem and 10-15 minutes on LPs [1][52]. The HM also judges team fit and, with the bar raiser, makes the final call [53].
- **What the L5 bar looks like here.** You take a vague problem from the team's domain and frame it end to end: data and labels, model, offline and online metrics, launch, and monitoring. You make the trade-off calls yourself [17][52]. Depth on the team's own stack, and awareness of what the team has published, separate a "strong hire" from a "hire" in this round. That point is an inference, not a published rubric.
- **The team can change after the loop.** Some candidates pass the loop and are then team-matched or "recycled" to another org's opening [54]. Prepare a 2-minute pitch on how your work maps to at least two orgs.
- **GenAI everywhere.** Even on non-LLM teams, 2025-2026 loops treat transformers and LLMs as core breadth. An Ads candidate reported being rejected after not preparing for them [5][50].
- **Org churn (treat as unverified).** One GitHub aggregator reports that Amazon's AGI org was reorganized in 2025-2026 and that Nova 2 Lite, Nova 2 Sonic, Nova Forge and Nova Act remain active products [21]. An earlier chapter's search extract says Rufus was renamed "Alexa for Shopping" in May 2026 (see [27]). Confirm the current org and product names with your recruiter.

Tag legend: **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science or ML interview. Items from aggregators are marked "(aggregator)", and items from Research Scientist or MLE loops are marked "(RS loop)" or "(MLE loop)". **[Likely]** means a grounded inference from the team's products, job postings or Amazon Science publications. It has not been seen verbatim.

A. Cross-team "why this team" and team-fit questions

1. Why Amazon, why Applied Science, and why this particular team? [Likely]
    - Which of our team's products have you used as a customer, and what would you change first?
    - Which of our team's papers or launches have you read?
2. How would you apply your research or project to a problem at Amazon? [Reported] [55]
    - Which team's metric would your work move, and by how much, roughly?
3. (Recruiter or HM screen for a GenAI team.) What hands-on experience do you have with LLM post-training or with building agents? [Reported] (aggregator) [59]
    - Which part did you personally own: data, reward design, training infrastructure or evaluation?
4. Here is a high-level ML problem from our team. Propose data sources, an annotation strategy, a modeling technique, evaluation metrics, and the trade-offs and risks.

[Reported] [17]
    - What is the version you would ship in one quarter, and what would v2 add?
5. Answer an ML application question about one of our Amazon products. [Reported] [18]
6. Pick an agentic product we could build. Who is the customer, and how do you measure success? [Reported] (Seattle, 2026) [2]
    - Write the press-release headline (Working Backwards).
7. If the loop goes well but this team's headcount closes, which other Amazon science org would you want, and why? [Likely] [54]
8. What science problem in our org do you think is unsolved, and how would you start on it? [Likely]
    - How would you write the 6-page document proposing it?

B. Search (Stores Foundational AI / Search Science, formerly M5)

9. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [13]
    - What position, presentation and trust biases contaminate the clicks?
    - How would you estimate relevance from click logs while accounting for query intent? [Likely] (Amazon Science work on intent-based relevance from click logs [29])
10. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [56]
    - Pointwise, pairwise or listwise learning-to-rank: which, and why?
11. Walk through a ranking project you shipped: the customer pain point, why that model, which metrics you used, and how you diagnosed a disappointing A/B result. [Reported] (Amazon MLE loop project deep dive, aggregator) [20]
12. Compare a small and a large model from the same text-embedding family. What do you gain and what do you pay, for first-stage product retrieval? [Reported] (RS breadth round, aggregator) [19]
13. Compare approaches to building joint image-text embeddings, for example for visual search or for matching product photos to queries. [Reported] (RS online assessment, aggregator) [19]
    - Implement a numerically stable InfoNCE loss that is symmetric across both directions (image to text and text to image) for a batch of paired embeddings. [Reported] (science coding round, aggregator) [19][20][59]
    - How does batch size interact with in-batch negatives, and how would you mine hard negatives from the catalog?
14. Answer a search-relevance application question for Amazon search. [Reported] [3]
15. Compare bi-encoders and cross-encoders. Where does each sit in a product-search stack? [Reported] [3]
    - How would you distill the cross-encoder into something that meets a retrieval latency budget?
16. You have 10k product listings in 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [13]

    - How do you handle a long tail of categories with very few examples?
17. Design an end-to-end retrieval and ranking pipeline with candidate generation, serving, offline metrics versus online guardrails, A/B testing and position bias. [Reported] (Amazon search/recs/ads MLE loop, aggregator) [20][59]
18. Design a multi-turn search agent that handles long context. [Reported] (RS loop, aggregator) [19]
    - When would you summarize earlier turns, and when would you retrieve them instead?
19. The ESCI Shopping Queries dataset labels query-product pairs as Exact, Substitute, Complement or Irrelevant. Why is a 4-way label more useful than binary relevance for Amazon search? [Likely] [34]
    - How would you use Substitute and Complement labels differently in ranking and in recommendations?
    - The dataset covers English, Japanese and Spanish. How would you transfer a relevance model to a new marketplace?
20. How would you rank with several objectives at once (relevance, conversion, margin, delivery speed)? [Likely]
    - Would you use multi-task learning-to-rank, and how would you weight the tasks (for example uncertainty weighting, PCGrad or Nash-MTL)? [Likely] (Amazon's DeepMTL2R library lists 21 multi-task weighting methods plus a single-task baseline [45])
21. (Search/IR postings list learning-to-rank and recommender work [26].) How would you select a set of results that is both relevant and diverse from millions of candidates, faster than MMR? [Likely] (MUSS, UAI 2026 [39])
    - When does diversity hurt a high-intent query like a model number?
22. How would you use an LLM as a relevance judge for product search, and how would you validate it against human annotators? [Likely]
23. How would you detect and fix wrong attribute values (size, color, material) in catalog listings that hurt search? [Likely]
    - SynthAVE labels (attribute, value) pairs as Correct, Incorrect or Unknown, using majority vote over 21 LLM-judge configurations (7 model families with 3 prompts each). When would you trust an LLM arena as a labeler? [Likely] (EMNLP 2026 Industry [38])
    - In SynthAVE about 39% of pairs are Unknown (the text cannot decide). How would you treat "not determinable" in training and in your metric? [Likely] [38]
24. Design a system that detects duplicate or variant listings of the same product from different sellers (title, attributes and images), so search does not show the same item five times. [Likely]
    - How do you set the merge threshold when a false merge is costlier than a missed one?

C. Advertising (Sponsored Products, DSP, Ads Trust, Ads measurement)

25. Explain how transformers and attention work. How does a decoder-only LLM generate text at inference time? (An Ads candidate reported being rejected after not preparing this.) [Reported] [5]

26. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] (Advertising Trust) [8]
    - Walk through the full data pipeline from collection to deployment. [Reported] [8]
    - How do you get labels for rare policy violations?
27. Science Depth on an Ads loop: your resume, plus GenAI fundamentals, plus the business impact of your work. [Reported] [8]
28. Amazon Ads generates image creatives for advertisers. Design an evaluation framework that checks whether a generated image spells the brand name correctly and places it naturally. [Reported] (RS and MLE science-application / ML design round, aggregator) [19][20][59]
    - How would you combine OCR, a learned "naturalness" judge and human review, and how would you calibrate the judge?
29. Design a click counter over a sliding k-minute window, then an advertiser-metrics service with campaign and event entities, windowed pre-aggregation and idempotency for duplicate events. [Reported] (Ads MLE loop, aggregator) [20]
30. Design click-through-rate or conversion-rate prediction for Sponsored Products. [Likely]
    - Why does calibration matter for auction pricing?
    - How do you handle delayed conversions?
31. How would you learn a bidding policy offline from logged auction data? [Likely] (AuctionGym, KDD 2023 ADS track [37])
    - Why is offline evaluation of learning-to-bid hard (unobserved confounding, costly experiments)?
    - How does optimal bidding differ between second-price and first-price auctions, and what is bid shading?
32. How would you measure the incremental lift of an ad campaign for an advertiser? [Likely]
    - Ghost ads, geo experiments or observational causal methods: what are the trade-offs?
33. How would you blend sponsored products into a generative shopping answer without eroding trust, and how would you measure the effect? [Likely]
34. Design a budget-pacing algorithm for advertisers. How would you frame it as constrained optimization or as a bandit? [Likely]

D. Alexa AI, AGI and Amazon Nova (foundation models)

35. Expect several "modern LLM training technique" questions, with follow-ups on each stage (pretraining, SFT, preference tuning). [Reported] (AGI-team role) [6][7] (see chapter 07)
36. You are handed a Transformer implementation that trains but performs badly. Debug it. [Reported] (Amazon AGI, aggregator) [21]
    - Which bugs in masking, scaling, positional encoding or normalization produce silent quality loss?
37. Build and train a small classifier live. [Reported] (Amazon AGI, aggregator) [21]
38. Reinforcement-learning fundamentals plus coding. [Reported] (Amazon AGI, aggregator) [21]

    - Why does GRPO remove the critic, what does that cost, and how would you design the reward? [Reported] (RS phone screen, aggregator) [19]
    - How would you parallelize GRPO training? [Reported] (MLE loop, aggregator) [20]
    - Explain PPO's clipped objective. [Likely] [19]
39. Write the rotary position embedding (RoPE) formula and explain how it keeps relative position inside attention. [Reported] (MLE loop, aggregator) [20][59]
    - Compare RoPE with ALiBi, and explain KV caching and pipeline parallelism. [Likely] (listed as round topics, not as asked questions, in the RS aggregator [19])
40. For a residual MLP of width 4096 in BF16, is batch size 1 or 256 compute-bound, and what does that imply for serving an assistant model? [Reported] (MLE breadth round, aggregator) [20]
41. Architect an AI chatbot, covering intent detection, dialogue state, generation, fallback behavior, evaluation and continuous learning. [Reported] (RS and MLE ML-design rounds, aggregator) [19][20]
    - Where does a classic NLU intent model still beat an LLM for an Alexa-style assistant?
42. (Stretch, reported from a staff-level loop, so above the L5 bar.) Design a distributed pipeline that ingests, tokenizes, deduplicates and filters pretraining data for a foundation model. [Reported] (aggregator) [20]
    - How would you measure the effect of a quality filter without a full pretraining run?
43. Discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] (AS II, LLM team, Santa Clara) [12]
    - How does beam search work, and when does it hurt open-ended generation? [Reported] [12]
44. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] (Alexa International Tech) [10]
    - Where would you get diverse accent data, and how would you handle phonetic variation? [Reported] [10]
45. The Alexa International loop is breadth, depth, science application, coding and a bar raiser, with LPs throughout. What would you expect the science-application problem to look like for a multilingual assistant? [Reported for the format] [9]; the follow-up is [Likely]
46. How would you align a multilingual assistant with SFT, DPO, RLHF or RLAIF when preference data is scarce in some locales? [Likely] (the Alexa International posting lists these methods [23])
47. Instruction-tuned LLMs are trained on text preferences, but Alexa speaks its answers. What makes a response "speechworthy", and how would you collect preferences for it? [Likely] (Speechworthy Instruction-tuned LMs, EMNLP 2024 [42])
48. AlexaTM 20B was a multilingual seq2seq model pretrained with denoising plus causal LM. When would an encoder-decoder beat a decoder-only model for an Alexa task? [Likely] [40]
49. Training LLMs with MXFP4: why can the backward pass run in 4-bit with stochastic rounding and a random Hadamard transform and still give unbiased gradients? [Likely] (AISTATS 2025 [41])
    - What breaks if rounding is biased?

50. Moving from an ASR, then LLM, then TTS cascade to an end-to-end speech model (in the spirit of Nova Sonic): what changes in latency, evaluation and failure modes? [Likely] [21]
51. Design a browser-operating agent (in the spirit of Nova Act) that completes multi-step web tasks reliably. [Likely] [21]
    - How do you build its evaluation set and measure step-level versus task-level success?
52. An Alexa+ style assistant calls real device and service APIs (timers, smart home, ordering). How do you make tool calls reliable and safe? [Likely]
53. How would you make an on-device wake-word or NLU model 10x smaller with minimal quality loss? [Likely]

E. AWS AI (Bedrock, SageMaker, Q / coding agents, AI research)

54. Explain transformers, then explain LoRA. [Reported] (AWS AS loop) [4]
    - Where are the LoRA adapters inserted, and what does the rank control?
    - Explain the main parameter-efficient fine-tuning techniques, which parameters each trains, and when each beats full fine-tuning. [Reported] (RS and MLE online assessment, aggregator) [19][20]
55. Explain how an inference engine such as vLLM raises LLM serving throughput (paged KV cache, continuous batching). [Reported] (MLE online assessment, aggregator) [20]
    - Amazon Science's ammo uses multiple agents to optimize vLLM GPU kernels for one deployment. How would you check that an auto-generated kernel is both correct and faster? [Likely] [49]
56. Design an LLM quality-validation system that gates model or prompt updates before they reach customers. [Reported] (title listed for the RS science-application round, aggregator) [19]
57. Design an enterprise RAG service. [Reported] [51]
    - How do you enforce document-level access control in retrieval?
58. Design a GitHub Copilot-style coding agent. [Reported] (Senior AS, Boston) [3]
    - SWE-PolyBench scores agents on repository-level tasks in Java, JavaScript, TypeScript and Python with pass rate and file- and node-retrieval metrics. Why measure retrieval separately from pass rate? [Likely] (arXiv 2025 [46])
59. Evaluate an LLM agent with metrics beyond accuracy: tool use, hallucination and calibration. [Reported] [3]
60. RAGChecker separates retriever metrics (claim recall, context precision) from generator metrics (faithfulness, noise sensitivity, hallucination, context utilization). How would you use that split to debug a customer's RAG system? [Likely] (NeurIPS 2024 Datasets and Benchmarks [35])
61. RefChecker checks LLM outputs at the level of (subject, predicate, object) claim triplets in zero, noisy and accurate context settings. Why triplets rather than sentences? [Likely] (2024 [36])
    - Where does triplet extraction itself fail?
62. A customer's Bedrock-hosted workload costs too much. How would you cut inference cost sharply without unacceptable quality loss? [Likely] [21] (see chapter 07)

63. Chronos turns a time series into tokens by scaling and quantization, then trains a language model on them. Why does that work, and what does Chronos-Bolt change (patches, direct multi-step quantile output)? [Likely] (TMLR 2024; Chronos-Bolt Nov 2024 [33])
    - Chronos-2 (Oct 2025) adds multivariate and covariate-informed zero-shot forecasting. How would you evaluate it fairly against a tuned, dataset-specific model?
64. Design an agent that migrates a whole Java 8 repository to Java 17. How do you decide that the migration succeeded? [Likely] (MigrationBench, KDD 2026 [60])
    - MigrationBench approximates functional equivalence: the build passes, compiled class versions match the target, test methods are unchanged and the test count does not fall. What can still go wrong under these checks?
65. Coding agents run inside customer workspaces. How would you red-team one for jailbreaks that hide malicious intent in a half-finished multi-file project rather than in the prompt? [Likely] (JAWS-Bench, arXiv 2510.01359 [61])
66. A customer asks a Bedrock-hosted assistant about AWS services whose behavior changed after the model's training cutoff. How would you detect and handle stale knowledge? [Likely] (evolveQA, EMNLP 2026 [62])
67. Design a hallucination detector for RAG answers over very long contexts (around 90K characters), with sentence-level labels (Supported, Contradicted, Not Mentioned, Supplementary). [Likely] (TRIVIA+, ACL 2026 [63])
    - Why would you inject controlled label noise into the benchmark?
68. Synthesize realistic tabular data, or detect anomalies in tabular data, with generative models. What are the evaluation pitfalls? [Likely] (amazon-science TabSyn and AnoLLM repositories [49])

F. Rufus (generative shopping assistant)

69. Design Rufus: answer "What do I need for cold weather golf?" using the catalog, reviews and community Q&A. [Likely] [22][28]
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you evaluate offline before an A/B test, and what is the online success metric?
70. Design an agentic recommendation system for Amazon, covering components, model choices, and quality and safety evaluation. [Reported] [16]
71. Rufus runs on several LLMs, including Amazon Nova, on Bedrock. How would you route each query between a small and a large model? [Likely] [27]
72. How would you personalize answers with a customer's purchase and browse history while respecting privacy? [Likely] [27]
73. You have one latency budget for an assistant answer (say, first token in under one second). How do you split it across retrieval, reranking, generation and safety filters? [Likely] [22]
74. How would you summarize thousands of reviews into a short "Customers say" blurb, and measure faithfulness and coverage of minority opinions? [Likely]
75. Personalized autocomplete for a chat interface: how would you rank candidate completions, and would you report saved-keystrokes@k or acceptance-rate@k? [Likely] (amazon-science personalized chat autocomplete experiments [49])

76. How would you red-team a shopping assistant for unsafe, biased or policy-violating answers? [Likely]

G. Personalization and recommendations (homepage, Amazon Music, explanations)

77. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [16]
    - What is the north-star metric, and what are the guardrails?
78. A user goes from viewing to clicking to purchasing. Model the probability of each action given a user and a product. [Reported] [14]
    - Three independent models, or a joint model? [Reported] [14]
    - How would you build the dataset and down-sample (for example with reservoir sampling)? [Reported] [14]
79. Design a recommender system end to end. [Reported] (reported as the design round in an AS loop) [57]
    - How do you handle cold start for new users and new ASINs?
80. You have logged rankings from the Amazon Music home page with logging-policy propensities. How would you estimate a new ranking policy's value offline? [Likely] (Music off-policy evaluation benchmark [44])
    - Compare IPS, self-normalized IPS and doubly-robust estimators for slate rankings.
    - How do you check whether your OPE estimate matches the online A/B result?
81. You have a pool of LLM-generated explanations for a recommendation. Would you choose among them with RL fine-tuning (PPO, GRPO, DPO) or with a pairwise ranker? [Likely] (RecSys 2026 finding that LambdaRank beat single-action RL [47])
82. How would you avoid feedback loops that recommend only already-popular items? [Likely]
83. How would you personalize email or push notifications as uplift modeling rather than response modeling? [Likely]

H. SCOT (Supply Chain Optimization Technologies), forecasting and inventory

84. What is the newsvendor problem, and how would you use it to set inventory levels? [Reported] (Amazon supply-chain interviews) [15]
    - How does a quantile forecast map to the critical ratio?
85. Model an optimization problem from your own work: objective, decision variables and constraints. [Reported] [52]
    - When would you use a heuristic instead of an exact solver?
86. How would you forecast demand for millions of products, including new ones with no history? [Likely] [30]
    - Why do buying decisions need probabilistic or quantile forecasts rather than point forecasts?
    - Would you use a global model or per-item models?

87. Design a system that decides how much of each product to buy, from which vendor, and when. [Likely] (SCOT Optimal Sourcing / Buying Science posting [24])
88. Design inventory placement across fulfillment centers. How do forecast errors propagate into transportation cost? [Likely] [31]
89. How do you forecast through Prime Day when history is sparse and demand was censored by stock-outs? [Likely]
90. Could a pretrained foundation model like Chronos replace your team's per-domain forecaster? What evidence would convince you? [Likely] [33]
91. Design a delivery-promise prediction (the date shown on the detail page). What loss would you use if late costs more than early? [Likely] (Last Mile candidate context [11])
92. How would you run an experiment on a buying policy when products share warehouses and trucks (interference)? [Likely]

I. Pricing and causal inference

93. Design a detector that catches pricing errors (for example a $500 item listed at $5) before they go live, across millions of ASINs and sellers. [Likely]
    - How do you trade off blocking a real deal against letting a costly error through?
94. Design a system that recommends competitive prices for millions of products. [Likely]
    - How do you estimate price elasticity when your own algorithm set the prices (endogeneity)?
    - What guardrails prevent price spirals and loss of customer trust?
95. How would you measure the effect of a pricing algorithm when prices interact across products? [Likely]
    - Switchback, cluster-randomized or synthetic-control designs: when would you use each?
96. How would you validate a causal model before trusting it? [Likely] (Amazon's causal-validation package simulates data with a known effect and supports placebo tests [48])
    - Design a placebo test for a synthetic-control estimate of a price change.
97. How would you design an A/B test for a model launch, and what would you do if it is underpowered? [Reported] [58]
98. How would you decide whether to discount a product for a deal event, accounting for cannibalization and pull-forward demand? [Likely]
99. When would you use double ML, instrumental variables or difference-in-differences for a promotion's effect? [Likely]

J. Prime Video (and other Amazon content and devices teams)

No first-person Prime Video AS reports were found in this run. Everything below is [Likely].

100. Design recommendations for Prime Video, where viewing is sequential and feedback is implicit. [Likely] - Is a show watched for 3 minutes a positive or a negative label?

101. How would you choose artwork or thumbnails per user, and evaluate the policy offline? [Likely]
102. How would you detect video-quality problems (buffering, artifacts, audio-video sync) at scale from playback telemetry? [Likely]
103. Design ad insertion for ad-supported streaming: where to place breaks, and how to measure viewer drop-off. [Likely]
104. How would you forecast live-sports streaming demand for capacity planning? [Likely]
105. How would you detect and skip intros and recaps, or generate scene-level metadata, with video models? [Likely]

K. Payments, trust, fraud and abuse

106. Design a model to predict payment-method failure and pick the best fallback for the customer. [Likely] (Amazon Payment Products L5 posting [25])
107. Train and evaluate a fraud model under extreme class imbalance, delayed chargeback labels and adaptive adversaries. [Likely] - Amazon's Fraud Dataset Benchmark groups public datasets by fraud type (card-not-present, bots, malicious traffic, content moderation, credit risk). How would you check that a method generalizes across them? [Likely] [43]
108. Report precision, recall and AUC for imbalanced fraud data, and say which you would optimize. [Likely] (the RS aggregator lists imbalanced metrics as a round topic, not as an asked question [19])
109. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [18]

L. "Have you read our work?" Recent Amazon Science publications as interview questions

These are [Likely] prompts. HMs often ask "What do you think of X?" about their own team's work. Read the abstract and the evaluation section of anything in your target org.

110. Chronos (TMLR 2024), Chronos-Bolt (Nov 2024) and Chronos-2 (arXiv, Oct 2025; released to SageMaker JumpStart in Dec 2025 and AutoGluon-Cloud in Jun 2026): what is the weakest part of the "language of time series" argument? [Likely] [33]
111. Shopping Queries Dataset (ESCI, arXiv 2022, used in a KDD Cup): what does ESCI miss that real Amazon search needs (personalization, freshness, session context)? [Likely] [34]
112. RAGChecker (NeurIPS 2024 D&B) and RefChecker (2024): how would you meta-evaluate an automatic hallucination checker against human judgment? [Likely] [35] [36]
113. AuctionGym (KDD 2023 ADS; AdKDD 2022 best paper): what does a simulator buy you that logged data cannot, and what does it cost you? [Likely] [37]
114. SynthAVE (EMNLP 2026 Industry): what are the risks of evaluating on synthetically generated products with anonymized brands? [Likely] [38]
115. MUSS (UAI 2026): MMR is a special case of MUSS with one partition. What does the constant-factor approximation guarantee buy you in practice? [Likely] [39]

116. Training LLMs with MXFP4 (AISTATS 2025): why target the backward pass first? [Likely] [41]
117. Speechworthy instruction-tuned LMs (EMNLP 2024): how would you extend a listening-based preference dataset to multi-turn voice? [Likely] [42]
118. Multimodal chain-of-thought (mm-cot) and Auto-CoT: are these still relevant once reasoning models are trained with RL? [Likely] [49]
119. Pairwise ranking beats single-action RL for offline explanation selection (RecSys 2026): when would you expect that result to flip? [Likely] [47]
120. Amazon Science's 2026 open-source releases include agent self-improvement recipes (Ratchet, Double Ratchet), RL-training kernels (DualKV shared-prompt FlashAttention), multi-agent vLLM kernel optimization (ammo) and long-context hallucination benchmarks (ACL 2026). Pick one relevant to this team and critique it. [Likely] [49]
121. Select, Label, Evaluate: Active Testing in NLP (EMNLP 2026) reports performance estimates within 1% of the full test set using up to 95% fewer labels. With a fixed annotation budget, how would you choose which examples to label to estimate a new model's accuracy? [Likely] [64]
122. What Fits (Into Few Tokens) Doesn't Overfit (arXiv 2026) argues that strategies which compress into a short description overfit a reused validation set less. Your team has tuned against the same offline benchmark for two years. How would you check whether it is overfit? [Likely] [65]
123. ReSkill (arXiv 2026) treats creating reusable skills as part of the agentic RL loop rather than a separate step. What could go wrong when skills and the policy evolve together, and how would you evaluate it? [Likely] [66]
124. Adaptively looped diffusion language models (ALoDLM, arXiv 2610.04198) give hard tokens more refinement passes than easy ones. Would you consider a diffusion LM for a latency-bound assistant, and what would you measure first? [Likely] [67]
125. WebIE (ACL 2023) targets faithful information extraction from web text. How would you extract product attributes from noisy seller descriptions without inventing facts that are not in the text? [Likely] [68]
126. CaML (WWW 2023) maps retail products to emission-factor categories with zero-shot text similarity. How would you evaluate a zero-shot product-to-taxonomy matcher when you have only a few hundred human labels? [Likely] [69]
127. Earlier LLM distillation work, "Learning with less: knowledge distillation from LLMs via unlabeled data": when is distilling an LLM into a small classifier the right call for a production team? [Likely] [32]

M. L5-bar probes that recur across all teams

128. Your model's online A/B result fell short of its offline gain. How do you find out why? [Reported] (RS depth round, aggregator) [19] - Separate the causes: offline metric mismatch, train-serve skew, position or presentation effects, novelty, and an underpowered test.
129. Of everything you proposed for our problem, what is the riskiest assumption, and how would you de-risk it in two weeks? [Likely]
130. Your design improves our team metric but hurts a sister team's metric (for example, conversion up and returns up). How do you decide? [Likely]

131. Which parts would you build, and which would you reuse from existing Amazon services or other science teams? [Likely]
132. How would you hand this system to engineering for long-term ownership, with monitoring and retraining triggers? [Likely]
133. Tell me about a time you influenced a team outside your own with your science. How would that look here? [Likely] (L5 scope: influence on peers and sister teams; see chapter 01)

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12. ML System Design Reference Diagrams: Six Canonical Amazon Prompts and a 45-Minute Whiteboard Template

Scope: this chapter adds the diagrams that chapter 06 leaves out. It gives one labelled block diagram for each of six canonical Amazon Applied Scientist (L5) design prompts, a whiteboard template for drawing and narrating any of them in 45 minutes, and the questions an interviewer is likely to ask about each block. The diagrams are skeletons to draw from memory. The latency figures in them are illustrative budgets to defend or revise, not Amazon numbers. The question sections contain questions only, with no answers.

Research note (please read): no new web research was possible for this chapter. The run's shared web-search budget was already used up before this chapter started, so every planned search was refused. Test fetches to aws.amazon.com and docs.aws.amazon.com failed DNS resolution in this environment. Every [Reported] tag below therefore comes from the sourced findings already recorded in chapter 06 (and through it, chapters 01-04) of this guide, and each one cites the original page. Those findings were built from search-result extracts, not full page reads. The AWS service references [23]-[29] are official product or documentation pages. They are cited only to name the components, were not re-fetched in this run, and are not interview reports. A follow-up session with a fresh search budget could add first-person design-round reports, especially from Blind, 1point3acres and Prepfully.

Review pass (2026-10-14): an adversarial review tried to re-fetch about 20 of the cited URLs. Only GitHub-hosted pages could be reached. [20] (Chronos) and [21] (ESCI) were confirmed live, and their content matches how this chapter uses them. Every other host returned a DNS failure or a proxy 403. That includes amazon.jobs, amazon.science, aws.amazon.com, docs.aws.amazon.com, Glassdoor, Blind, Medium, LeetCode, Quora, 1point3acres, Dataford, IGotAnOffer, OA VO Service, IEEE Spectrum and About Amazon. Those references could be confirmed neither live nor dead, so they were kept as recorded in chapter 06. The web-search budget was also used up. The review made three kinds of change. (1) It added four new verified sources [37]-[40], all read in full from GitHub: two TrueInterview round guides that summarise 2026 candidate reports, the TrueInterview Amazon question index, and a compiled set of candidate posts. It also added about 25 new questions sourced from those pages or grounded in the diagrams. (2) It retagged the newsvendor question as [Likely], because its source is a supply-chain interview and not a science or ML one. It upgraded the offline-wins-online-neutral probe to [Reported]. (3) It removed four duplicate or generic items. Questions that the index [39] attributes to Amazon without naming the role are tagged [Likely] with a "(see [39])" pointer, not [Reported].

How the [Reported] / [Likely] tag system works

Every question in this guide carries one of two tags:

- **[Reported] [n]**: a cited source (a candidate post on Glassdoor, Blind, LeetCode Discuss, Medium or 1point3acres, a prep-vendor write-up such as IGotAnOffer, Dataford or Interview Kickstart, or a round guide compiled from candidate reports such as TrueInterview) attributes this question, or a close paraphrase, to an Amazon science interview. An Amazon Machine Learning Engineer loop also counts, and the question is then marked "(MLE loop)". A question the source places above L5 (for example in a staff loop) is marked as such. The number points to the References list at the end of the chapter. A follow-up probe under a [Reported] question is not itself reported unless it carries its own [Reported] tag.
- **[Likely]**: a grounded inference from the team's products, from Amazon job postings or Amazon Science publications, or from the standard way ML system design rounds are probed. Nobody has been seen asking it verbatim. A "(see [n])" after a [Likely] tag points to the product or posting the inference is based on, not to an interview report.

What the tags do not mean:

- [Reported] does not mean the question is common. One report is enough for the tag.
- [Likely] does not mean the question is rare. Most block-level probes ("how do you detect drift?") are [Likely] because candidates rarely write them down, not because interviewers skip them.
- No tag was upgraded to [Reported] without a citation. Where this chapter could not trace a question to a source, it is [Likely].

Format notes for this round

- **Duration and slot.** The design round is one of about four to six interviews of 55-60 minutes each in the L5 Applied Scientist loop. One 2025-2026 Applied Scientist II report lists "1 Sys Design (Team-specific)" next to breadth, depth, coding and Bar Raiser rounds [2]. Amazon's official prep page describes interviews of about 55 minutes with members of the science community, each also covering Leadership Principles [1]. Plan for roughly 45 minutes of design and 10-15 minutes of LP questions [1][16].
- **Interviewer.** Usually the hiring manager or a senior scientist on the hiring team, with a prompt from that team's domain [2][8]: Ads trust [9], Alexa speech [10], search and recommendations [3][12][13], or LLM agents and RAG [3][14].
- **L5 bar.** Blind commenters say the design or application round is typical for L5 and may be missing from L4 loops [5]. At L5 you are expected to own the system end to end: framing a vague business problem, choosing data, labels and models, picking offline and online metrics, and planning launch and monitoring, with sound trade-off calls along the way [5][16]. IGotAnOffer describes the round as proposing data sources, annotation strategy, modeling technique, evaluation metrics, and trade-offs and risks [7]. Candidates report being asked to walk the data pipeline "from collection to deployment" [9].

- **2026 reports.** A TrueInterview round guide built from 20 candidate reports (mostly 2026 Q1-Q3) describes the science application round as "an open-ended product problem" graded on framing, evaluation design and launch realism. It adds that "Ads and search teams ask for ranking pipelines instead" [37]. The matching MLE guide says prompts "follow the team's domain" and lists train-serve skew, drift, duplicate events and rollback paths as the failure modes candidates are expected to raise [38]. Several 2026 reports say Leadership Principle questions ate much of the hour, so rehearse a compressed version of the template [37].
- **Science over boxes.** The weight is on framing, labels, modeling and metrics rather than on distributed-systems detail. Candidates still report questions about scale, latency and deployment [9][22]. So the diagram exists to anchor the science discussion, not to replace it.

Part 1. The whiteboard template (draw this first, for any prompt)

1.1 The universal skeleton

Draw the same seven zones every time, then fill them for the specific prompt. Left to right is the online path; the bottom band is offline; the arrow back from the right closes the loop.

```
+-------------------------------------------------------------------------------+
| (1) PROBLEM FRAME:  customer | decision being made | north-star | guardrails   |
+-------------------------------------------------------------------------------+

 ONLINE PATH  (write the latency budget above each arrow, total p99 in the corner)
 ------------------------------------------------------------------------------
 [Client/Request] --> [(4a) Feature fetch] --> [(4b) Stage 1: recall / score]
                                                --> [(4c) Stage 2: rank / decide]
                                                --> [(4d) Policy / business rules]
                                                --> [Response]  ----+
                                                                    |
 [(6) LOGGING: impressions, actions, model version, features served] <+
          |
 ---------|---------------------------------------------------------------------
 OFFLINE  v
 [(2) DATA SOURCES] --> [(3a) Labels + sampling] --> [(3b) Training + offline eval]
        |                                                    |
        v                                                    v
 [(3c) FEATURE STORE  offline <--same definitions--> online]   [Model registry]
                                                              |
                                                              v
                                        [(5) Deploy: shadow -> canary -> A/B]

 [(7) MONITORING: data drift | prediction drift | business KPI | latency/cost]
      --> retraining trigger --> back to (3b)
```

Zone checklist (say each heading out loud as you draw it):

1. Problem frame: customer, decision, north-star metric, guardrails, constraints (latency, cost, scale).
2. Data sources: logs, catalog, transactions, third-party or human labels.
3. Offline: label definition, sampling, training, offline evaluation, feature store, model registry.
4. Online serving: stages with a latency budget per stage.

5. Launch: shadow mode, canary, A/B test with guardrails.
6. Feedback and logging loop: what is logged, at what granularity, with which model version.
7. Monitoring and retraining triggers.

1.2 The 45-minute narration clock

```
min 0-5    Clarify: who is the customer, what decision, what scale, what latency, what "good" means
min 5-8    State north-star + 2 guardrails + offline proxy metric; write them in zone (1)
min 8-12   Draw the empty skeleton (all 7 zones) before going deep anywhere
min 12-20  Data + labels + sampling (zone 2-3a)          <- science depth expected here
min 20-30  Models per stage, features, offline eval (3b, 4b-4c)
min 30-35  Serving path with latency budget, AWS components (4, 3c)
min 35-40  Launch, A/B design, feedback loop, monitoring + retrain triggers (5-7)
min 40-45  Risks, v1 vs v2, what you would cut, questions for the interviewer
```

1.3 AWS component cheat-sheet (where each one goes on the board)

```
Zone                Component an interviewer expects you to be able to name
------------------  ------------------------------------------------------------
Event ingestion     Amazon Kinesis Data Streams (clickstream, transactions) [25]
Stream features     Kinesis -> Managed Service for Apache Flink / Lambda -> online store
Data lake           Amazon S3 + AWS Glue catalog; Athena / EMR for batch features
Feature store       SageMaker Feature Store (online + offline store) [24]
Low-latency lookup  Amazon DynamoDB (user state, counters, precomputed candidates) [26]
Vector / text index Amazon OpenSearch Service (BM25 + k-NN vector search) [27]
Training            SageMaker Training jobs, SageMaker Pipelines, Model Registry [23]
Serving             SageMaker real-time endpoints / batch transform [23]
Monitoring          SageMaker Model Monitor + CloudWatch alarms [23]
LLM / RAG           Amazon Bedrock (foundation models, Knowledge Bases, Guardrails) [28]
Managed recsys/fcst Amazon Personalize [29]; Chronos open-source forecasting models [20]
```

Questions on the template itself

1. Here is a high-level ML problem from our team. Propose a solution, covering data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Reported] [7]
    - Which parts of the problem actually need ML, and which could be rules?
    - What is your v1 that could ship in one quarter?
2. Walk me through the full data pipeline for your design, from collection to deployment. [Reported] [9]
    - At which arrow on your diagram can training and serving features diverge?
3. Before you draw anything: what decision does this system make, and how often? [Likely]
    - Is it a batch decision (nightly) or a per-request decision, and how does that change the diagram?
4. What is the end-to-end p99 latency budget, and how have you split it across the stages you drew? [Likely]
    - Which stage would you cut first if the budget were halved?
5. Which boxes on your diagram are precomputed offline, and which must run per request? [Likely]
    - What runs on SageMaker real-time endpoints, and what runs as batch transform?
6. Which boxes would you build, and which would you reuse from existing AWS services or other Amazon teams? [Likely]

    - Why SageMaker Feature Store rather than DynamoDB alone for your features?
7. Point to where you log. What exactly is in a log record, and why do you need the served feature values rather than recomputing them? [Likely]
8. Here is a problem from a domain you know but are not expert in. Propose a product that solves it, then explain how you would collect data, which models you would compare, and how you would evaluate it with both engineering and business metrics. (AWS AI Applied Scientist loop) [Reported] [40]
9. Leadership Principle questions took the first 25 minutes, and you have about 30 minutes for the design. Which zones of the template do you compress or skip, and which must you still cover? [Likely] (see [37])
10. You are drawing the streaming feature path (Kinesis to a stream processor to the SageMaker Feature Store online store). How do you make sure the training set reads each feature exactly as it was at prediction time, with no future leakage? [Likely] (see [24][25])

Part 2. Recommendations: candidate generation -> ranking -> re-ranking

Diagram

```
FRAME: homepage / detail-page widgets | north-star: downstream purchases (or GMS) per session
       guardrails: diversity, returns, latency p99 ~150-200 ms (illustrative)

ONLINE
[Request: customer_id, page, device]
    | ~10 ms
    v
[Feature fetch]  DynamoDB / SageMaker Feature Store online:
    |            recent views, purchases, category affinities, session events
    | ~30 ms
    v
[CANDIDATE GENERATION  (thousands, high recall)]
    |  - two-tower ANN (user emb -> item emb index, OpenSearch k-NN)
    |  - item-to-item co-view / co-purchase lists (precomputed, DynamoDB)
    |  - trending / popular in category (cold-start fallback)
    | ~50 ms
    v
[RANKING  (hundreds -> ~50)]  GBDT or deep multi-task model
    |  heads: p(click), p(add-to-cart), p(purchase)   on SageMaker endpoint
    | ~30 ms
    v
[RE-RANKING / POLICY]  diversity, dedupe variants, out-of-stock filter,
    |                  business rules, exploration slots
    v
[Widget response] ---> impressions + clicks + purchases ---> Kinesis
                                                              |
OFFLINE                                                       v
[S3 data lake: impressions, actions, catalog, orders]  <-------+
    |
    +--> labels (view->click->purchase), negative sampling, position-bias correction
    +--> train retrieval towers (weekly) + ranker (daily)  [SageMaker Training]
    +--> rebuild item embedding index (OpenSearch) + co-purchase tables (DynamoDB)
    +--> offline eval: recall@K (retrieval), NDCG / AUC (ranker), replay/off-policy est.
    v
[Model registry] -> shadow -> A/B (unit: customer)

MONITOR: CTR / purchase rate by widget, coverage of catalog, embedding drift,
         popularity concentration -> retrain or re-index triggers
```

Questions

11. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [6]
    - Draw candidate generation, ranking and re-ranking. Why three stages and not one model?
    - What is the north-star metric, and what are the guardrails?
12. Design a recommender system. [Reported] [15]
    - Two-tower retrieval, matrix factorization, or a sequence model for candidate generation: which, and why?
    - How do you handle cold start for new customers and new ASINs, and where on the diagram does that path sit?
13. A user goes from viewing to clicking to purchasing a product. Model the probability of each action given a user and a product. [Reported] [11]
    - Would you build three independent models, or model view, click and purchase jointly? [Reported] [11]
    - How would you build the dataset, choose features, and down-sample (for example with reservoir sampling)? [Reported] [11]
    - Purchases are only observed after clicks. How do you correct for that selection bias?
14. Design an agentic recommendation system for Amazon, covering component architecture, model choices, and quality and safety evaluation. [Reported] [6]
    - Which box in the classic three-stage diagram does the LLM agent replace, and which does it call as a tool?
15. How do you build the negatives for the two-tower model: random, in-batch, or hard negatives? [Likely]
    - How does the choice show up in recall@K offline versus in the ranking stage online?
16. How often do you rebuild the item embedding index, and how do new ASINs get into it between rebuilds? [Likely]
17. Which ranking features must be real-time (session events), and how do they reach the online feature store within seconds? [Likely]
    - What happens to ranking quality if the streaming path lags by 10 minutes?
18. What does the re-ranking stage optimize that the ranker cannot? [Likely]
    - How do you trade relevance against diversity and novelty on one page of widgets?
19. How do you evaluate a new ranker offline when your logs were generated by the old ranker? [Likely]
    - When would you use inverse propensity scoring or a replay estimator, and what logging does each require?
20. How do you avoid a feedback loop in which the system only recommends already-popular products? [Likely]
    - Where do exploration slots sit on your diagram, and what do they cost?
21. Design "Frequently bought together". Where does it sit in the diagram: candidate generator, ranker feature, or separate widget? [Likely]
22. Which monitoring signal would tell you the embedding model has gone stale before CTR drops? [Likely]

23. Predict the click probability for music recommendations. [Likely] (see [39]: reported at Amazon in a September 2026 phone screen, role not stated)
    - Which features and labels carry over from the product-widget diagram, and what changes for streaming media (skips, play duration, repeat plays)?
24. Design a "Popular products" or module-ranking layer for the homepage that decides which widgets appear and in what order. [Likely]
    - How do you compare a widget that drives clicks with one that drives purchases on a common scale?
25. When would you use a managed recommender such as Amazon Personalize instead of building the three-stage stack yourself? [Likely] (see [29])
    - What would you lose in control over the re-ranking and exploration stages?
26. The ESCI labels separate exact matches, substitutes and complements. How would you use the substitute-versus-complement distinction differently in a "similar items" widget and in a "frequently bought together" widget? [Likely] (see [21])

Part 3. Product search: retrieval -> ranking

Diagram

```
FRAME: query -> ranked product list | north-star: purchases per search / search conversion
       guardrails: zero-result rate, relevance (human-judged), latency p99 ~200-300 ms (illustrative)

ONLINE
[Query "running shoes men size 10", customer, locale]
    | ~15 ms
    v
[QUERY UNDERSTANDING]  spell-correct, product-type + attribute extraction,
    |                  intent classifier, query embedding
    | ~40 ms
    v
[RETRIEVAL (hybrid, thousands)]   OpenSearch:
    |   lexical BM25 on title/attributes  +  k-NN on bi-encoder embeddings
    |   filters: locale, in-stock, eligibility
    | ~60 ms
    v
[RANKING L1]  GBDT / light neural LTR on query-product features
    | ~50 ms
    v
[RANKING L2]  cross-encoder or distilled transformer on top ~100
    |
    v
[BLENDING]  organic results + sponsored slots + widgets; business rules
    v
[SERP] ---> impressions (with position), clicks, adds, purchases, reformulations ---> Kinesis

OFFLINE
[S3: search logs, catalog, human relevance judgments (ESCI-style labels [21])]
    +--> click model / position-bias correction -> training labels
    +--> train bi-encoder (retrieval), LTR ranker, cross-encoder teacher -> distill
    +--> re-embed catalog -> OpenSearch index refresh; incremental updates for new listings
    +--> offline eval: recall@K, NDCG on judged sets, LLM-judge vs human agreement
    v
[Registry] -> interleaving / A/B (unit: customer or query-session)

MONITOR: zero-result rate, null-click rate, reformulation rate, index freshness lag,
         per-locale metrics -> retrain / re-index triggers
```

Questions

27. You have a large collection of click data for many queries and product documents. Is click data alone sufficient? How would you train a retrieval model with the data at hand? [Reported] [12]
    - How do you correct for position bias in clicks? (see [19])
    - What other biases (presentation, trust, query intent) contaminate click labels?
28. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [13]
    - How would you estimate true relevance from clickstream logs with position bias? [Reported] [13]
    - Pointwise, pairwise or listwise learning-to-rank: which, and why?
    - Design the search retrieval and ranking pipeline including the online experiments and the position-bias correction, with candidate generation drawn explicitly. [Reported] [37][38]
29. Today the Amazon search bar only supports general retail products. How would you design it to also support Amazon Fresh grocery search? [Reported] [40]
    - What changes in retrieval and ranking when availability depends on the delivery zone and time slot?
    - Would you train one ranker across both catalogs or a separate one for Fresh, and how would you decide?
30. Design faceted product search at large scale, with brand, price and attribute filters and accurate facet counts. [Likely] (see [39]: reported at Amazon in an August 2026 onsite, role not stated)
    - Where do the attribute-extraction models that fill the facets sit on your diagram, and how do their errors show up to customers?
31. How would you turn the four ESCI labels (Exact, Substitute, Complement, Irrelevant) into training targets for the ranker? [Likely] (see [21])
    - Should a substitute rank above or below an exact match that is out of stock?
32. Compare bi-encoders and cross-encoders. Where would each sit in a product-search stack? [Reported] [3]
    - What is your latency budget per stage, and how do you distill the cross-encoder?
33. Why hybrid lexical plus vector retrieval rather than vector alone? [Likely]
    - Which queries does each one fail on (part numbers, brand names, vague intent)?
34. How do you keep the OpenSearch index fresh for new listings and for price and stock changes? [Likely]
    - Which updates can wait for a batch rebuild, and which must be incremental?
35. Draw the query-understanding block. How would you launch it in a new marketplace and language with little labeled data? [Likely]
36. How would you detect and reduce zero-result or low-quality-result searches? [Likely]
    - When should you relax the query, and when should you show "did you mean"?
37. How do you combine relevance with conversion, margin and sponsored placement in the blending box? [Likely]
    - How do you stop business boosts from eroding relevance over time?
38. Design an evaluation framework for search relevance using human annotation and LLM judges. [Likely]
    - How do you measure annotator agreement and calibrate an LLM judge against humans?
39. Interleaving or A/B test for a new ranker: when would you use each? [Likely]

40. How would you build semantic search over hundreds of millions of products within the latency budget you drew? [Likely]
    - What ANN index parameters trade recall against latency, and how do you pick them?

Part 4. Sponsored ads: CTR prediction + auction

Diagram

```
FRAME: which ads to show, in which slot, at what price | north-star: long-run advertiser
       + shopper value (revenue subject to relevance/trust); guardrails: shopper CTR on organic,
       ad relevance, advertiser ROAS; latency p99 ~100 ms for the ads call (illustrative)

ONLINE
[Search / page context + shopper]
    | ~10 ms
    v
[AD CANDIDATE RETRIEVAL]  keyword / product targeting match, semantic match (OpenSearch k-NN)
    |                     budget + eligibility filters (budget state in DynamoDB)
    | ~20 ms
    v
[AD TRUST / POLICY FILTER]  creative moderation results (precomputed), brand safety
    | ~30 ms
    v
[pCTR x pCVR MODELS]  calibrated, on SageMaker endpoint; features from online store
    |
    v
[AUCTION]  rank by bid x pCTR (x quality); reserve price; pricing rule (GSP / first-price)
    |       pacing multiplier from budget-pacing service
    v
[Ad slots in SERP / detail page] ---> impressions, clicks, spend events ---> Kinesis
                                                                              |
    +--> real-time spend aggregation -> DynamoDB budget state (pacing)  <----+
    +--> S3 logs -> conversions joined later (attribution window, delayed labels)

OFFLINE
[S3: impressions, clicks, delayed conversions, advertiser metadata, creatives]
    +--> label join with attribution window; delayed-feedback correction
    +--> train pCTR / pCVR (daily or incremental); calibration layer
    +--> offline eval: log-loss, AUC, calibration (predicted vs observed CTR by bucket)
    +--> auction simulator / counterfactual replay for reserve and pricing changes
    v
[Registry] -> A/B (unit: shopper; budget-split or advertiser-side designs for marketplace effects)

MONITOR: calibration drift, spend pacing error, advertiser ROAS, policy-violation rate
         -> recalibrate / retrain / pacing alerts
```

Questions

41. Design a system that recommends trustworthy ads to Amazon users while keeping brand safety and compliance. [Reported] [9]
    - Walk through the full data pipeline from collection to deployment. [Reported] [9]
    - Where does the trust filter sit relative to pCTR scoring, and why there?
    - How do you get labels for rare policy violations?
42. Design click-through-rate prediction for Sponsored Products ads. [Likely] (see [18])
    - Why does calibration matter in the auction box, and how do you measure it?
    - How do you handle delayed conversions in pCVR training?
43. What breaks in your pCTR model when a new advertiser or ASIN has no history? [Likely]

44. Generalized second price or first price: how does the choice change what the model must output? [Likely]
45. Where does budget pacing sit in the diagram, and what happens if the spend stream lags? [Likely]
    - How would you frame pacing as constrained optimization or as a bandit?
46. Why is a plain shopper-level A/B test biased for auction changes, and what design fixes it? [Likely]
47. Revenue is flat, cost-per-click is up and conversions are down after a pacing change. Use your diagram to diagnose it. [Likely]
48. Design an ad-moderation classifier for images and text in creatives. Is it online or offline in your diagram? [Likely] (see [9])
    - How do you handle advertisers who iterate on rejected creatives?
49. How would you measure the incremental lift of an ad campaign for an advertiser? [Likely]
50. How do you set the number of ad slots on a page without hurting organic search quality? [Likely]
51. Design an evaluation framework that measures whether a text-to-image system spells a requested brand name correctly and places it naturally in a generated ad image. [Reported] [37][38]
    - Which checks can be automated, and which need a human review protocol?
    - How do you reconcile automatic scores with human ratings when they disagree?
52. Design an advertiser platform that records impressions, clicks and conversions and returns rates over custom time windows. (MLE loop) [Reported] [38]
    - Which of these aggregates would you also reuse as features for the pCTR model, and how do you keep the two definitions identical?
53. How would you evaluate adding video ads to the search results page? [Likely] (see [39]: reported at Amazon in a November 2025 phone screen, role not stated)
    - Which metrics guard the shopper experience, and how would you design the experiment when advertisers are also affected?

Part 5. Fraud detection with streaming features

Diagram

```
FRAME: approve / challenge / block a transaction or account action | north-star: fraud loss
       prevented at a fixed customer-friction budget; guardrails: false-positive rate on good
       customers, review-queue capacity; latency p99 ~50-100 ms in the checkout path (illustrative)

ONLINE
[Checkout / login event]
    |
    +--> Kinesis Data Streams ---> stream processor (Flink / Lambda)
    |                              velocity counters: orders per card per 1h,
    |                              new-device-per-account per 24h, address changes
    |                              ---> online feature store / DynamoDB (seconds fresh)
    | ~10 ms
    v
[Feature fetch]  streaming counters + batch features (account age, history)
    |            + graph features (shared device / card / address links, precomputed)
    | ~20 ms
    v
[RISK MODEL]  GBDT on SageMaker endpoint (+ rules engine for known patterns)
    |
    v
[DECISION POLICY]  thresholds by cost: approve | step-up auth | hold for review | block
    v
[Outcome] ---> decision + score + features logged ---> Kinesis / S3
                    |
                    v
[HUMAN REVIEW QUEUE]  prioritized by expected loss; investigator labels --> label store

OFFLINE
[S3: transactions, chargebacks (weeks late), investigator labels, account graph]
    +--> label maturity window; handle labels only seen for blocked/reviewed cases
    +--> train risk model (frequent), graph embeddings (batch)
    +--> offline eval: precision/recall at operating point, $ loss curves, by segment
    v
[Registry] -> shadow scoring -> canary -> champion/challenger

MONITOR: score distribution drift, approve/decline rates, chargeback rate (lagging),
         review-queue backlog, new-pattern alerts -> retrain / rule hot-fix triggers
```

Questions

54. Design a fraud detection system for Amazon payments transactions. [Likely] (see [17])
    - How do you deal with extreme class imbalance and delayed chargeback labels?
    - Which of precision, recall and AUC would you report on imbalanced fraud data, and why? [Reported] [37]
    - How do you pick the operating threshold given the costs of false positives and false negatives?
55. Which features have to be streaming, and why can't they be batch? [Likely]
    - How do you compute a "1-hour velocity" counter identically in the stream and in the training data?
56. Given a stream of clicks with an ad ID and a timestamp, return a given ad's clicks in the last k minutes. (MLE loop, phone screen) [Reported] [38]
    - Now move that counter onto your diagram. Where does it run, how does it handle late or duplicate events, and how is the same window rebuilt for training data?
57. Design an end-to-end spam detection system. [Likely] (see [39]: reported at Amazon in a phone screen, role not stated)
    - Which boxes from the fraud diagram carry over unchanged, and which change when the "transaction" is a review, message or listing?
58. Where in your diagram can training/serving skew appear, and how would you detect it? [Likely]

59. You only observe true labels for transactions you approved. How does that bias training, and what would you do about it? [Likely]
    - Would you let a small random slice through for unbiased labels, and how would you justify the cost?
60. What do you do with the uncertain band: friction, step-up auth, or human review? [Likely]
61. How would you prioritize a human-investigator queue when you can review only 1% of flagged cases? [Likely]
    - How do investigator decisions feed back into training without biasing it?
62. How would you add graph signals (shared devices, cards, addresses) within the latency budget? [Likely]
63. How do you handle adversaries adapting to your model, and which monitor catches it first? [Likely]
64. Your fraud model's precision drops sharply over two weeks. Walk me through the investigation on your diagram. [Likely]
65. How would you detect fake reviews or abusive sellers with the same architecture? What changes? [Likely]
66. Design account-takeover detection at login time under a strict latency budget. [Likely]

Part 6. SCOT demand forecasting

Diagram

```
FRAME: how many units of each product to buy, where, and when | north-star: in-stock
       at minimum holding + shortage cost; guardrails: forecast bias, excess inventory;
       batch system: daily/weekly run, SLA in hours, not milliseconds

DATA SOURCES
[Sales history (censored by stockouts)] [Prices, promotions, Prime Day calendar]
[Catalog attributes / embeddings]        [Glance views / search demand signals]
[Holidays, weather, macro]               [Lead times, vendor constraints]
    |
    v
[S3 data lake + Glue] --> [FEATURE PIPELINE (EMR / Spark, SageMaker Processing)]
    |                     demand de-censoring, hierarchy (item -> category -> region)
    v
[TRAINING]  global probabilistic model across millions of series
    |       (quantile / distribution outputs; deep global models, Chronos-style
    |        pretrained models [20]); cold-start from attributes of similar items
    v
[BACKTEST]  rolling-origin evaluation: quantile loss (P50/P90), bias, by segment
    v
[BATCH INFERENCE]  SageMaker batch transform / Processing: quantile forecasts per
    |               item x location x week, horizon of weeks to months
    v
[FORECAST STORE (S3 / DynamoDB for lookup)]
    |
    v
[DOWNSTREAM OPTIMIZATION]  buying (newsvendor-style quantile choice), placement
    |                      across fulfillment centers, transportation
    v
[Purchase orders]  ---> realized sales, stockouts, overrides ---> back to data sources

MONITOR: forecast error vs realized demand by segment, bias drift, in-stock rate,
         holding cost, manual overrides -> retrain / recalibrate triggers
         (A/B is hard: shared warehouses and trucks create interference)
```

Questions

67. How would you forecast demand for millions of products, including new ones with no history? [Likely] (see [32][33][34])
    - Why use probabilistic or quantile forecasts rather than point forecasts for buying decisions?
    - Would you use a global model or per-item models, and why?
68. What is the newsvendor problem, and how would you use it to set inventory levels? [Likely] (see [36]: reported in Amazon supply-chain interviews, not a science loop)
    - How does a quantile forecast map to the newsvendor critical ratio?
69. How would you model an optimization problem from your work, including objective, decision variables and constraints? [Reported] [16]
    - When would you use a heuristic instead of an exact solver?
70. Sales are censored when items are out of stock. Where in your pipeline do you correct for that? [Likely]
71. How would you forecast demand during Prime Day or other promotions, when history is sparse and distorted? [Likely]
72. How do you design the backtest so that it does not leak future information? [Likely]
73. How do forecast errors propagate into placement and transportation costs? [Likely] (see [33])
74. How do you evaluate the forecasting system against in-stock rate and holding cost rather than MAPE alone? [Likely]
    - How would you run an experiment on a buying policy when products share warehouses and trucks?
75. What triggers a retrain in a batch forecasting system, and how do you guard against a bad model reaching purchase orders? [Likely]
76. Buyers override forecasts by hand. How do you use the overrides, and how do you measure whether they help? [Likely]
77. Design a system that decides how much of each product to buy from which vendor and when. [Likely] (see [34])
78. Chronos-style pretrained models forecast zero-shot, and newer versions accept covariates. When would you use one as the cold-start or fallback forecaster in your diagram, and when would you train a global model on Amazon data instead? [Likely] (see [20])
79. Design an end-to-end regression system for demand in a domain with strong weather and calendar effects (energy demand was the reported example). Then map it onto the SCOT diagram. [Likely] (see [39]: reported at Amazon in an onsite, role not stated)
80. Build a package-allocation model for couriers, then prove that the new allocation beats the current manual baseline. [Likely] (see [39]: both reported at Amazon in onsites, role not stated)
    - Why is a simple A/B split of packages biased here, and which design (switchback, geographic or time clusters) would you use?
81. Inventory counts reach the forecasting system from a distribution-center inventory update service. What happens to your forecasts and buying decisions when those updates arrive late or are duplicated? [Likely] (see [39])

Part 7. Rufus / Alexa-style RAG assistant

Diagram

```
FRAME: answer shopping questions grounded in catalog, reviews, Q&A | north-star: task
       success / downstream purchase without regret (returns); guardrails: factuality,
       safety, latency (time-to-first-token ~1 s, illustrative), cost per query

ONLINE
[User turn "What do I need for cold weather golf?" + conversation state + customer context]
    | ~50 ms
    v
[INTENT / ROUTER]  small classifier or LLM: product search | product Q&A | comparison |
    |              order status (tool) | out-of-scope
    | ~100-200 ms
    v
[RETRIEVAL]  query rewrite -> hybrid search in OpenSearch (BM25 + k-NN embeddings)
    |        over catalog attributes, review snippets, Q&A; product search API as a tool
    |        re-rank top-k with cross-encoder
    | ~500 ms+ (streaming)
    v
[GENERATION]  LLM via Amazon Bedrock or a self-hosted fine-tuned model on SageMaker;
    |         prompt with retrieved evidence + citations to ASINs
    v
[GUARDRAILS]  grounding check vs evidence, safety / policy filter (Bedrock Guardrails),
    |         price / availability facts taken from live systems, not the model
    v
[Streamed answer + product cards] ---> thumbs, clicks, adds, purchases, follow-up turns
                                        ---> logs (with retrieved doc IDs + prompt version)

OFFLINE
[Catalog + reviews + Q&A] --> chunk + embed --> OpenSearch vector index (incremental refresh)
[Logs + human annotations + LLM-judge labels]
    +--> eval sets: retrieval recall, answer faithfulness, helpfulness, safety red-team
    +--> fine-tuning (SFT, preference tuning) / distillation to a smaller serving model
    v
[Registry: model + prompt + retriever versions] -> offline eval gate -> A/B

MONITOR: hallucination rate (sampled judge), refusal rate, latency / cost per turn,
         retrieval hit rate, safety incidents -> prompt fix / re-index / retrain triggers
```

Questions

82. Design an enterprise RAG service. [Reported] [14]
    - How do you chunk, embed and index heterogeneous documents?
    - How do you enforce document-level access control in retrieval?
    - How do you evaluate retrieval quality separately from generation quality?
83. Evaluate an LLM agent with metrics beyond accuracy, such as tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you build an offline evaluation set for multi-step agent tasks?
    - When would you trust an LLM-as-judge?
84. Pick an agentic product we could build. Who is the customer, and how do you measure success? [Reported] [2]
    - Propose agentic AI solutions for a hypothetical project and defend their viability. [Reported] [37]
85. Architect an AI chatbot, covering intent detection, dialogue state, retrieval from a knowledge base, generation, fallback behaviour, evaluation and continuous learning. [Reported] [37][38]
    - Where in the Rufus diagram does dialogue state live, and what does the router do when it is uncertain?
    - What is the fallback when retrieval returns nothing relevant: refuse, ask a clarifying question, or hand off?

86. Design a multi-turn search agent, and explain how it handles long conversation context. [Reported] [37][38]
    - What do you keep verbatim, what do you summarize, and what do you re-retrieve on each turn?
    - How would you evaluate a whole multi-turn session rather than single answers?
87. Design an LLM quality validation system that checks assistant answers before and after launch. [Likely] (see [39]: reported at Amazon in an onsite, role not stated)
88. How would you measure product-market fit for a voice or chat shopping assistant such as Alexa Shopping? [Likely] (see [39]: reported at Amazon in an onsite, role not stated)
    - Which behaviour separates retained, satisfied users from novelty use?
89. Bedrock Knowledge Bases or your own chunking, embedding and OpenSearch retrieval pipeline: what would push you to build rather than reuse? [Likely] (see [27] [28])
90. Design a distributed pipeline that ingests, tokenizes, deduplicates and filters the training data for the assistant's fine-tuning. (staff-level loop; above the L5 bar, so treat it as a stretch) [Reported] [37][38]
91. On an LLM team, discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] [22]
    - Greedy, beam or sampling decoding for a customer-facing assistant: which, and why?
92. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] [10]
    - Where would you get diverse accent training data? [Reported] [10]
    - Where does ASR sit in front of the RAG diagram, and how do its errors propagate into retrieval?
93. Design Rufus: answer "What do I need for cold weather golf?" from the catalog, reviews and Q&A. [Likely] (see [35][30][31])
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you mix organic and sponsored products in a generative answer?
94. Why does the router exist? What would break if every turn went to the large model? [Likely]
95. Which facts must come from live systems (price, stock, delivery date) rather than from the index or the model? [Likely]
96. How do you keep the vector index fresh as reviews and listings change? [Likely]
97. Bedrock-hosted model or a fine-tuned model on SageMaker: how do you decide? [Likely]
    - How would you serve the assistant to millions of users at acceptable cost: distillation, quantization, caching, speculative decoding, or model routing?
98. What is your time-to-first-token budget, and which stage consumes most of it? [Likely]
99. How would you summarize thousands of reviews into a short "Customers say" blurb, and where does that job sit in the diagram (online or offline)? [Likely]
    - How do you stop manipulated reviews from steering the summary?
100. How would you detect and mitigate unsafe, biased or policy-violating responses? [Likely]
    - Which red-teaming process and which guardrail models would you use?
101. What does your feedback loop log, and how do you turn thumbs-down and follow-up turns into training data? [Likely]

102. How do you version prompts, retrievers and models together so a regression can be rolled back? [Likely]

Part 8. Cross-cutting probes on any diagram (feedback, monitoring, retraining)

103. How do you monitor this model in production, and how do you detect data or concept drift? [Likely]
    - Where on the diagram does the retraining trigger fire, and what threshold fires it?
    - How do you validate the retrained model before rollout?
104. How would you roll back safely if the new model misbehaves on a subset of traffic? [Likely]
105. Your labels arrive weeks late (returns, chargebacks, retention). How do you train and evaluate under label delay? [Likely]
106. How would you design an A/B test for a model launch? [Reported] [4]
    - What do you do if the test is underpowered or the metric is heavy-tailed?
107. Your model's online A/B result fell short of the offline gain. How do you find out why? [Reported] [37]
    - Which arrows on your diagram would you inspect first?
108. How would you cut the cost of this system by 50% with minimal quality loss? [Likely] (Frugality)
109. Where does personally identifiable information flow in your diagram, and how do you minimize it? [Likely]
110. Design a causal study to measure the impact of a customer reminder or notification that your model triggers, when you cannot randomize everyone. [Likely] (see [39]: reported at Amazon in a phone screen, role not stated)
111. Which monitoring alarm would page you at 3 a.m., and which would only go on a weekly dashboard? [Likely]

Part 9. L5-bar probes inside the design round

112. Of everything on your diagram, what is the single riskiest assumption, and how would you de-risk it in two weeks? [Likely]
113. A product manager wants to launch in a month and your offline evaluation is incomplete. What do you do? [Likely] (Bias for Action versus Insist on the Highest Standards)
114. Your design improves conversion but increases returns. How do you decide whether to ship? [Likely]
115. Which metric would you refuse to optimize directly, and why? [Likely]
116. If you had 10x the data, or one-tenth of it, which boxes on the diagram would change? [Likely]
117. How would you explain this diagram to a non-scientist stakeholder in two minutes? [Likely] (Earn Trust)

118. How would you hand this system over to an engineering team for long-term ownership? [Likely]

References

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2. Amazon Applied Scientist II Interview Experience & Questions, Glassdoor, 2025-2026 (includes the "1 Sys Design (Team-specific)" loop report and the agentic-product prompt). https://www.glassdoor.com/Interview/Amazon-Applied-Scientist-II-Interview-Questions-EI_IE6036.0,6_KO7,27.htm
3. Amazon Applied Scientist Interview Experience & Questions, Glassdoor, 2025-2026 (LLM agent evaluation, bi-encoder vs cross-encoder). https://www.glassdoor.com/Interview/Amazon-Applied-Scientist-Interview-Questions-EI_IE6036.0,6_KO7,24.htm
4. Amazon applied scientist interview-onsite, Blind, circa 2024. https://www.teamblind.com/post/amazon-applied-scientist-interview-onsite-t01hpnc4
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13. "What are some Amazon Research Scientist interview questions?", Quora. https://www.quora.com/What-are-some-Amazon-Research-Scientist-interview-questions
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16. Amazon Applied Scientist Interview Debrief: Five-Round Loop + LP, OA VO Service (vendor-published candidate debrief; promotional). https://oavoservice.com/en/articles/amazon-applied-scientist-vo-loop-lp-ml-coding
17. Applied Scientist II, Amazon Payment Products (L5) (Job ID 3168758), amazon.jobs (posting may have expired). https://www.amazon.jobs/en/jobs/3168758/applied-scientist-ii-amazon-payment-products-l5
18. "Applied scientist interview at Amazon Ads", Blind. https://www.teamblind.com/post/applied-scientist-interview-at-amazon-ads-zoerec5t
19. "Intent based relevance estimation from click logs", Amazon Science. https://www.amazon.science/publications/intent-based-relevance-estimation-from-click-logs
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22. Amazon interview review RVW96068026 (Applied Scientist II, LLM team, Santa Clara), Glassdoor, 2025. https://www.glassdoor.com.hk/Interview/Amazon-Interview-E6036-RVW96068026.htm

AWS components (official product and documentation pages, cited only to name components; not interview reports; not re-fetched in this run):

23. Amazon SageMaker (training, endpoints, Pipelines, Model Registry, Model Monitor), AWS. https://aws.amazon.com/sagemaker/
24. Amazon SageMaker Feature Store, AWS documentation. https://docs.aws.amazon.com/sagemaker/latest/dg/feature-store.html
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Product, science and team context (basis for [Likely] inferences):

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Added in the 2026-10-14 review (each read in full from GitHub):

37. "Amazon Research Scientist Interview Process", TrueInterview, mirrored in kevin-2023-code/Tech-Interview-Questions on GitHub (CC BY 4.0). Built from 20 candidate-reported experiences, mostly 2026 Q1-Q3, which the page says come mostly from Applied Scientist loops. Covers science application prompts (agentic proposals, multi-turn search agent, AI chatbot, brand-name image evaluation, ranking pipelines), the online-versus-offline A/B probe, and fraud metrics in ads phone screens. https://github.com/kevin-2023-code/Tech-Interview-Questions/blob/main/companies/amazon/interview-process-research-scientist.md
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40. goldin2008/Note, "interview_ds.md", GitHub. A compiled set of candidate posts, mostly in Chinese and undated. Its "Amazon AS" section includes the Amazon Fresh search application question and an AWS AI Applied Scientist "Machine Learning Design" round report. https://github.com/goldin2008/Note/blob/HEAD/interview_ds.md

13. ML System Design: Product Tagging, Catalog Attribute Extraction and PDF/Document Understanding

Scope: design-round questions on two systems the rest of this pack does not cover in depth. (1) Product tagging and catalog attribute extraction. (2) PDF and document understanding pipelines (OCR and Textract-style layout parsing, key-value and table extraction, classification and routing, human review). The focus is the "Science Application / ML System Design (team-specific)" round of the Amazon L5 Applied Scientist (Applied Scientist II) loop, 2025-2026.

Research note (please read): no new web research was possible for the first draft of this chapter. The run's shared web-search budget was already used up before it started. Every [Reported] tag in that draft came from sourced findings already gathered and cited in chapters 01-11 of this guide. A question is tagged [Reported] only where a source attributes it, or a close paraphrase, to an Amazon science or ML interview. No candidate report found so far names a "PDF pipeline" or "Textract" design prompt verbatim, so every document-understanding question is [Likely]. Those questions are grounded in Amazon's products (Textract, Augmented AI/A2I, Comprehend, Bedrock, the product catalog) and in the round's reported format.

Review pass (2026-10-14): web search was again unavailable because the shared budget was used up, so no new Blind, Glassdoor or 1point3acres reports could be found. Fourteen cited or candidate URLs were fetched successfully, all on github.com: refs [6], [7], [9]-[12] and the new refs [15]-[22]. Each one supported the claims made for it. Refs [1]-[5], [8], [13] and [14] could not be re-fetched: amazon.jobs, igotanoffer.com, leetcode.com, medium.com, teamblind.com, amazon.science and enginebogie.com failed DNS resolution in this environment. A refused connection is not evidence that a page is dead, so those references are kept, with this caveat. Question 1 cited only a prep-vendor description of the round, so it was retagged [Likely]. The 26 new questions come from a dated Amazon question aggregator ([15], 2025-2026 entries), the Amazon MLE aggregate [7], and public AWS and Amazon Science repositories [16]-[22]. Where a reported prompt's role or round is not stated, the item says so. A later session with a search budget should look for first-person reports using the terms "attribute extraction", "catalog quality", "Textract" and "IDP".

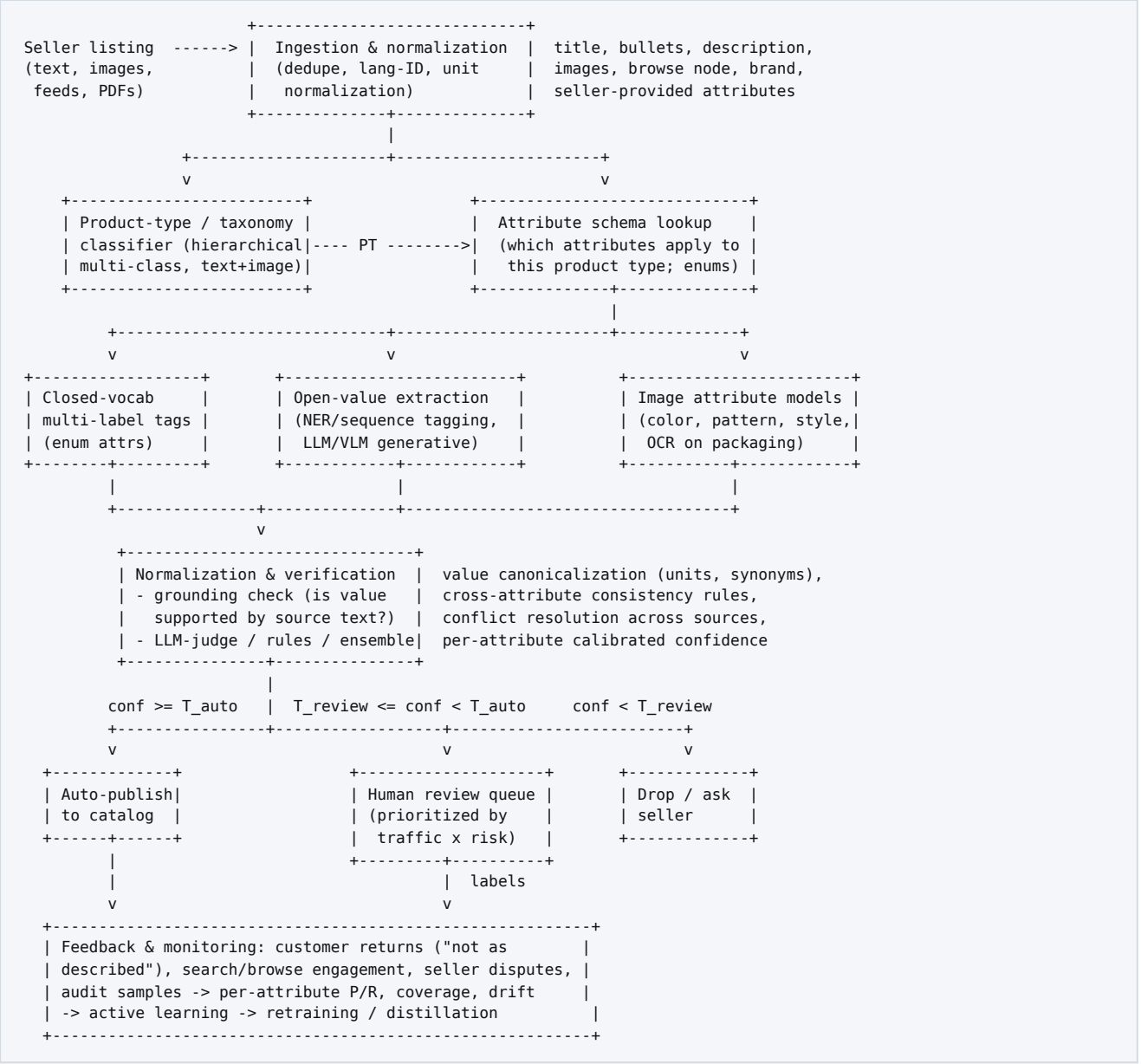
Additions beyond the questions-only format: the user asked for "ML system design diagrams" and "details about the tag/PDF system". For that reason this chapter includes two whiteboard reference diagrams and a checklist of the components, metrics and failure modes interviewers probe. These are scaffolds for drawing your own design. They are not model answers.

Format notes for this round

- **Where it sits.** L5 AS loops have about 4-6 interviews of roughly 55-60 minutes. One 2025-2026 Applied Scientist II loop was reported as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Amazon's official prep page describes about four 55-minute interviews with members of the science community. Each interviewer is assigned Leadership Principle competencies and asks 2-3 behavioral questions [1].
- **Interviewer.** Usually the hiring manager or a senior scientist on the hiring team. The prompt comes from the team's domain [2]. A tagging or document prompt is most likely on catalog, Selling Partner, Product Graph, AWS AI services (Textract, Comprehend, Bedrock), Finance or Payments, and Compliance teams. That part is an inference.
- **What is assessed.** IGotAnOffer describes the round as being given a high-level ML problem and proposing data sources, annotation strategy, modeling technique, evaluation metrics, and trade-offs and risks [3]. Candidates report being drilled on the data pipeline "from collection to deployment" [8], and on building with very little labeled data [14].
- **L5 bar.** Blind commenters say L5 loops usually include a design or application round and L4 loops may not [4]. At L5 you are expected to own the problem end to end: frame a vague problem, design the labels and taxonomy, choose models, set per-attribute or per-field metrics and thresholds, plan human review, launch, and monitor. You should also make the cost-accuracy trade-off calls yourself.
- **Closest reported prompts.** "Categorize 10k product listings across 100 categories, each with a title, description and photo" [5]. "Design an evaluation framework that checks whether a text-to-image system spells a brand name correctly, combining OCR accuracy, embedding checks and a human review protocol" [6][7]. "Design an LLM quality validation system" [6][15]. A dated Amazon aggregator also lists "Design Faceted Product Search at Large Scale" and "Design a Product Image Upload Pipeline for Sellers" (both August 2026, role not stated) and "Design an end-to-end spam detection system" (undated) [15]. These are the nearest catalog-side prompts on record.

Tag legend: **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science or ML interview. **[Likely]** means a grounded inference from the role, team and Amazon products, not seen verbatim.

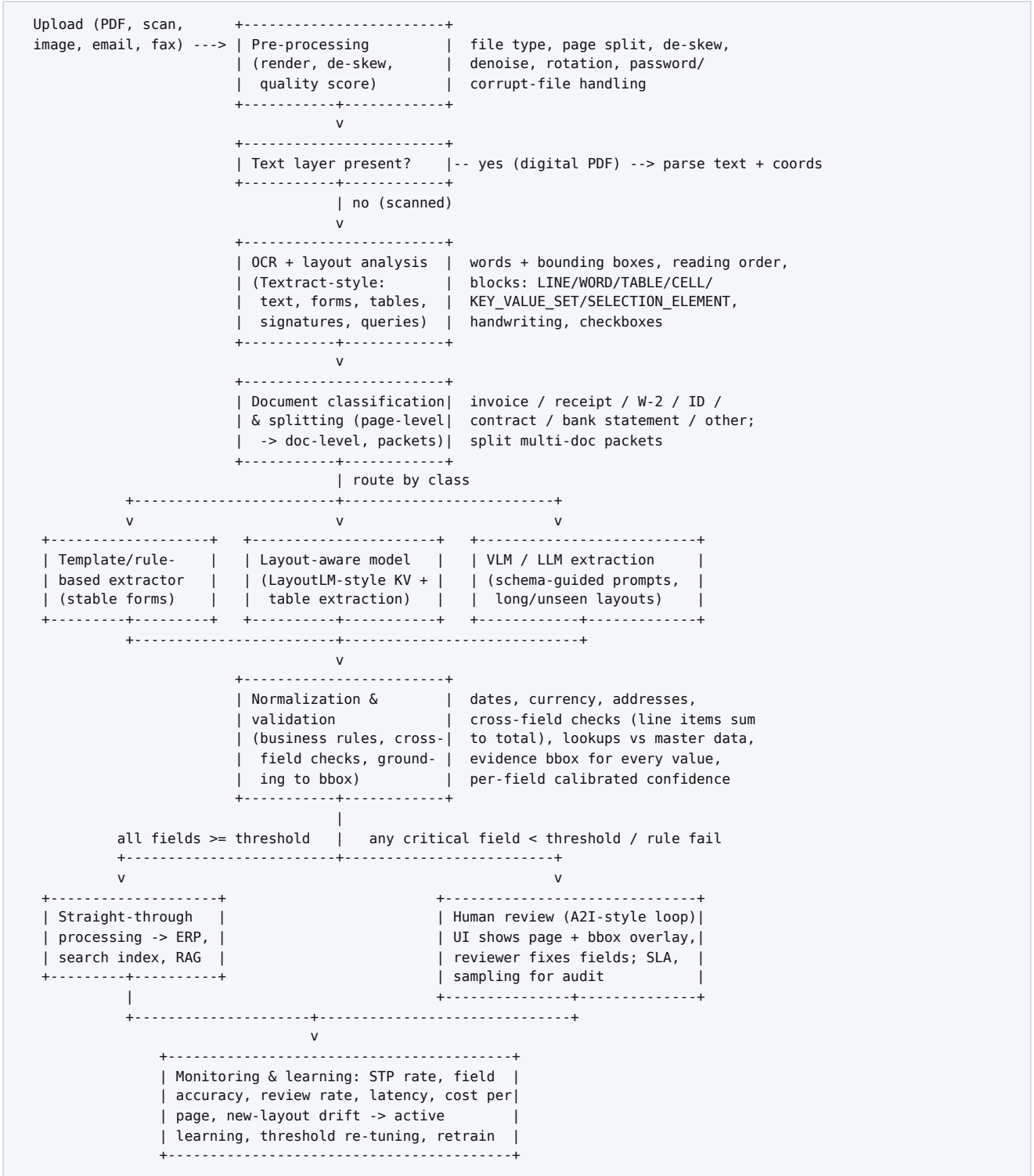
Reference diagram 1: Product tagging and attribute extraction (whiteboard scaffold)



Details interviewers usually expect you to cover, as a checklist rather than an answer:

- **Problem shape.** Product type is a hierarchical, single-path classification over a large, changing taxonomy. Tags are multi-label. Attribute values are either closed-vocabulary (enum) or open (free text, numeric with units).
- **Metrics.** Per-attribute precision, recall and F1. Coverage (fill rate) per product type. Precision at the auto-publish threshold. Traffic-weighted accuracy. Human review rate and cost per item. Downstream metrics: search relevance and conversion, "item not as described" returns.
- **Typical failure modes.** Hallucinated values. Values taken from the wrong product in a bundle or "compatible with" text. Unit and format errors. Taxonomy drift. Long-tail product types with no labels. Seller-provided values that are wrong or adversarial. Miscalibrated confidence on new categories.

Reference diagram 2: PDF and document understanding pipeline (whiteboard scaffold)



Details interviewers usually expect you to cover, as a checklist rather than an answer:

- **Problem shape.** OCR and layout, then classification and routing, then field, key-value and table extraction, then validation, then human-in-the-loop review. The pipeline is cascaded, so errors compound from stage to stage.
- **Metrics.** Character and word error rate (CER/WER) for OCR. Document classification accuracy and confusion. Per-field precision, recall and exact match. Table cell and structure accuracy (for example TEDS-style metrics). Straight-through-processing (STP) rate. Human review rate. Accuracy after review. Latency and cost per page.

- **Typical failure modes.** Low-quality scans, rotated or multi-column pages. Tables that span pages. Handwriting. Lookalike fields (ship-to versus bill-to). Multi-document packets. Unseen layouts. VLM hallucination of missing fields. Overconfident scores. PII exposure in logs and to reviewers. Reviewer fatigue and automation bias.

A. Framing and scoping (both systems)

1. Here is a high-level ML problem from our team (for example, filling missing product attributes, or extracting fields from supplier invoices). Propose a solution covering data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Likely] (round format as described by a prep vendor [3]; the same wording is cited as a candidate-reported prompt in chapters 03, 06 and 11)
    - Which parts need ML, and which can be rules, templates or seller-facing forms?
    - What is the v1 you would ship in one quarter?
2. Walk me through the full data pipeline for your design, from collection to deployment. [Reported] [8]
    - Where do labels come from, and how noisy are they?
    - How do you version data, taxonomy and models together?
3. How would you build this with very little labeled data? [Reported] [14]
    - Where would weak supervision, distant supervision from existing catalog values, or LLM pseudo-labels come in?
4. Who is the customer of the tagging system: shoppers, search ranking, sellers, or internal compliance? How does the answer change your metric? [Likely]
    - Which single metric would you report to leadership, and which guardrail metrics would sit next to it?
5. What scale are we designing for (items per day, pages per day, number of attributes and document types), and how do the answer and the cost budget change your architecture? [Likely]
    - Batch or real-time? What is the latency SLA at listing creation compared with backfill?
6. How would you write the 6-page or PR-FAQ document that proposes this system? [Likely]
    - What would the "customer FAQ" say about wrong tags or wrong extracted fields?
7. How would you decide that an LLM-based attribute or document extractor is ready to launch to millions of customers? [Likely] (adapted from a launch-gating prompt in a synthesized Amazon question bank [16])

    - Which per-attribute or per-field gates must pass, and on which slices?
    - What does shadow mode tell you that an offline gold set cannot?

8. Your extraction model performs worse in production than offline. Diagnose and fix it. [Reported for the general prompt "Diagnose and fix underperforming ML model" (Amazon MLE aggregate, project deep dive) [7]]; extraction application [Likely]

    - Check first for train-serve skew: a new OCR engine version, different HTML cleaning, or a different product-type mix.
    - How would you tell label noise in the gold set apart from real model regressions?

B. Product taxonomy and product-type classification

9. You have 10k product listings across 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [5]
    - How do you fuse text and image signals?
    - How do you handle a long tail of categories with very few examples?
    - How do you evaluate when the category taxonomy itself is noisy?
10. Scale that to a taxonomy of tens of thousands of nodes and hundreds of millions of products. What changes? [Likely]

- Flat classifier, top-down hierarchical, or retrieval over node embeddings? Why?
- How do you enforce hierarchy consistency in predictions?

11. The taxonomy changes: nodes are split, merged and added every month. How do you keep models and labels valid? [Likely]

- How do you remap historical labels without full relabeling?
- How would you detect that a node should be split?

12. How would you evaluate a zero-shot product-to-taxonomy matcher when you only have a few hundred human labels? [Likely] (CaML, Amazon Science [11])

    - How would you compute confidence intervals per node with so few labels?

13. Is a product-type error near the root (electronics versus apparel) worse than one near a leaf? How would you build that into the loss and the metric? [Likely]

14. How would you use an LLM to categorize millions of products into a large taxonomy, combining LLM labeling, a distilled classifier and human audit? [Likely] (LLM distillation work at Amazon Science [13])

    - When is the LLM in the serving path, and when only in the labeling path?
    - What is the cost per million items of each option?

15. Map every catalog product to an external standard taxonomy (for example NAICS industry codes, used to look up emission factors for carbon footprinting) with no training pairs. [Likely] (CaML, Amazon Science [11])

- Zero-shot semantic similarity with top-k candidates and human verification: where does each fit?
- How does a mapping error propagate into the downstream estimate, and which errors matter most?

16. For a search query, classify each retrieved product as Exact, Substitute, Complement or Irrelevant. How would better catalog attributes change this classifier, and how would you measure the effect? [Likely] (Shopping Queries / ESCI dataset, Amazon Science [18])

- Which attribute mismatches (size, compatibility, brand) turn an Exact into a Substitute?

C. Multi-label tagging and attribute schema

17. Design a multi-label tagging system for product listings (for example "waterproof", "vegan", "gift idea", "for kids"). [Likely]

    - Sigmoid per tag or a label-set model? How do you set per-tag thresholds?
    - How do you handle label co-occurrence and mutually exclusive tags?

18. How do you decide which attributes apply to which product type, and who owns that schema? [Likely]

19. Closed-vocabulary attributes (color family, material) compared with open values (model number, wattage): how does your modeling differ? [Likely]

    - When would you turn an open attribute into an enum?

20. How would you handle extreme label imbalance across thousands of tags? [Likely]

    - Is macro or micro averaging the right summary, and for whom?

21. Missing tags are not negatives. How does partial labeling (positive-unlabeled data) affect training and evaluation? [Likely]

    - How do you estimate true recall when most items have never been checked for a tag?

22. How do you canonicalize extracted values ("Navy", "navy blue", "dark blue"; "1.5 L" compared with "1500 ml")? [Likely]

    - Is normalization a model, a dictionary, or both?

23. Two attributes conflict (size "XL" with a chest measurement of 34 inches). How does your system detect and resolve cross-attribute inconsistency? [Likely]

24. Design faceted product search at large scale. [Reported] (Amazon aggregate, August 2026, role and round not stated) [15]

- ML angle [Likely]: where do facet values come from when sellers leave attributes empty, and what extraction coverage does a facet need before you show it?

- How do extraction precision and coverage show up as filter recall for shoppers?
- Which facets would you show for a given query, and how would you learn that?

25. A shopper filters by "Material: cotton" and many cotton items are missing from the results. Is that a precision or a recall problem in your attribute system, and how would you measure it from logs alone? [Likely]

D. Attribute value extraction (classical and LLM)

26. Design a system to extract attribute values (brand, color, size, material, wattage) from product titles and descriptions. [Likely] (attribute extraction at catalog scale also appears in chapters 06 and 11)
    - Sequence tagging (BIO), QA-style span extraction, or generative extraction? What drives the choice?
    - How do you scale to thousands of attributes without one model per attribute?
27. How would you detect and fix wrong attribute values (size, color, material) on listings at catalog scale? [Likely]
    - How would you use an LLM for attribute extraction, and how do you verify its outputs?
28. How would you extract product attributes from noisy seller descriptions without inventing facts that are not in the text? [Likely] (WebIE, faithful extraction, Amazon Science [10])
    - How do you check that each extracted value is grounded in a span of the source?
    - What is your policy when the value is implied but not stated?
29. Use an LLM to generate or enrich product listings (titles, bullet points, attributes) from seller input. How do you prevent made-up attributes? [Likely]
30. Text says "compatible with iPhone 15", but the product is a case, not a phone. How do you stop the model from assigning the wrong brand or model? [Likely]
31. A bundle listing contains three products. How do you attach attributes to the right component? [Likely]
32. Which attributes would you extract from images instead of text (color, pattern, neckline), and how do you resolve text-image disagreement? [Likely]
    - When would you OCR the packaging in product images?
33. How would you serve an LLM extractor over hundreds of millions of items within budget? [Likely]
    - Batching, caching, distillation into a small model, or extracting only for high-traffic items?
    - How do you decide which items to re-extract when the model improves?

34. How do you design prompts and output schemas for structured LLM extraction (JSON schema, enums, "unknown" option)? [Likely]
    - How do you measure and reduce schema-invalid outputs?
35. How would you get calibrated confidence from an LLM extractor? [Likely]
    - Token log-probabilities, self-consistency across samples, verifier models, or agreement with a classical model?
36. Attribute evidence is spread across the title, description, bullet features and images, and the sources sometimes disagree. How do you combine them, and which source wins a conflict? [Likely] (multi-source attribute value extraction, MAVE dataset built from Amazon product profiles [19])
37. Your extraction benchmark spans more than a thousand product categories. How would you report results so that a few large categories don't hide failures on the long tail? [Likely] [19]
38. Design a product image upload pipeline for sellers. [Reported] (Amazon aggregate, August 2026, role and round not stated) [15]

- ML angle [Likely]: which image-quality and policy checks run at upload, and which attributes would you extract from the images at the same time?
- How would you detect an image that does not match the listing's title or product type?

E. LLM verification and LLM-as-labeler

39. Label (attribute, value) pairs as Correct, Incorrect or Unknown using a majority vote over many LLM-judge configurations (several model families and prompts). When would you trust such an "LLM arena" as a labeler? [Likely] (SynthAVE, Amazon Science [9])
    - How would you measure judge agreement with humans, and how many human labels do you need?
    - What correlated errors survive majority voting?
40. Design an LLM quality validation system for extracted attributes. [Reported for the general prompt "Design an LLM quality validation system"] [6]; attribute application [Likely]
    - Which checks are deterministic (schema, units, ranges), and which are model-based?
41. Generator and verifier are the same model family. What goes wrong, and how would you fix it? [Likely]
42. How would you use LLM-generated synthetic listings to train a small extractor, and how do you avoid training on the LLM's own biases? [Likely]

43. When should the verifier's "Unknown" send an item to humans, and when should it leave the attribute empty? [Likely]

44. Your LLM-arena majority vote disagrees with the label produced by the synthetic-data pipeline. Who breaks the tie, and how do you size the human adjudication budget? [Likely] (SynthAVE, Amazon Science [9])

45. A stratified audit of a few hundred LLM-arena labels shows high agreement with an expert (Cohen's kappa around 0.9). Is that enough to release the labels as a benchmark or training set? What would you still check? [Likely] [9]

- How large must the audit be to bound the label error rate per language or per attribute?

46. One locale's labels were judged by a different judge-model version than the others. How does that affect cross-locale comparisons, and what would you do about it? [Likely] [9]

F. Cold-start tagging

47. A new product type launches with zero labeled examples. How do you tag and extract attributes on day one? [Likely]

    - Zero-shot LLM, nearest-neighbor transfer from similar product types, or seller-provided values?

48. A new marketplace and language launches (for example a new EU locale). How do you reuse models trained on English data? [Likely]

    - Translate-train, translate-test, or multilingual encoders? How do you handle locale-specific units and sizes?

49. A brand-new attribute is requested by the search team ("pet-friendly"). How would you backfill it across the catalog? [Likely]

    - How do you estimate precision before a full human audit?

50. A new seller with no history uploads 50,000 listings. How does your system treat their provided attributes compared with established sellers? [Likely]

    - How do you avoid penalizing honest new sellers?

51. With a fixed annotation budget, how would you choose which examples to label to estimate a new tagger's accuracy? [Likely] (active testing, Amazon Science [12])

52. You are estimating a new tagger's accuracy with active testing. When would you stop labeling with an adaptive stopping rule instead of fixing the budget up front? [Likely] (active testing, Amazon Science [12])

- How do you report the uncertainty of an estimate built from non-uniformly selected samples?

G. Human-in-the-loop labeling and review queues

53. Design the human-in-the-loop component: who labels, what they see, and how labels flow back to training. [Likely]
    - How do you measure inter-annotator agreement and resolve disagreement?
    - How do you write labeling guidelines for subjective tags ("casual", "luxury")?
54. You can afford 5,000 human labels per month. How do you spend them between training data, evaluation audits and production review? [Likely]
55. How do you prioritize the review queue? [Likely]
    - Uncertainty, traffic, risk of harm (safety and compliance attributes), or revenue impact?
56. Reviewers accept model suggestions 98% of the time. Is that good news? [Likely]
    - How would you detect automation bias, and would you hide suggestions for a sample?
57. How do you use review outcomes for active learning without biasing your evaluation set? [Likely]
    - Why do you need a random audit sample separate from the uncertainty-sampled queue?
58. How do you set and maintain SLAs for review when volume spikes (Prime Day onboarding)? [Likely]
    - What degrades gracefully: thresholds, coverage, or latency?
59. One managed human-review service is shared by text classification, document extraction and image moderation. What should be shared across tasks (workforce, UI framework, audit sampling), and what must be specific to each task? [Likely] (Amazon Augmented AI sample workflows for Textract, Comprehend custom classification, Rekognition and custom SageMaker models [21])

- How would reviewer corrections on a custom classification endpoint flow back into retraining?

H. Metrics and evaluation for tagging

60. Which offline metrics would you use for attribute extraction, and why per attribute rather than a global F1? [Likely]
    - How would you weight attributes by business importance?
61. Define coverage and precision for a catalog attribute. How do you trade one against the other when setting the auto-publish threshold? [Likely]
    - Draw the precision-coverage curve. Where would you operate for "brand" compared with "color"?

62. How do you build a gold evaluation set that is representative of traffic and of the long tail? [Likely]
    - Stratified by product type, traffic tier and source?
63. How would you show that better attributes caused more sales or fewer returns? [Likely]
    - How would you design the A/B test when the treatment is catalog data that every surface reads?
    - What about interference between treatment and control through shared search ranking?
64. Which online signals indicate an attribute is wrong (filter abandonment, "not as described" returns, customer Q&A), and how noisy are they? [Likely]

I. Document ingestion, OCR and layout

65. Design a system that ingests millions of PDFs per day (invoices, receipts, contracts, forms) and extracts structured data. [Likely] (Amazon Textract product context)
    - Which steps are synchronous and which are asynchronous?
    - How do you scale per page compared with per document?
66. How do you decide whether to OCR a PDF or use its embedded text layer? [Likely]
    - How do you detect a "fake" text layer, such as a bad OCR from a scanner?
67. Walk me through layout analysis: words and bounding boxes, reading order, blocks, tables and key-value pairs. [Likely]
    - How do you get reading order right on multi-column pages and pages with sidebars?
68. How would you handle handwriting, stamps, signatures and checkboxes? [Likely]
69. How do you measure OCR quality (CER/WER) without ground-truth transcriptions for production documents? [Likely]
    - Which image-quality signals predict OCR failure before you run it?
70. Design an evaluation framework that checks whether a text-to-image system spells a requested brand name correctly and places it naturally. [Reported] [6][7]
    - How would you combine OCR accuracy, embedding-based semantic checks and a human review protocol? [Reported] [6]
    - Where does OCR itself mislead your evaluation (stylized fonts, curved text)?
71. Design receipt and expense extraction (vendor, date, total, tax, line items) for an expense-reporting product. [Likely] (Textract AnalyzeExpense summary fields and line items [20])

- How do you evaluate summary fields separately from line items?
- How do you handle crumpled phone photos of receipts, or foreign currencies?

72. Design identity-document extraction (driver's licenses, passports) for an identity-verification flow. What accuracy, latency and fraud constraints change your design? [Likely] (Textract AnalyzeID [20])

- Which errors are worse here: a false reject or a false accept?

J. Key-value, table and field extraction

73. Design key-value extraction for forms (invoice number, date, total, vendor). [Likely]
    - Template-based, layout-aware model (text + position + image), or VLM? When would you use each?
74. How would you extract line items from invoice tables whose layout differs across thousands of vendors? [Likely]
    - How do you handle tables spanning pages, merged cells and tables without borders?
    - What metric would you use for table structure compared with cell content?
75. How do you tell lookalike fields apart (bill-to compared with ship-to, invoice date compared with due date)? [Likely]
76. How would you attach evidence (page and bounding box) to every extracted value, and why does it matter for review and audit? [Likely]
77. What cross-field validation rules would you add (line items sum to subtotal, tax rate in plausible range, dates ordered), and how do rules interact with model confidence? [Likely]
78. How would you add "queries" support, where a customer asks a natural-language question of a document ("What is the policy end date?")? [Likely]
    - How do you evaluate answers that are absent from the document?
79. Extracted tables must export cleanly to spreadsheets and data frames for downstream users. How would you evaluate table output for usability, not only cell accuracy? [Likely] [20]

- Header detection, merged cells, numeric typing: which of these break downstream analysis most often?

K. Document classification, splitting and routing

80. Design document classification for an intake pipeline (invoice, receipt, W-2, ID, bank statement, contract, other). [Likely]
    - Text only, image only, or multimodal? How do you treat "other"?
81. A single uploaded PDF contains several documents (a mortgage packet). How do you split it into documents? [Likely]

    - Page-pair boundary classification or sequence labeling?

82. How do you route documents to different extractors, and what happens when the classifier is wrong? [Likely]
    - How do you limit the damage of a misrouted document downstream?
83. A new document type appears in traffic that the classifier has never seen. How do you detect it (open-set or out-of-distribution detection)? [Likely]
84. Documents arrive in 20 languages. How do classification and extraction change? [Likely]
85. Lay out an intelligent document processing pipeline in five stages: classification, extraction, enrichment, validation and human review. What does enrichment add, and why does validation come before human review? [Likely] (AWS intelligent document processing guidance [22])
86. Design an agentic document-processing system in which an LLM agent decides, per document, which tools to call (OCR, table parser, master-data lookup, human review). How do you bound cost and latency, and how do you make each decision auditable? [Likely] [22]

- When is a fixed pipeline better than an agent?

87. Orchestrate a document workflow that mixes LLM prompt steps, OCR, queues and human review. How do you handle retries, idempotency and partial failure across stages? [Likely] [22]

L. Confidence thresholds, straight-through processing and human review

88. How would you set per-field confidence thresholds that send a document to human review, given a target accuracy after review? [Likely] (Textract with Augmented AI human-review product context)
    - Why per field and per document type, rather than one global threshold?
    - How do you recalibrate when the model is retrained?
89. Define straight-through-processing (STP) rate. How do you trade STP against error cost? [Likely]
    - Write the expected-cost expression you would minimize (error cost, review cost, latency).
90. Model confidence is badly calibrated on new layouts. How do you detect and fix it? [Likely]
    - Temperature scaling per document type, conformal prediction, or a separate error predictor?
91. Design the reviewer UI and workflow. What does the reviewer see, and what do you log? [Likely]
    - How do you capture reviewer corrections as training data with field-level granularity?

92. Some fields are high-stakes (payment amount, bank account number) and others low-stakes (memo). How does review policy differ? [Likely]
93. How do you sample documents that passed automatically for audit, and how big must the sample be to bound the error rate? [Likely]

M. Multimodal VLM and LLM extraction for documents

94. When would you replace an OCR-plus-layout-model pipeline with a vision-language model that reads the page image directly? [Likely]
    - What are the cost, latency and accuracy trade-offs per page?
    - How do you get bounding-box evidence from a VLM?
95. A VLM fills in a field that is not in the document ("hallucinated invoice number"). How do you detect and prevent it? [Likely]
    - How would you cross-check VLM outputs against OCR tokens?
96. How do you handle 300-page contracts that exceed the model's context window? [Likely]
    - Chunking by layout section, retrieval, or a map-reduce extraction?
97. Design a hybrid: a cheap specialist model for common layouts and a VLM fallback for hard or unseen ones. How do you decide which path each document takes? [Likely]
98. How would you distill a large VLM extractor into a smaller model for high-volume document types? [Likely] (LLM distillation at Amazon Science [13])
99. Extracted documents feed a RAG assistant. How does extraction quality affect retrieval and answers, and how would you evaluate end to end? [Likely]
    - How do you represent tables and key-value pairs in the index?
100. Design an enterprise assistant that answers questions over a company's internal documents (PDFs, scans, spreadsheets) while respecting each user's document permissions. [Likely] (synthesized Amazon question bank, not a leaked question [16])
    - Where in ingestion and retrieval do you enforce permissions?
101. Evaluate that document assistant at the claim level, separating retriever errors (claim recall, context precision) from generator errors (faithfulness, noise sensitivity, hallucination). How would a badly extracted table show up in those metrics? [Likely] (RAGChecker, Amazon Science [17])
102. A customer's VLM-based document extraction costs too much per page. Cut the cost substantially without unacceptable quality loss. [Likely] (adapted from a cost-reduction prompt in a synthesized Amazon question bank [16])
    - Page triage, cheaper OCR-plus-text model on easy pages, caching, batching, smaller distilled models: in which order would you try them?

N. Data, labeling and privacy for documents

103. How do you get labeled training data for document extraction when customer documents are confidential? [Likely] - Synthetic document generation, customer opt-in, on-premise labeling, or federated approaches?
104. How do you handle PII (SSNs, account numbers) in training data, logs and reviewer views? [Likely] - Which fields would you mask from reviewers, and how does that affect review quality?
105. How would you generate synthetic invoices or forms that actually help the model on real layouts? [Likely] - How do you measure the synthetic-to-real gap?
106. A customer wants a custom extractor for their own form with 20 examples (adapter-style fine-tuning). How would you design that feature? [Likely] - How do you stop one customer's adapter from leaking another customer's data?

O. Failure modes, monitoring and drift (both systems)

107. What are the top five failure modes of your tagging or document system, and how would you detect each in production? [Likely]

108. A vendor changes its invoice template overnight and extraction accuracy for that vendor collapses. How do you notice and recover? [Likely]

109. Precision on "brand" dropped from 97% to 91% after a model update, but overall F1 went up. What do you do? [Likely] - How would per-attribute launch gates have caught it?

110. How do you monitor drift in input distribution (new product types, new document layouts, new scanners) without labels? [Likely]

111. Errors compound across the cascade (OCR, then classification, then extraction). How do you attribute an end-to-end error to a stage? [Likely] - Would you train the pipeline end to end, or keep stages separate? Why?

112. Sellers learn to game your tagger (keyword stuffing to get tagged "organic"). How do you make it robust? [Likely]

113. How would you roll back a bad batch of auto-published attributes that already reached search and detail pages? [Likely] - What lineage and versioning do you need for that?

114. Design an end-to-end spam detection system. [Reported] (Amazon aggregate, undated, role and round not stated) [15]

- Listing application [Likely]: apply it to keyword-stuffed titles, fake attribute values and duplicate listings. Which signals come from your tagging system?
- How do you keep adversarial sellers from probing your thresholds?

P. Trade-offs, cost and the L5 bar

115. Classical extractor, fine-tuned small LLM, or a hosted frontier LLM: for each, what is the cost per million items or pages, the accuracy and the latency? How would you choose? [Likely]

116. Rules and templates still beat your model on the top 50 vendors. Do you keep them? [Likely]

117. Where would you invest the next quarter: better models, better labels, better taxonomy, or better review tooling? Defend the choice with data you would collect. [Likely]

118. How would you explain a precision-coverage trade-off to a product manager who wants "100% of attributes filled"? [Likely]

119. Tell me about a time you shipped a model whose labels were noisy or whose taxonomy was in flux. What would you do differently? [Likely] (LP blend: Dive Deep, Insist on the Highest Standards)

120. Tell me about a time you had to simplify an over-engineered extraction or labeling pipeline. [Likely] (LP: Invent and Simplify)

121. For a classification or extraction model you built, explain the backbone choice, how you handled class imbalance, and what you would do with more resources. [Reported] (Amazon MLE aggregate, project deep dive on a detection system) [7]

122. Write a symmetric, numerically stable InfoNCE loss in PyTorch for paired image and text embeddings, then explain how you would use such a model to check that a listing's images match its text attributes. [Reported for the coding prompt "Handwritten InfoNCE in PyTorch" (Amazon MLE aggregate and dated aggregator entry, February 2026) [7][15]]; tagging application [Likely]

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Note on product context: Amazon Textract, Amazon Augmented AI (A2I), Amazon Comprehend custom classification and Amazon Bedrock are public AWS products that ground the [Likely] document questions. The official docs.aws.amazon.com pages could not be fetched in either pass. Refs [20]-[22] are AWS's own public sample repositories and stand in for them.

14. End-to-End GenAI System Design: Whiteboard Diagrams and Probe Questions for RAG, Agents, Evaluation and Guardrails, and Fine-Tune-to-Serve

Scope: chapter 07 covers GenAI concepts. This chapter covers GenAI architectures in the format of the design round. It has four whiteboard skeletons and the probe questions an interviewer is likely to ask about each block:

1. a RAG shopping assistant in the style of Rufus
2. an LLM agent that uses tools
3. an LLM evaluation and guardrail pipeline
4. a fine-tuning-to-serving platform on Bedrock and Nova

Cross-cutting sections cover caching, cost and latency budgets, hallucination monitoring, and the online A/B test plus human-evaluation loop. The diagrams are empty frames for you to fill in out loud. They are not reference answers, and every box is keyed to the question numbers below. Any latency or cost figures you write on them are budgets to defend, not Amazon numbers. The question sections hold questions only, with no answers.

Research note (please read): web search was unavailable for this chapter, both in the first run and in the October 2026 review pass, because the shared search budget was already used up. Direct fetches worked only for GitHub-hosted pages. Fetches to amazon.jobs, aboutamazon.com, aws.amazon.com, 1point3acres, Blind, Glassdoor, Medium, IGotAnOffer, Dataford, OA VO Service and IEEE Spectrum all failed DNS resolution in this environment.

Verification status (October 2026 review pass):

- **Re-fetched and checked line by line:** the GitHub references [5]-[10] and [17]-[20]. Every [Reported] tag that cites them was checked against the page text. Wording that went beyond the source was corrected: question 5 now uses the source title, question 97 no longer calls video ads a GenAI feature, and the interviewer note in the format notes was rewritten. A sub-probe under question 18 duplicated question 36 and was removed.
- **Newly fetched and added:** references [28]-[35], all on GitHub. They support the new Part 7 (questions 108-134).
- **Not re-checked:** the non-GitHub references [1]-[4], [11]-[16] and [21]-[27] could not be opened. They are carried over from search extracts gathered for chapters 06, 07, 11 and 12. They are kept and not confirmed dead, but they are not confirmed live either. Re-verify them before relying on any single [Reported] item that cites only them (questions 2, 8, 15, 22, 34, 35 and 48).

How the [Reported] / [Likely] tag system works

Every question in this guide, including this chapter, carries one of two tags.

- **[Reported] [n]**: a cited source attributes this prompt, or a close paraphrase, to an Amazon interview. The source can be a candidate post on Glassdoor, Blind, 1point3acres or Medium, or a question aggregator that compiles candidate reports. The aggregator pages [5]-[7] say they are built from candidate-reported interviews. They cover Amazon Applied Scientist, Research Scientist and MLE loops together, so the round or role is noted where the source gives it. A follow-up probe under a [Reported] question is not itself reported unless it has its own tag.
- **[Likely]**: a grounded inference from Amazon's GenAI products (Rufus or Alexa for Shopping, Bedrock, Nova), from job postings and Amazon Science or AWS open-source work, or from the standard way GenAI design rounds are probed. Nobody has been seen asking it verbatim. A "(see [n])" after [Likely] points to the product or repository the inference is based on, not to an interview report.
- **"[Reported for the topic]"** means the source names the theme (for example, "alert thresholds and rollback") but not the exact wording.
- Reference [8] marks every one of its questions as synthesized ("not leaked"), so it only ever supports [Likely] items.

What the tags do not mean:

- [Reported] does not mean the question is common.
- [Likely] does not mean it is rare. Block-level probes are rarely written down by candidates.
- No tag was upgraded without a citation.

Format notes for this round

- **Slot and length.** The round is one hour-long interview in a loop of about three to six, each about 55-60 minutes, and each includes Leadership Principle questions [1][5][6][8]. The aggregators report three to five rounds for scientists [5] and four or five for ML engineers [6]. One 2025-2026 Applied Scientist II loop listed "1 Sys Design (Team-specific)" next to breadth, depth, coding and Bar Raiser rounds [2]. Plan for about 40-45 minutes of design.

- **Name of the round.** For scientists it often appears as the *Science Application* round. It is an open-ended product problem, often paired with Think Big or Invent and Simplify [5]. On ML-engineering-flavoured loops it is called *ML system design*, and the prompt comes from the team's domain: ranking, a chatbot, or an evaluation framework [6].

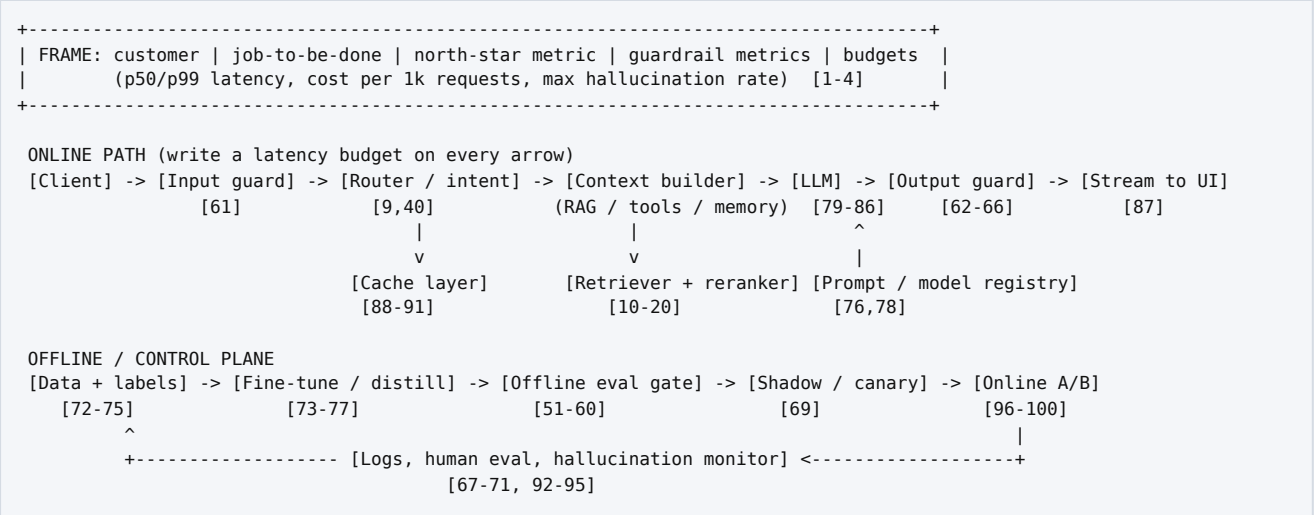
- **Interviewer and prompt.** Hiring is team by team, so the prompt follows the hiring team's domain: search and ranking, ads, generative-AI post-training or training infrastructure [5][6]. One Glassdoor loop listed the design round as "team-specific" [2]. Generative-AI recruiters ask specifically about post-training and agent experience before the loop [5][6].

- **What is graded.** Reported criteria [5]:
  - framing
  - an evaluation design that combines automatic metrics with a human protocol and reconciles the two
  - retrieval, agents or fine-tuning used with judgment, with their failure modes named
  - launch realism: latency, cost, monitoring, and what ships first

  Strong answers cover the lifecycle through monitoring. One candidate spent much of the hour on alert thresholds and rollback [6]. "There is no single right architecture" [5].
- **L5 bar.** L5 (Applied Scientist II) means "owning a science problem from framing to launch" [5]. Candidates can be rejected when their project scope falls short of the target level, even with strong technical answers [5]. On generative-AI teams, LLM depth is the most common technical reason for rejection [6]. A Blind poster reported being rejected from Amazon Ads after not preparing transformers and LLMs [27].
- **Other loop reports.** Amazon AGI loops for ML roles report an onsite with "ML application/design", and the Nova team reportedly follows the standard Applied Scientist process [9]. IGotAnOffer describes the design round as proposing data sources, an annotation strategy, a modeling technique, evaluation metrics, and trade-offs and risks [12]. Blind threads describe a design or application round as typical for L5 loops [15][16]. A vendor-published debrief says L5 follow-ups go deeper than L4 ones [26]. A Medium write-up describes the Applied Scientist II Science Depth round as resume plus GenAI fundamentals plus business impact [14].
- **Practice advice from the aggregator.** "Practice an agent, a generation-quality evaluation and a ranking system end to end in 40 minutes each, starting with the metric" [5].

Part 0. The universal GenAI whiteboard skeleton

Draw this frame first for any GenAI prompt, then specialize it with Parts 1-4. The numbers in brackets are question numbers in this chapter.



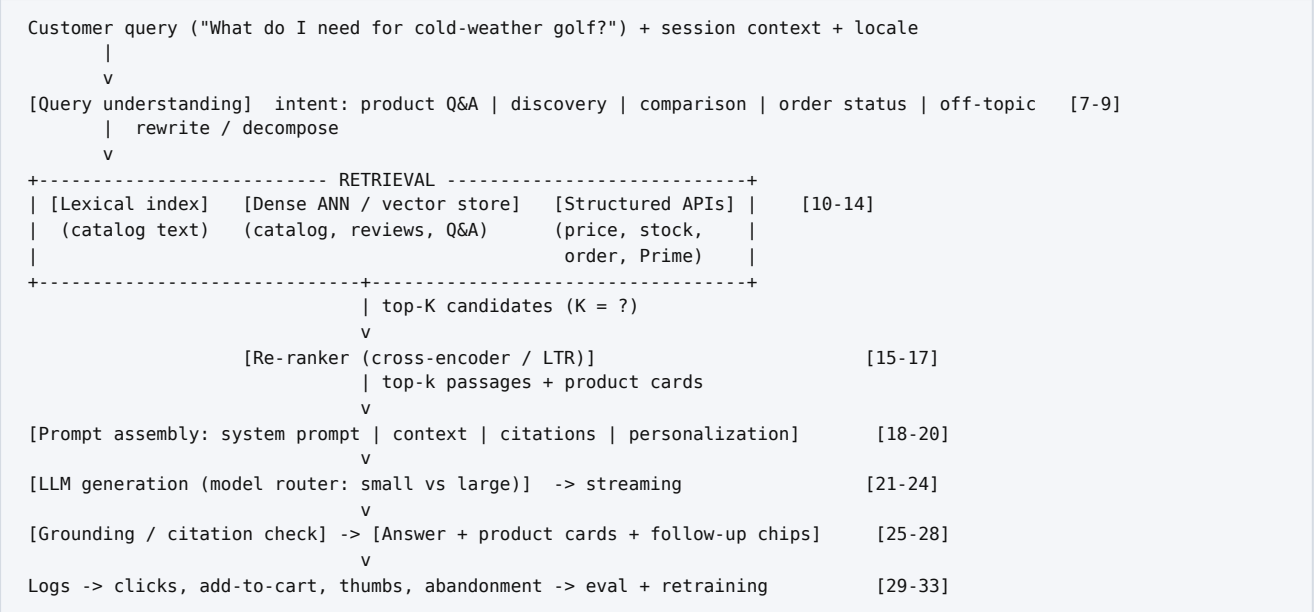
1. Before drawing anything, what clarifying questions do you ask about the customer, the surface (app, voice, web), and traffic? [Likely]
    - Which single north-star metric would you commit to, and which two or three guardrail metrics? [Likely]
2. Pick an agentic or GenAI product we could build. Who is the customer, and how do you measure success? [Reported] [2]
    - Why is this a GenAI problem at all, and what would the non-LLM baseline be? [Likely]
3. Propose agentic AI solutions for a hypothetical project and defend their viability. [Reported] (Science Application round, Amazon RS/AS aggregate) [5][6]
    - What would ship first as the smallest launchable version, and what would wait for v2? [Likely]
4. Write your latency and cost budget on the board before choosing a model. How do you split p99 latency across guard, retrieval, rerank, generation and streaming? [Likely]
    - How would the budget change for a voice surface (Alexa) compared with a text surface? [Likely]
5. Measure product-market fit for Alexa Shopping ("Measure PMF for Alexa Shopping"). [Reported] (question title in Amazon question list, undated; role not stated) [7]
    - How would the PMF plan change if Alexa Shopping is an LLM-driven conversational assistant rather than voice ordering? [Likely]
    - Which leading indicators would you trust in the first two weeks, before retention data matures? [Likely]
6. Design ChatGPT. [Reported] (Amazon question list, May 2026; role not stated) [7]
    - What changes if the same system must answer only from Amazon's catalog and reviews? [Likely]

Part 1. RAG shopping assistant (Rufus / Alexa for Shopping style)

Context for [Likely] items: Rufus is Amazon's generative shopping assistant. It answers from catalog, review and community Q&A data, and Amazon describes it as built on several LLMs, including Amazon Nova, on Bedrock [21][22][23]. An earlier chapter's search extract

says it was renamed Alexa for Shopping in May 2026 (see chapter 11). Confirm the current name with your recruiter.

1.1 Skeleton to draw



1.2 Framing and scope

7. Design an LLM shopping assistant that answers "What do I need for cold-weather golf?" from the catalog, reviews and community Q&A. [Likely] (see [21][22])
    - Which query types should the assistant decline or hand to classic search instead? [Likely]
    - How do you separate the assistant's success from search's success when both are on the same page? [Likely]
8. Design an enterprise RAG service. [Reported] (Amazon ML onsite) [4]
    - How would per-user permissions (ACLs) be enforced at retrieval time instead of after generation? [Likely] (see [8])
9. Design an AI chatbot architecture covering intent detection, dialogue state, retrieval, generation, fallback and continuous learning ("Design a Reliable AI Chatbot Architecture"). [Reported] (Science Application / ML design, Amazon RS, MLE aggregates; title sighted Aug 2026) [5][6][7]
    - What is the fallback when retrieval returns nothing relevant, and how is that decided? [Reported for the topic: fallback] [5]
    - How does the system learn continuously from production conversations without learning from its own mistakes? [Reported for the topic: continuous learning] [5] [6]

1.3 Retrieval and vector store

10. What sources would you index (titles, bullets, descriptions, reviews, Q&A, images), and how would you chunk each? [Likely] (see [24])
    - How do you keep the index fresh when price, stock and reviews change every minute? [Likely]

11. Lexical, dense or hybrid retrieval for product questions: when does each fail? [Likely]
    - How would you fuse the scores, and how would you tune the fusion weights? [Likely]
12. Compare a small and a large model from the same text-embedding family for this retriever. What do you gain, and what do you pay? [Reported] (Amazon RS aggregate, breadth round) [5]
    - What are the index memory and re-embedding costs when you upgrade the embedding model? [Likely]
13. How would you choose the ANN index type and its recall/latency settings for hundreds of millions of product vectors? [Likely]
    - How do you measure ANN recall loss against exact search in production? [Likely]
14. How would you fine-tune the embedding model on Amazon shopping data (query-product pairs, hard negatives)? [Likely] (see the ESCI shopping-queries dataset [24])
    - How do you mine hard negatives without pulling in false negatives? [Likely]

1.4 Re-ranking

15. What is the difference between a cross-encoder and a bi-encoder, and where does each sit in this pipeline? [Reported] [3]
    - How many candidates can the cross-encoder afford to score within the latency budget? [Likely]
16. Design a search retrieval and ranking pipeline with online experiments and position-bias correction. [Reported] (Amazon MLE/AS aggregate) [6]; a faceted product-search variant appears as "Design Faceted Product Search at Large Scale" [Reported] (Amazon question list, Aug 2026; role not stated) [7]
    - What position bias enters the logged training data when the LLM, rather than a ranked list, presents the products? [Likely]
17. Should the re-ranker optimize relevance to the question, purchase likelihood, or both? How do you avoid an assistant that just pushes sponsored items? [Likely]

1.5 Prompt assembly and generation

18. How do you budget the context window across the system prompt, conversation history, retrieved passages and product attributes? [Likely]
    - When do you summarize, retrieve or truncate conversation history? [Likely] (the reported multi-turn search agent prompt is question 36)
19. How do you add personalization (purchase history, preferences) to the prompt without privacy leaks or filter bubbles? [Likely]
20. How do you make the answer cite specific products and reviews so that each claim can be traced? [Likely]
21. How would you route each query between a small and a large model, and what signal triggers escalation? [Likely] (see [21])
    - How do you check offline that the router does not send hard queries to the small model? [Likely]
22. Which decoding settings (greedy, beam, temperature, top-p) would you use for a customer-facing shopping answer, and why? [Reported] (Applied Scientist II, LLM team) [13]

23. The model must answer in 10+ marketplaces and languages. Do you translate the query, retrieve multilingually, or use per-locale models? [Likely]
24. How would you stream the answer while product cards and prices are still being fetched? [Likely]

1.6 Grounding, hallucination and failure handling

25. What would you do if the assistant states a wrong price, wrong compatibility or a wrong return policy? [Likely]
    - Which facts must never come from the LLM's parametric memory, and how do you enforce that? [Likely]
26. How would you detect, at serve time, an answer that is not supported by the retrieved context? [Likely]
    - What is the latency cost of a separate groundedness checker, and when would you run it asynchronously instead? [Likely]
27. When should the assistant say "I don't know" or ask a clarifying question instead of answering? [Likely]
28. A seller stuffs reviews or listings with text written to manipulate the assistant (indirect prompt injection). How do you defend against it? [Likely]

1.7 Metrics for the RAG assistant

29. Which offline metrics would you track for the retriever, the re-ranker and the generator separately? [Likely] (see Amazon's RAGChecker, which splits retriever and generator diagnostics [17])
    - How do you build the labelled evaluation set, and how do you stop it from going stale? [Likely]
30. How would you evaluate a RAG system automatically on a new domain with no labelled data, for example with generated exams? [Likely] (see Amazon's auto-rag-eval, ICML 2024 [18])
31. Which online metrics show that the assistant helped shopping, rather than just being used? [Likely]
    - How do you attribute a purchase that happened two sessions after an assistant conversation? [Likely]
32. Offline metrics improved but the A/B test fell short. How do you find out why? [Reported] (Amazon RS/MLE aggregates, depth round) [5][6]
33. How would you measure and report the assistant's hallucination rate to leadership every week? [Likely]

Part 2. LLM agent with tool use

2.1 Skeleton to draw

```
User goal ("reorder my usual coffee, but cheaper if possible")
       |
       v
[Planner LLM] <---- [Memory: session state | user profile | episodic summaries]     [36-38]
       |  plan / next action (structured tool call, JSON schema)
       v
[Policy + permission gate] -- irreversible? --> [Human confirmation]                [44-46]
       |
       v
+---------------------- TOOL LAYER --------------------------+
| search() | get_product() | compare() | cart.add() | order() |                     [39-42]
| (timeouts, retries, idempotency keys, rate limits)          |
+-----------------------------+-----------------------------+
                              | observation (untrusted text)
                              v
[Verifier / critic] -- fail --> replan (step budget N, token budget T)              [43,47]
       | pass
       v
[Final response + action summary]  -> trajectory logs -> agent eval harness         [48-50]
```

2.2 Prompts

34. Design an agentic recommendation system for Amazon, covering the component architecture, model choices, and quality and safety evaluation. [Reported] [11]
    - Where in the architecture does the agent add value over the existing recommender, and where does it only add latency? [Likely]
35. Design a GitHub Copilot-style coding agent. [Reported] (Senior AS loop) [3]
    - How would you scope it down for an L5 answer in 40 minutes? [Likely]
36. Design a multi-turn search agent and explain how it handles long conversation context. [Reported] (Science Application, Amazon RS aggregate) [5]
    - What goes into short-term memory, what into long-term memory, and how is stale memory expired? [Likely]
37. Design a browser-operating agent (in the spirit of Nova Act) that completes multi-step web tasks reliably. [Likely] (see [8], synthesized; chapter 11)
38. Should the agent be one LLM with many tools, or a router with specialist sub-agents? What evidence would decide it? [Likely]

2.3 Tools, planning and reliability

39. How do you define tool schemas so the model calls them correctly, and how do you validate arguments before execution? [Likely]
40. With 200 available tools, how does the agent pick the right one without stuffing every schema into the prompt? [Likely]
41. A tool times out or returns an error halfway through a multi-step plan. What happens next? [Likely]
    - How do you make tool calls idempotent so that a retry does not place two orders? [Likely] (see [8], synthesized)
42. How would you detect and suppress loops or repeated tool calls within one run? [Likely]
43. How do you set and enforce step, token and wall-clock budgets per task, and what does the user see when the budget runs out? [Likely]
44. Which actions require explicit human confirmation, and how is that decided? [Likely] (see [8], synthesized)
45. Tool outputs can carry prompt-injection text (a product page that says "ignore previous instructions"). How do you isolate untrusted observations? [Likely]

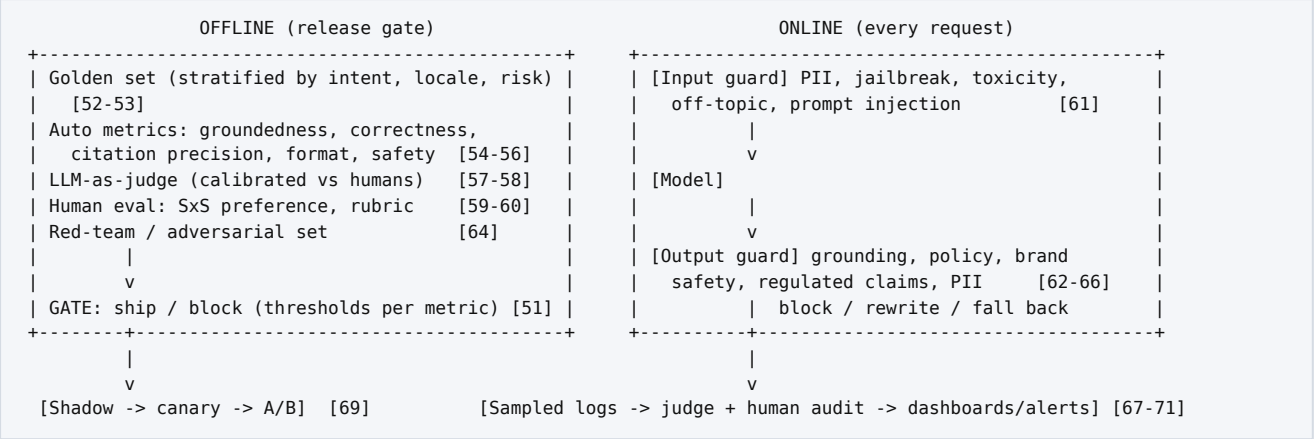
46. How do you enforce that the agent acts only with the user's own permissions, for example on payments and addresses? [Likely]
47. Would you add a verifier or critic model, and how do you stop the verifier from sharing the planner's blind spots? [Likely]

2.4 Agent evaluation

48. Evaluate an LLM agent with metrics beyond accuracy: tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you score a trajectory that reached the right outcome through an unsafe step? [Likely]
49. Agents are stochastic. How do you report reliability (pass@k, pass^k, variance across seeds), and how many runs per task do you need? [Likely]
    - How would you run multi-turn, LLM-driven simulated-user tests in CI before every agent release? [Likely] (see AWS Labs agent-evaluation [19])
50. How would you build a failure taxonomy from production trajectories and turn it into regression tests? [Likely]

Part 3. LLM evaluation and guardrail pipeline

3.1 Skeleton to draw



3.2 Evaluation pipeline prompts

51. Design an LLM quality validation system. [Reported] (Amazon RS/MLE aggregates; title in Amazon question list) [5][6][7]
    - What do you measure offline, and which online guardrails do you set? [Reported for the topic] [6]
    - How do automated checks and human review divide the work for generated output? [Reported for the topic] [6]
52. How would you build and maintain a golden evaluation set for a shopping assistant across intents, locales and risk levels? [Likely]
    - How do you stop the golden set from leaking into fine-tuning data? [Likely]
53. Design an evaluation framework that checks whether a text-to-image system spells a brand name correctly and places it naturally. [Reported] (Science Application; Amazon

RS/MLE aggregates) [5][6]
    - How would OCR accuracy, embedding-based semantic checks and a human review protocol fit together? [Reported for the topic] [5]
54. How would you measure groundedness (faithfulness to retrieved context) at the claim level? [Likely] (see [17])
55. Which metrics catch a model that is fluent and confident but wrong? [Likely]
56. How do you evaluate format and instruction-following, for example JSON for product cards, length limits and tone? [Likely]
57. Would you use an LLM as a judge? How do you calibrate it against human labels and detect its biases (position, verbosity, self-preference)? [Likely]
    - How often must the judge be re-validated when the judged model or the judge model changes? [Likely]
58. When is perplexity misleading for comparing checkpoints, and how do pairwise preference evaluations (win rate, Elo or Bradley-Terry) complement it? [Reported for the topic] (Amazon AGI candidate summaries; see chapter 07, [8]); exact wording [Likely]
59. Design the human-evaluation protocol: rubric, side-by-side preference, number of raters, agreement and cost. [Likely]
    - How do you size the human-eval sample to detect a 2-point win-rate difference? [Likely]
60. Offline judge scores say model B is better, but human raters prefer model A. What do you do? [Likely]

3.3 Guardrails

61. What does the input guard check (PII, jailbreak, toxicity, off-topic, injection), and should it be a classifier, rules, or an LLM? [Likely] (see [20], the responsible-AI and securing-RAG examples)
    - How do you trade false blocks (lost customers) against misses (safety incidents), and who sets the thresholds? [Likely]
62. What does the output guard check before an answer reaches a customer: grounding, policy, regulated claims (health, safety), competitor or brand issues? [Likely]
63. When the output guard trips, do you block, rewrite, regenerate or fall back to a template? What does each cost in latency? [Likely]
64. How would you red-team the assistant before launch, and how do red-team findings become regression tests? [Likely]
65. How do you run guardrails on a streamed response, where tokens reach the user before the full answer exists? [Likely]
66. Should guardrails be configurable per tenant or per marketplace (as in a managed service such as Bedrock), and how do you test each configuration? [Likely] (see [8], synthesized; [25])

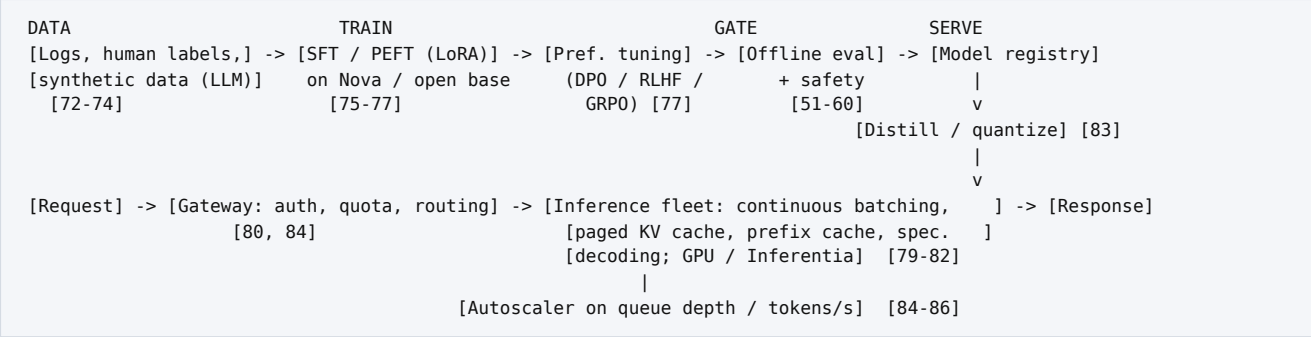
3.4 Hallucination monitoring in production

67. Design a production hallucination monitor: what do you sample, who or what labels it, and what triggers an alert? [Likely]
    - How would you set alert thresholds and a rollback rule for a quality metric that is noisy day to day? [Reported for the topic: alert thresholds and rollback] [6]

68. Which cheap online proxies (thumbs-down, rephrase, abandonment, returns) correlate with hallucination, and how would you validate them? [Likely]
69. Walk through shadow mode, canary and full A/B for a new model version. What has to be true to move from each stage to the next? [Likely] (see [8], synthesized)
70. Catalog data changed, and the assistant now confidently cites a discontinued product. How would monitoring catch it? [Likely]
71. How do you detect drift in the query mix (new seasonal intents, new product categories) that makes the offline evaluation set unrepresentative? [Likely]

Part 4. Fine-tuning to serving infrastructure (Bedrock / Nova)

4.1 Skeleton to draw



4.2 Data and training

72. For a shopping assistant, when do you fine-tune, when do you use RAG, and when is prompting enough? What evidence would decide? [Likely]
73. How would you build the SFT dataset: which sources, how labelled, how deduplicated, and how checked for quality? [Likely]
    - Design a distributed pipeline that ingests, tokenizes, deduplicates and filters training data. [Reported] (Amazon MLE aggregate, staff loop) [6]
74. How would you use a large model to generate or label training data for a smaller one, and how would you control label noise? [Likely]
75. Explain the main parameter-efficient fine-tuning techniques and when each beats full fine-tuning. [Reported] (online assessment, Amazon RS/MLE aggregates) [5][6]
    - When would you serve many LoRA adapters on one base model, and what does that do to batching? [Likely]
76. How would you customize a Nova model on Bedrock compared with self-hosting an open-weights model on SageMaker? What drives the choice? [Likely] (see [20][25])
77. Why does GRPO remove the critic, what does that cost, and how would you design the reward for the assistant? [Reported] (phone screen, Amazon RS aggregate) [5]
    - How would you parallelize GRPO training? [Reported] (ML depth round, Amazon MLE aggregate) [6]
    - How do you design a preference dataset or reward from customer feedback without rewarding sycophancy or upselling? [Likely]

78. How do you version prompts, adapters, retrieval indexes and guardrail configs together so that a rollback is consistent? [Likely]

4.3 Serving, latency and cost

79. Describe how an inference engine such as vLLM raises serving throughput for large language models. [Reported] (online assessment, Amazon RS/MLE aggregates) [5][6]
    - How do continuous batching and a paged KV cache interact with p99 time-to-first-token? [Likely]
80. For a residual MLP block in BF16, is it compute-bound or memory-bound at batch size 1 and at batch size 256? Do the analysis from bandwidth and peak throughput. [Reported] (manager screen, Amazon RS/MLE aggregates; title sighted Aug 2026) [5] [6][7]
    - What does this imply for the batch size you would serve the assistant at? [Likely]
81. Break the end-to-end latency into queue time, time-to-first-token and per-token time. Which would you optimize first for a chat surface, and which for voice? [Likely] (see [8], synthesized)
82. When would you use speculative or parallel decoding, and how do you check that it does not change output quality? [Likely]
83. You must cut serving cost by 50% with no measurable quality loss. Walk through distillation, quantization, routing, caching and batch inference in the order you would try them. [Likely] (see [8], synthesized; chapter 11)
    - How do you prove that quality did not regress after each cost change? [Likely]
84. Prime Day brings 10x traffic. How do you plan capacity (provisioned versus on-demand), and what degrades gracefully first? [Likely]
85. Design a multi-tenant inference platform: control plane versus data plane, per-tenant throttling, and noisy-neighbor isolation. [Likely] (see [8], synthesized)
86. The p99 latency doubled after a deploy. How do you triage it? [Likely] (see [8], synthesized)
87. How do you stream tokens to the client, and how do you stop upstream generation quickly when the user leaves? [Likely]

4.4 Caching

88. Where would you cache in this system: query-to-answer, retrieval results, embeddings, prompt prefixes, KV prefixes? What does each save? [Likely]
89. How would you design a semantic cache for shopping questions, and how do you stop it from serving a stale price or another user's personalized answer? [Likely] (see [8], synthesized)
    - How do you measure the cache's hit rate and its error rate? [Likely]
90. How does prefix caching relate to the KV cache, and how would you lay out the prompt to maximize prefix reuse? [Likely] (see [8], synthesized)
91. What is the cross-tenant leakage risk of prompt caching in a shared service, and how do you prevent it? [Likely] (see [8], synthesized)

Part 5. Online A/B and human-evaluation loop (closing the system)

```
[New model / prompt / index] -> [Offline gate] -> [Human SxS eval] -> [Shadow] -> [A/B: treatment vs control]
                                                                                          |
    [Decision memo: ship / iterate / kill]  <- [Guardrail check] <- [Metric readout] <-+
                   |
                   +-> [Failure mining -> new golden items + training data]  (loop back)
```

92. Which logged signals (explicit feedback, rephrases, add-to-cart, returns) would you collect for evaluation and retraining, and how do you de-bias them? [Likely]
93. How do you sample production conversations for human review: uniform, risk-weighted or uncertainty-weighted? [Likely]
94. How do you keep human raters consistent across marketplaces and languages, and how do you audit rater quality? [Likely]
95. How do you stop the feedback loop from training the model on its own hallucinations? [Likely]
96. Design the A/B test for a new assistant model: unit of randomization, primary metric, guardrails, duration and power. [Likely]
    - Conversations span sessions and users share devices. How do you handle interference and carry-over effects? [Likely]
97. How would you evaluate adding video ads? [Reported] (Amazon question list, Nov 2025; role not stated) [7]
    - How would the evaluation change if the feature added to the page were an LLM-generated element, such as an AI review summary, instead of video ads? [Likely]
98. The treatment lifts engagement with the assistant but lowers conversion on the page. Ship or not? [Likely]
99. How would you use interleaving or counterfactual (off-policy) evaluation to compare rankers or answer strategies faster than an A/B test? [Likely]
100. Write the one-paragraph launch recommendation you would send to a director after the test. Which numbers must it contain? [Likely]

Part 6. L5-level cross-cutting probes (any of the four designs)

101. Which part of this design is the riskiest science bet, and how would you de-risk it in the first month? [Likely]
102. What would you build if you had one-tenth of the GPU budget? [Likely]
103. Which component would you buy (Bedrock managed features) and which would you build, and why? [Likely] (see [25])
104. Which component fails silently, and how would you find out? [Likely]
105. How would this design change for Alexa voice, where there is no screen for product cards or citations? [Likely]
106. Design Amazon's recommendation system, then explain where an LLM would and would not improve it. [Reported for the base prompt] (Amazon system-design list, SDE-leaning aggregate) [10]; the LLM extension is [Likely]

107. Walk me through the GenAI system you built that is closest to this one. What would you change now? [Likely] - Explain an AI-agent project you built and how it was adopted. [Reported] (Amazon MLE aggregate, project deep dive) [6]

Part 7. Additional probes (October 2026 review pass)

These questions were added in the review pass. They are grouped by the part of the design they extend, and numbering continues from Part 6 so that the question numbers on the diagrams stay valid. A [Reported] tag here cites a question title or sample question on a fetched GitHub aggregator page. Where only the base prompt is reported, the GenAI extension is tagged [Likely] separately.

7.1 Extends Part 1 (RAG shopping assistant)

108. Design a RAG system. [Reported] (Amazon question list, page 2; System Design; undated; role not stated) [29]
    - Walk the board from ingestion and chunking through retrieval, re-ranking, generation and evaluation in 40 minutes. Which block would you spend the most time on for a shopping use case? [Likely]
109. Compare approaches to building joint image-text embeddings. [Reported] (online assessment, Amazon RS aggregate) [5]
    - A customer uploads a photo of a broken part and asks "what replaces this?" How would you add image queries to the retriever in Part 1? [Likely]
110. Compare a dual-encoder model with a vision-language model that uses a resampler. [Reported] (science breadth round, Amazon RS aggregate) [5]
    - Which of the two would you use for multimodal retrieval, and which for answering a question about a product image? [Likely]
111. Using the Exact, Substitute, Complement and Irrelevant labels from the Shopping Queries dataset, how would you get the assistant to recommend a substitute when the exact item is out of stock, and how would you evaluate those recommendations? [Likely] (see [24])
112. Design typeahead suggestions. [Reported] (multi-company list that includes Amazon, April 2026; role not stated) [30]
    - How would you generate the assistant's follow-up question chips with an LLM within the same latency budget as typeahead? [Likely]
113. When would a long-context model replace retrieval for a product-comparison query, and how would you test whether facts in the middle of a long context are lost? [Likely]
114. Online review content moderation: flag reviews that should not be published. [Reported] (title "Online Review Content Moderation", Amazon question list, page 2; listed as an AI-coding task, not a design round; undated) [29]; a related design title, "Design an end-to-end spam detection system", appears on page 1 of the list [Reported] [7]
    - How would you extend the moderation system to catch reviews written to manipulate an LLM assistant that retrieves them (see question 28)? [Likely]

7.2 Extends Part 2 (LLM agent with tool use)

115. An intent classifier sends each turn to a specialist agent (product Q&A, orders, returns). How do you keep conversation context consistent across agents, and how do you measure routing errors? [Likely] (see the classifier-routed orchestrator in [33])
116. A model-driven agent loop or an explicit workflow graph: when do you let the model plan freely, and when do you hard-code the steps? [Likely] (see [35])
117. Which agent concerns would you hand to a managed runtime (memory, identity, a tool gateway, policy, tracing), and which would you keep in your own code? [Likely] (see Bedrock AgentCore [34])
118. You expose existing services, such as order history and returns, to the agent as tools through a protocol such as MCP. How do you version tool contracts so that a schema change does not break agents already in production? [Likely] (see [34])
119. What would you record in an agent trace (spans for each LLM call and tool call) to debug a failed trajectory, and how would you sample traces at Amazon scale? [Likely] (see [34])
120. Walk through a concrete AI-assisted engineering workflow: what you delegated, what context you protected, and how you verified the output. [Reported] (new-grad SDE screens, Aug 2026; not an Applied Scientist report) [10]
    - If you were designing the Copilot-style coding agent from question 35, how would you measure whether its users verify its output? [Likely]

7.3 Extends Part 3 (evaluation, guardrails and hallucination monitoring)

121. Would a hallucination checker extract claims as sentences, sub-sentences or knowledge triplets? How does the granularity change precision, recall and cost? [Likely] (see Amazon's RefChecker [31])
    - How does checking differ when the reference is accurate retrieved context, noisy context, or no context at all? [Likely] (see [31])
122. How would you test semantic robustness, that is, whether the answer changes when the query has typos, a paraphrase or different casing? [Likely] (see [32])
123. How would you measure stereotyping or toxicity across demographic groups in shopping answers before launch, and which result would block the launch? [Likely] (see [32])
124. How would you run evaluators online on sampled production agent sessions, and how would you keep the evaluators' cost within budget? [Likely] (see the online evaluation in [34])
125. Design a metrics system. [Reported] (Amazon question list, page 2; System Design, Hard; undated; role not stated) [29]
    - Which per-request fields (model version, prompt version, retrieved document IDs, guard verdicts, tokens, latency) must the metrics system store so that a hallucination spike can be traced to one change? [Likely]

7.4 Extends Part 4 (fine-tuning to serving)

126. Explain parallelism and the collective operations used in training. [Reported] (Amazon question list; System Design, Hard; undated) [7]
    - Which parallelism would you use to fine-tune a 70B-parameter model for the assistant on a fixed GPU budget, and what does each choice cost in communication? [Likely]

127. Calculate the memory size for a given model structure. [Reported] (title "Memory Size Calculation for DDN Model Structure", Amazon question list, page 2; AI Coding) [29]
    - Estimate the serving memory for the assistant: weights, KV cache per request at your context length, and activations. How many concurrent requests fit on one accelerator? [Likely] (the aggregator advises being able to explain "the memory arithmetic behind LoRA, quantization and KV caching" [6])
128. The base model you fine-tuned is deprecated or its roadmap narrows. How would you design for model portability across adapters, prompts and evaluation suites? [Likely] (see the reported Nova roadmap note in [8])

7.5 Extends Parts 5 and 6 (A/B loop and cross-cutting probes)

129. Design a causal study for the impact of reminders. [Reported] (Amazon question list; System Design, Hard; undated) [7]
    - The assistant sends a proactive message ("the price on your saved item dropped"). How would you estimate its causal effect on purchases when randomization is limited? [Likely]
130. Predict click probability for music recommendations. [Reported] (Amazon question list, Sep 2026; role not stated) [7]
    - How would you add a conversational LLM layer on top of this recommender, and how would you evaluate whether it beats the click model alone? [Likely]
131. Diagnose and fix an underperforming ML model. [Reported] (title linked from the Amazon RS and MLE aggregates) [5][6]
    - The assistant's weekly groundedness score fell three points with no deploy. Walk through your diagnosis across data, retrieval index, model and guardrails. [Likely]
132. Tell me about a time generative AI gave you a wrong answer. How did you catch it, and what did you change? [Reported] (described as close to standard across Amazon loops since 2025) [28]
133. Walk me through your post-training or agent experience. [Reported for the topic] (recruiter screens for generative-AI teams) [5][6]
    - Which of those projects would you present as an L5 example of owning a science problem from framing to launch? [Likely]
134. Your hiring team owns a classic ranking or ads system. Where would you add an LLM first, and what evidence would make you remove it? [Likely]

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Entries marked "fetched" were read in this chapter's first run and re-fetched and checked in the October 2026 review pass. Entries marked "fetched (review pass)" were first read in that pass. Entries marked "carried over" come from earlier chapters of this guide. They were not re-read, because their hosts failed DNS resolution in this environment, so they are unverified rather than confirmed dead.

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